

Geometry of Observation

Frozen Typed Corpus and Reproducibility Master

Eighteen reference modules, one scoped contract, and an immutable release map

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P12 corpus freeze v1.3.0 — 28 July 2026

Release status. This master freezes 18 normative modules with 158 component pages, 347 typed expressions, and zero GO Core findings. It is a provenance and reproducibility object. Concatenation does not turn the component results into one theorem, does not establish physical equivalence between application domains, and does not claim a new theory of general relativity, information theory, mechanics, quantum chemistry, accelerator physics, or astrodynamics.

Abstract

This release freezes the strict Geometry of Observation corpus after the P0-P11 migration sequence. The common mathematical discipline is not a universal physical dynamics but a typed separation of hidden state, invertible representation change, reduction, observation channel, sampling protocol, estimator, and declared invariant or recoverable target. The frozen object contains eighteen independently scoped reference modules, the GO Core v0.2 type and dimension contract, a machine-readable corpus ledger, an acyclic dependency ledger, canonical release metadata, exact component hashes, and executable validation. Every component PDF is embedded without textual revision; its original hash remains the authority for component identity. Short identifiers are document-scoped and receive fully qualified release names. Strong, model, hypothesis, diagnostic, and empirical claims remain separated. The master therefore records a clean corpus state while preserving the scientific boundaries of each module.

Аннотация. Релиз фиксирует строгий корпус геометрии наблюдения после миграций P0-P11. Общей структурой является не единая физическая динамика, а типизированное разделение скрытого состояния, обратимой смены представления, редукции, канала наблюдения, дискретизации, оценителя и явно объявленного инварианта или восстанавливаемой величины. В master включены восемнадцать модулей без переписывания их содержимого, машинные реестры, зависимости, контрольные суммы и воспроизводимые проверки. Совпадение структуры между областями не трактуется как доказательство физической эквивалентности.

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1 The frozen release object

Definition 1.1 (Corpus freeze). Let

$$\mathfrak{F} = (C, L, O, D, S, H, M) \quad (1)$$

denote, respectively, the core contract, corpus ledger, canonical module order, dependency ledger, component source map, component hash map, and release metadata. A release is frozen when all seven entries are versioned and every file named by S has a verified digest in H . Two freezes are identical precisely when their canonical serializations and every referenced byte stream agree.

The baseline ledger is go-corpus-ledgers@1.2.0, with digest 3f50f7941f8549c4b9de626fdb107431c07238582cdf8ebce90203500189f68e. P12 does not mutate that ledger. Instead it adds a release-level ledger whose order and dependency semantics are explicit. A change to a component byte, ordering rule, dependency edge, author identity, licence, or release metadata requires a new release version. A filename change alone is not sufficient to create a new scientific object, while a changed byte stream cannot retain the old freeze identity.

Proposition 1.2 (Hash-addressed component identity). *Assume collision resistance of SHA-256 for the finite release set. If two release records have the same ordered component paths, byte counts, and SHA-256 digests, then the probability of an undetected component substitution is negligible under the stated cryptographic assumption.*

Boundary 1.3. This is an integrity statement, not a mathematical proof that the scientific claims are true. Hash agreement proves preservation of the audited object; it does not replace peer review.

2 Common typed architecture

The minimal release-wide observation pattern is

$$\mathcal{X} \xrightarrow{\Phi_t} \mathcal{X}_t \xrightarrow{F_g} \mathcal{X}'_t \xrightarrow{R_\lambda} \mathcal{X}_\lambda \xrightarrow{\mathcal{Q}_\lambda} \mathcal{Y}_\lambda \xrightarrow{\mathcal{S}_{\Delta t, T}} \mathcal{Y}_\lambda^M \xrightarrow{\mathcal{A}} \mathcal{I}. \quad (2)$$

Here F_g is an invertible frame, gauge, coordinate, or basis change. The reduction R_λ , observation $\mathcal{Q}_\lambda$, and sampler $\mathcal{S}_{\Delta t, T}$ may lose information. The estimator $\mathcal{A}$ is a decision rule, not a physical transformation of the hidden system. A module may omit stages only by declaring the corresponding identity or noiseless map.

For a declared group G , protocol λ , and representation ρ , the release distinguishes

$$I(g \cdot x; \lambda) = I(x; \lambda) \quad (\text{invariance}), \quad (3)$$

$$\mathcal{Q}_\lambda(g \cdot x) = \rho(g) \mathcal{Q}_\lambda(x) \quad (\text{covariance}), \quad (4)$$

$$T(x) = \bar{T}(\mathcal{Q}_\lambda(x)) \quad (\text{deterministic recoverability}). \quad (5)$$

In a stochastic finite-alphabet setting, exact recoverability of a target $T(X)$ from Y is equivalently stated as

$$H(T(X) \mid Y) = 0 \quad (6)$$

under the declared probability law. None of these statements is well-typed without its transformation group, domain, protocol, and unit context.

Boundary 2.1 (No cross-domain identity claim). The recurrence of equation (2) in geometry, mechanics, spectroscopy, molecular inference, or satellite networks records a shared inference architecture. It does not identify their state spaces, Hamiltonians, field equations, probability laws, or empirical content.

3 Corpus layers and canonical order

The master order is semantic rather than chronological. It preserves every component identity while placing interfaces before the laboratories that use them.

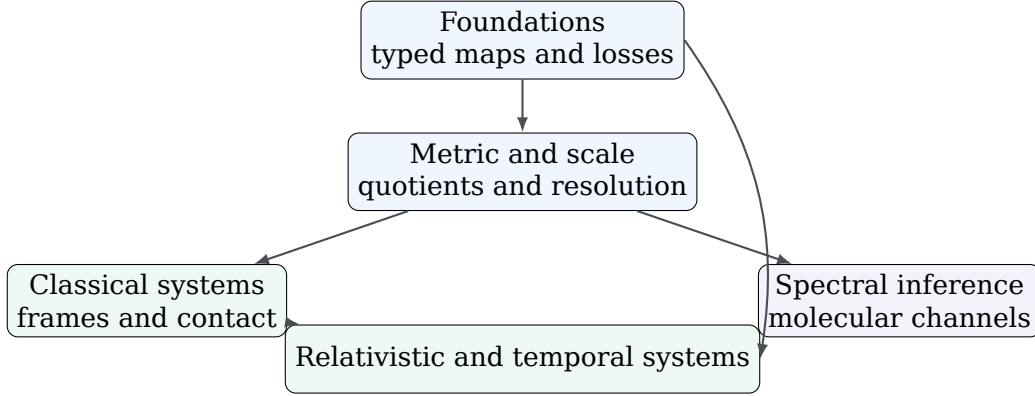


Table 1: Canonical semantic order of the frozen corpus. Counts are module-local.

No.	Layer	Module	v	pp.	Expr.	Claims
1	Foundation	Functional Interface GO-QS	0.2.0	12	3	2
2	Foundation	Tensorial Observation Geometry in GR	0.3.0	11	2	3
3	Information	Information-Theoretic Observation Geometry	0.2.0	9	9	9
4	Metric	Metric Entropy and Observational Entropy Defect	0.2.0	10	13	12
5	Metric	Distance and Scale Interface under Observation Maps	0.2.0	9	12	8
6	Scale	Planck-to-Cosmos Observation Rulers	1.1.0	12	19	10
7	Scale	Mandelbrot Rulers as Observation-Scale Geometry	1.1.0	11	12	9
8	Discrete geometry	Regular Polyhedra under Typed Observation Filters	1.1.0	7	42	10
9	Mechanics	Frames, Forces, Constraints, and Dissipation under Observation Maps	0.1.0	7	10	4
10	Mechanics	Foucault Networks on Celestial Bodies	1.1.0	6	7	4
11	Dynamics	Billiards as a Typed Geometry of Observation Laboratory	1.1.0	9	24	13
12	Contact mechanics	Bobsleigh Contact Geometry as a Typed Observation System	1.1.0	6	11	4
13	Mechanics	Roller-Coaster Geometry as a Typed Observation Laboratory	1.1.0	6	11	4
14	Contact mechanics	Gear Contact Geometry as a Typed Finite Observation System	1.1.0	9	20	9
15	Spectral	Conical Intersections as Typed Spectral Singularities under Observation Maps	1.1.0	8	36	11

No.	Layer	Module	v	pp.	Expr.	Claims
16	Molecular inference	Quantum Chemistry as a Typed Inference Stack under Observation Maps	1.1.0	9	47	11
17	Relativistic	Relativistic Beam Paths as Observation Geometry	1.3.0	9	16	4
18	Temporal networks	Satellite Networks under Typed Frames and Temporal Observation Channels	1.2.0	8	53	9

4 Inter-module dependency audit

Dependencies are typed. A *core-contract* edge is normative: the target ledger must validate against GO Core v0.2. An *extension-contract* edge binds a module to an auxiliary contract such as the SI-HEP passport or the mechanics interface. A *documented-interface* edge records an explicit source-level reference between modules. A *contextual* relation is excluded from the normative graph and cannot be used as a proof dependency.

Table 2: Explicit non-global dependency edges. The universal GO Core edge to all modules is omitted from the rows.

From	To	Edge kind
go-distance-scale-contract@0.3.0	Distance and Scale Interface under Observation Maps	extension contract
go-distance-scale-contract@0.3.0	Mandelbrot Rulers as Observation-Scale Geometry	extension contract
go-planck-cosmos-scale-contract@0.4.0	Planck-to-Cosmos Observation Rulers	extension contract
go-si-hep-quantity-passport@0.5.0	Relativistic Beam Paths as Observation Geometry	extension contract
go-frame-force-dissipation-contract@0.6.0	Frames, Forces, Constraints, and Dissipation unde...	extension contract
go-frame-force-dissipation-contract@0.6.0	Foucault Networks on Celestial Bodies	extension contract
go-frame-force-dissipation-contract@0.6.0	Bobsleigh Contact Geometry as a Typed Observation...	extension contract
go-frame-force-dissipation-contract@0.6.0	Roller-Coaster Geometry as a Typed Observation La...	extension contract
go-frame-force-dissipation-contract@0.6.0	Gear Contact Geometry as a Typed Finite Observati...	extension contract
go-gear-contact-contract@0.7.0	Gear Contact Geometry as a Typed Finite Observati...	extension contract

From	To	Edge kind
go-billiards-observation-contract@0.8.0	Billiards as a Typed Geometry of Observation Labo...	extension contract
go-conical-intersections-observation-contract@0.9.0	Conical Intersections as Typed Spectral Singulari...	extension contract
go-quantum-chemistry-observation-contract@1.0.0	Quantum Chemistry as a Typed Inference Stack unde...	extension contract
go-regular-polyhedra-observation-contract@1.1.0	Regular Polyhedra under Typed Observation Filters	extension contract
go-satellite-networks-observation-contract@1.2.0	Satellite Networks under Typed Frames and Tempora...	extension contract
Information-Theoretic Observation Geometry	Metric Entropy and Observational Entropy Defect	documented interface
Metric Entropy and Observational Entropy Defect	Distance and Scale Interface under Observation Maps	documented interface
Distance and Scale Interface under Observation Maps	Planck-to-Cosmos Observation Rulers	documented interface

The normative and documented-interface graph has 29 nodes and 36 edges. Its topological sort is unique only up to independent branches; acyclicity is the relevant release condition. Each edge endpoint resolves to a frozen module or a versioned contract, and each documented evidence fragment is checked against the cited source.

5 Namespaces and symbol discipline

Document-local identifiers are not silently promoted to global names. For document d and local identifier a , the master name is

$$\text{qid}(d, a) = d : : a. \quad (7)$$

This operation is injective on ordered pairs. The audit found 7 repeated short identifiers across expression and claim registers; all become distinct after qualification. These overlaps are reuse of local vocabulary, not ledger collisions. Map identifiers and quantity identifiers are already globally unique in the frozen ledger.

Symbols remain scoped records, not bare glyphs. Thus G may denote a symmetry group in one module and the gravitational constant in another only when the scope and quantity identifier are carried with the expression. The master does not create a universal symbol table by typographic coincidence.

6 Claim stratification and scientific boundary

The corpus claim register contains 136 entries: 103 theorem/proposition/corollary records, 14 models, 11 hypotheses, 7 diagnostics, and 1 empirical record. A strong label is accepted by the release only when the local ledger supplies the hypotheses and

the source states the result in its declared scope. P12 does not strengthen a component claim.

Layer	Release interpretation
Definition	Fixes notation or an object; no empirical claim.
Theorem, proposition, corollary	Mathematical statement under the local hypotheses.
Model	Declared idealization or computational benchmark.
Hypothesis	Unproved assumption or proposed condition.
Diagnostic	Protocol-dependent summary; not an invariant unless separately proved.
Empirical	Externally measured input with provenance and uncertainty requirements.

Boundary 6.1 (Master claim firewall). The release does not claim a theory of everything, a derivation of Standard Model parameters, a new electronic-structure method, a new ergodicity theorem, a new classification of regular polyhedra, or an operational orbit-determination system. Its defensible contribution is the typed audit infrastructure and the module-specific corrections recorded in the appended documents.

7 Metadata normalization and provenance

The canonical release author is **Stassis Stashkevichyus**, identified by ORCID 0009-0000-2294-705X. Earlier component PDFs contain the historical display strings *Stassis Research Program* or *Stas, Independent Research Program*. Their bytes are preserved; the container metadata supplies one normalized creator record without pretending that the old PDF metadata was rewritten.

No DOI is fabricated in this freeze. The release metadata is prepared for one corpus-level archival record; component modules are not assigned separate placeholder identifiers. If a DOI is minted, the immutable release metadata must be updated in a successor version or by an archive-controlled identifier field that does not alter the deposited files.

The root citation file follows Citation File Format 1.2.0. The Zenodo metadata file follows the documented GitHub-release override structure. The release is marked CC BY-NC-ND 4.0 as directed by the author. This licence permits non-commercial redistribution of unadapted material with attribution; it is not an open-source software licence and does not permit distribution of modified versions.

8 Validation and reproducibility

The P12 gate requires all of the following:

1. exact resolution of all 18 PDF and source paths;
2. agreement of component page counts, byte counts, and SHA-256 digests with the freeze ledger;
3. a clean GO Core replay: 18 PASS/0 FAIL/0 BLOCKED;
4. uniqueness of document IDs and fully qualified local IDs;
5. an acyclic, endpoint-complete dependency graph with verified source evidence;
6. successful execution of the P1-P11 release validators;

7. A4 geometry, no encryption or form fields, embedded fonts, and extractable text for every component and the master;
8. page-content preservation between each source PDF and its corresponding master page interval;
9. valid citation, archival, licence, and checksum records;
10. a deterministic ZIP archive with fixed timestamps, sorted members, and byte-identical rebuild.

The master contains 8 release-front-matter pages followed by the 158 unchanged component pages, for a total of 166 pages. PDF bookmarks and the external page map identify every component boundary. The component PDFs, not the master page numbering, remain the primary unit of scientific citation.

9 How to cite and review this object

Scientific review should proceed module by module. A reviewer should first inspect the claim register and protocol fields, then the local proofs or counterexamples, and finally the numerical and release checks. A criticism of one application module does not automatically invalidate the typed core; conversely, a clean core lint does not validate an application theorem whose hypotheses are false.

Until an archival DOI exists, cite the title, author, ORCID, version, date, and the SHA-256 digest recorded in `SHA256SUMS_v1_3.txt`:

Stassis Stashkevichyus, *Geometry of Observation: Frozen Typed Corpus and Reproducibility Master*, P12 corpus freeze v1.3.0, 28 July 2026, ORCID 0009-0000-2294-705X.

10 Release conclusion

P12 closes the migration sequence at the corpus level. The defensible result is an immutable, typed, hash-addressed collection of eighteen reference modules with explicit claim boundaries and reproducible validation. The release is suitable for archival provenance and external module-by-module review. It is not, by itself, evidence that the umbrella programme has become a single new physical theory.

Функциональный интерфейс геометрии наблюдения

Типизированный вариационный протокол GO-QS

Stassis Research Program

Строгая эталонная редакция v0.2 (P0) — 2026

Статус. Эталонный документ фазы P0 для системы GO-QS v0.1. Текст определяет математический интерфейс задачи наблюдения и оценивания. Он не вводит новую физическую теорию, не объявляет эвристические фильтры инвариантами природы и не отождествляет результат оптимизации с физической геометрией без отдельно заданной интерпретационной карты.

Аннотация

Задаётся типизированный функциональный интерфейс геометрии наблюдения. Симметрия или смена кадра, редукция разрешения, детерминированное наблюдение, стохастический канал и оценщик разделены как операторы разных типов. “Нулевая сцена” заменена явной калибровочной системой; геометрическое сечение определено через регулярное множество уровня и pullback на вложенную подповерхность; хроматическая уникальность ограничена конечным конфликтным графом. Все слагаемые вариационного функционала безразмерны, снабжены протоколом и политикой нулевого знаменателя. Доказана теорема существования минимизатора прямым методом и сформулирован критерий структурной восстановимости. Причинность введена через лоренцево отношение J^- , а Landauer-Bekenstein-Margolus-Levitin ограничения вынесены из универсального функционала в условные физические модули. Результат является интерфейсом для воспроизводимых inverse problems, а не самостоятельной динамической теорией.

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1. Область действия и статусы утверждений

В документе используются шесть статусов:

DEFINITION, THEOREM, PROPOSITION, MODEL, HYPOTHESIS, DIAGNOSTIC.

Определения и теоремы относятся к математическому ядру. Модель фиксирует конкретный статистический или вычислительный выбор. Гипотеза требует независимого обоснования. Диагностика является вычислимым признаком и не считается физическим инвариантом.

Ограничение 1.1 (Граница новизны). Карты наблюдения, марковские ядра, регуляризация и прямой метод вариационного исчисления являются стандартными конструкциями. Содержание этого текста — их строгая типовая сборка для корпуса Geometry of Observation и устранение несовместимых размерностей прежнего интерфейса.

2. Нормативный пакет наблюдения GO-QS

Определение 2.1 (Пакет наблюдения). Пусть Λ — множество допустимых протоколов. Пакетом наблюдения называется

$$\mathfrak{D} = \left(X, \Phi, \xi, \Lambda, \left\{ X_\lambda, Y_\lambda^0, Y_\lambda, R_\lambda, O_\lambda, K_\lambda, \hat{\theta}_\lambda \right\}_{\lambda \in \Lambda}, G, \mathcal{Q}, \mathcal{D} \right),$$

где:

1. X — скрытое пространство состояний с объявленной категорией;
2. $\Phi^t : X \rightarrow X$ — полная динамика, если она задана;
3. ξ — дескриптор собственных масштабов системы;
4. $R_\lambda : X \rightarrow X_\lambda$ — редукция или огрубление;
5. $O_\lambda : X_\lambda \rightarrow Y_\lambda^0$ — идеальная детерминированная карта наблюдения;
6. $K_\lambda(dy | y^0)$ — марковское ядро из Y_λ^0 в измеримое пространство данных Y_λ ;
7. $\hat{\theta}_\lambda : Y_\lambda \rightrightarrows \Theta_\lambda$ — возможно многозначный оценщик;
8. G действует на состояниях обратимыми отображениями $T_g : X \rightarrow X$;
9. $\mathcal{Q}$ — реестр типизированных величин;
10. $\mathcal{D}$ — профиль разнотипных дефектов.

Условный закон данных при скрытом состоянии x равен

$$P_\lambda(dy | x) = K_\lambda(dy | O_\lambda(R_\lambda x)).$$

Детерминированный датчик получается при ядре Дирака.

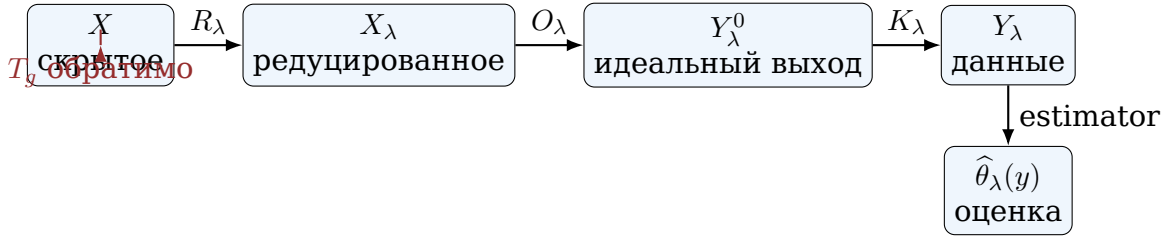


Рис. 1: Типы операторов не взаимозаменяемы. T_g не является каналом потери информации.

Предложение 2.2 (Композиционный тип). Если R_λ и O_λ измеримы, а K_λ — марковское ядро, то $P_\lambda(\cdot | x)$ задаёт марковское ядро из X в Y_λ .

Доказательство. Для каждого x это вероятностная мера как сечение K_λ . Измеримость $x \mapsto P_\lambda(A | x)$ для измеримого A следует из измеримости $O_\lambda \circ R_\lambda$ и определяющего свойства ядра. $\square$

2.1. Паспорт величины

Каждая величина q имеет тип

$$\text{Type}(q) = (D_q, V_q, \kappa_q, \mathbf{d}_q, U_q, C_q, \rho_q, \lambda_q, \xi_q, \text{Unc}_q, \text{Status}_q),$$

где $\mathbf{d}_q \in \mathbb{Q}^7$ — SI-вектор размерности, κ_q — семантический тип, C_q — контекст единиц, ρ_q — закон действия кадра или калибровочной группы. Физически безразмерные угол, вероятность, ранг, число элементов и информация не считаются автоматически арифметически совместимыми.

3. Калибровочная сцена и камера-трасса

Определение 3.1 (Калибровочная сцена). Калибровочная сцена есть пакет

$$C_0 = (o, E_0, \ell_0, t_0, \mathcal{W}_0),$$

где o — начало, $E_0 = (e_1, \dots, e_n)$ — ориентированный ортонормированный репер, $\ell_0 > 0$ и $t_0 > 0$ — эталонные длина и время, а $\mathcal{W}_0$ — область валидности калибровки.

Запись $0! = 1$ из v0.1 не является частью определения. Её можно использовать только как мнемонику “одна выбранная исходная калибровка”. Ни единственность репера, ни его физическая привилегированность из этой записи не следуют.

Определение 3.2 (Камера-трасса). При евклидовой пространственной кинематике камера-трасса есть измеримая или гладкая кривая

$$C_\lambda : I_{\text{cam}} \rightarrow SE(n).$$

Она входит в протокол λ . Если C_λ неизвестна, её параметры c включаются в пространство Θ_λ , а наблюдаемость c должна проверяться отдельно.

Смена координат, активное перемещение камеры и физическая симметрия объекта являются разными операциями. Их отождествление запрещено реестром GO-QS.

3.1. Строгое содержание ghost-mode

Определение 3.3 (Детерминированный слой совместимости). Для идеального выхода $y^0 \in Y_\lambda^0$ определяется

$$F_{y^0}^\lambda = \{x \in X : O_\lambda(R_\lambda x) = y^0\}.$$

При заданном prior $\pi(dx)$ и стохастическом канале скрытая неопределённость описывается posterior $\pi_\lambda(dx \mid y)$. Термин “ghost-mode” далее означает либо слой $F_{y^0}^\lambda$, либо posterior-распределение, но не новый неформальный кортеж невидимых объектов.

4. Геометрические сечения: корректный тип

Пусть M — гладкое d -мерное многообразие, а z_t — поле или геометрическая структура на M .

Определение 4.1 (Регулярное сечение уровня). Для гладкой функции $f_{j,\lambda} : M \rightarrow \mathbb{R}$ и её регулярного значения $c_{j,\lambda}$ положим

$$\Sigma_{j,\lambda} = f_{j,\lambda}^{-1}(c_{j,\lambda}), \quad \iota_{j,\lambda} : \Sigma_{j,\lambda} \hookrightarrow M.$$

Тогда $\Sigma_{j,\lambda}$ является вложенным подмногообразием размерности $d - 1$, а $\iota_{j,\lambda}$ — включение.

Двумерное сечение получается при $d = 3$. Для поля z_t корректная операция ограничения есть pullback $\iota_{j,\lambda}^* z_t$ тогда, когда тип поля допускает pullback.

Дискретный выход задаётся отдельно:

$$S_{j,\lambda} = Q_{j,\lambda} \circ \iota_{j,\lambda}^* : \mathcal{Z}(M) \longrightarrow \mathcal{Z}(Q_{j,\lambda}),$$

где $Q_{j,\lambda}$ — конечная сетка, квантователь или оператор выборки.

Замечание 4.2. Карта $M \rightarrow \Sigma$ сама по себе не называется сечением. Ретракция $r : M \rightarrow \Sigma$, сечение расслоения $s : B \rightarrow E$ и включение множества уровня $\iota : \Sigma \hookrightarrow M$ имеют разные типы.

Для графа $G = (V, E)$ аналогом является объявленный оператор выбора $V_{j,\lambda} \subseteq V$ и индуцированный подграф. Он не смешивается с гладким pullback.

5. Конечная хроматическая диагностика

Определение 5.1 (Конфликтный граф). После дискретизации пусть $V_{j,\lambda}$ конечно. Объявленная симметричная relation $\mathcal{R}_{\chi,\lambda}$ задаёт

$$G_{\chi,\lambda} = (V_{j,\lambda}, E_{\chi,\lambda}), \quad (v, w) \in E_{\chi,\lambda} \iff v \neq w \text{ и } \mathcal{R}_{\chi,\lambda}(v, w).$$

Для меток $\chi : V_{j,\lambda} \rightarrow \mathcal{C}$ нормированный collision defect равен

$$\delta_\chi = \begin{cases} \frac{1}{|E_{\chi,\lambda}|} \sum_{(v,w) \in E_{\chi,\lambda}} \mathbf{1}_{\{\chi(v)=\chi(w)\}}, & |E_{\chi,\lambda}| > 0, \\ 0, & |E_{\chi,\lambda}| = 0. \end{cases}$$

Это число из $[0, 1]$ и имеет статус DIAGNOSTIC. Для непрерывных цветовых признаков разрешена margin-loss только после объявления метрики d_C и масштаба $m_C > 0$.

Ограничение 5.2 (Континуальная патология v0.1). Требование различать все точки с рациональной разностью на континууме не принимается как базовый постулат. Рациональные разности плотны, а непрерывный RGB-код не обеспечивает заявленную глобальную уникальность. Допустимы конечная сетка, конечная ε_x -сеть или отдельная теорема о раскраске конкретного графа.

6. Операциональные масштабы и геометрические фильтры

6.1. Масштаб

Для положительной длины ℓ используются

$$\widehat{\ell} = \frac{\ell}{\ell_{\text{ref}}}, \quad \chi_L = \log_b \frac{\ell}{\ell_{\text{ref}}}, \quad b > 0, \quad b \neq 1.$$

Аргумент логарифма безразмерен. Спиральная параметризация $r(\theta) = r_0 e^{a\theta}$ допустима при безразмерном $a\theta$. Золотое сечение, рациональная сетка или спектральный масштаб являются опциональными model priors, а не универсальными линейками.

6.2. Пифагоров диагностический штраф

Пусть $u, v \neq 0$ — рёбра ячейки и

$$r_u = \frac{\|u\|}{\sqrt{\|u\|^2 + \|v\|^2}}, \quad r_v = \frac{\|v\|}{\sqrt{\|u\|^2 + \|v\|^2}}.$$

Для конечного набора примитивных троек

$$\mathcal{P}_A = \{(a, b, c) \in \mathbb{N}^3 : a^2 + b^2 = c^2, \gcd(a, b, c) = 1, c \leq A\}$$

определим

$$\delta_{\text{pyth}}(u, v) = \min_{(a, b, c) \in \mathcal{P}_A} \left[\left(r_u - \frac{a}{c} \right)^2 + \left(r_v - \frac{b}{c} \right)^2 + \beta_{\perp} \left(\frac{\langle u, v \rangle}{\|u\| \|v\|} \right)^2 \right].$$

Если порядок рёбер не фиксирован, минимум берётся также по перестановке a, b . Все слагаемые безразмерны. Нулевые рёбра исключены из допустимой области.

6.3. Герон и Кэли-Менгер

Формулы Герона и Кэли-Менгера являются тождествами евклидовой геометрии, а не самостоятельными физическими ограничениями. Если они используются как тест согласованности шумных данных, остаток нормируется:

$$\delta_{\text{Heron}} = \frac{|A_{\text{obs}}^2 - s(s-a)(s-b)(s-c)|}{\ell_{\text{ref}}^4}, \quad s = \frac{a+b+c}{2}.$$

Если вводится предпочтение целочисленных или рациональных площадей, оно помечается MODEL, задаётся конечным target set в переменной A/ℓ_{ref}^2 и исключается из функционала по умолчанию.

6.4. Симметрия

Для нормированного представления z в гильбертовом пространстве и объявленной группы G_{λ} допустима диагностика

$$\delta_{\text{sym}}(z) = \begin{cases} \inf_{g \in G_{\lambda} \setminus \{e\}} \frac{\|z - g \cdot z\|^2}{\|z\|^2}, & \|z\| > 0, G_{\lambda} \neq \{e\}, \\ 0, & \|z\| = 0 \text{ или } G_{\lambda} = \{e\}. \end{cases}$$

Группа, представление и нулевая политика являются частью протокола.

7. Динамический дефект замыкания

Пусть $y_v : [0, T_{\text{hor}}] \rightarrow (Y_{\lambda}, d_{Y, \lambda})$ — наблюдаемая траектория маркера, а $r_Y(\lambda) > 0$ — разрешение выхода. Для $0 < P_{\min} \leq P \leq P_{\max} < T_{\text{hor}}$ положим

$$\delta_{\text{cl}, v}(P) = \left[\frac{1}{T_{\text{hor}} - P} \int_0^{T_{\text{hor}} - P} \left(\frac{d_{Y, \lambda}(y_v(t+P), y_v(t))}{r_Y(\lambda)} \right)^2 dt \right]^{1/2},$$

$$\delta_{\text{cl}, v} = \inf_{P \in [P_{\min}, P_{\max}]} \delta_{\text{cl}, v}(P).$$

Эта величина безразмерна и протокол-зависима. Малое значение на конечном горизонте не доказывает периодичность полной скрытой динамики.

8. Причинность и физические бюджеты

8.1. Лоренцева причинность

Пусть (M, g) — time-oriented Lorentzian manifold. Данные, считанные в событии p_{obs} , могут зависеть от события q только если

$$q \in J^-(p_{\text{obs}}).$$

Это hard constraint допустимого множества $\mathcal{A}_\lambda$. В плоской пространственно-временной диаграмме с координатами (t, x) условие имеет вид

$$t_q \leq t_{\text{obs}}, \quad \|x_{\text{obs}} - x_q\| \leq c(t_{\text{obs}} - t_q).$$

Риманов шар $d(x, o) \leq c|t - t_0|$ не переносится в общий лоренцев случай.

8.2. Условные информационно-физические модули

Следующие формулы не входят в универсальный функционал:

$$\begin{aligned} Q_{\text{erase}} &\geq k_B T \ln 2 N_{\text{erased}}, \\ N_{\text{bits}} &\leq \frac{2\pi ER}{\hbar c \ln 2}, \\ \tau_{\perp} &\geq \frac{\pi \hbar}{2(E - E_0)}. \end{aligned}$$

Первая относится к логически необратимому стиранию при объявленной термодинамической модели; вторая требует условий применимости Bekenstein bound; третья — к времени перехода в ортогональное квантовое состояние при конечной средней энергии над основным состоянием. Биты, джоули и секунды не суммируются. При необходимости каждая формула задаёт отдельное допустимое множество или отдельный нормированный defect.

8.3. Нет универсальной неопределённости “сечение-путь”

Не существует выведенной в v0.1 теоремы $\Delta_\Sigma \Delta_\gamma \geq \kappa_{\text{obs}}$. Для конкретного прибора разрешена только модель

$$\hat{u}_\Sigma \hat{u}_\gamma \geq \kappa_\lambda, \quad \hat{u}_\Sigma = \frac{u_\Sigma}{u_{\Sigma,0}}, \quad \hat{u}_\gamma = \frac{u_\gamma}{u_{\gamma,0}},$$

где обе нормировки, метод оценки неопределённостей и область валидности получены из калибровки. Статус такого отношения — HYPOTHESIS или EMPIRICAL, но не квантовая теорема.

9. Безразмерный вариационный оцениватель

9.1. Единое пространство параметров

Пусть Θ_λ — объявленное пространство параметров, а $\mathcal{A}_\lambda \subseteq \Theta_\lambda$ — допустимое множество. Один и тот же символ $\theta \in \mathcal{A}_\lambda$ используется во всех членах. Например,

$$\theta = (z_0, \vartheta_\Phi, c, \vartheta_R, \vartheta_O, \chi),$$

но карта наблюдения или камера включается в θ только если она действительно оценивается из данных.

9.2. Согласование с данными

Для гауссовой модели с положительно определённой ковариацией $\Sigma_y(t)$:

$$\hat{D}_y(\theta; y) = \int \left\| \Sigma_y(t)^{-1/2} (y_\theta(t) - y(t)) \right\|^2 d\hat{\mu}_T(t), \quad d\hat{\mu}_T = \frac{dt}{T_{\text{hor}}}.$$

Для негауссового канала используется безразмерная отрицательная лог-правдоподобность с явно указанной базовой мерой.

9.3. Регуляризаторы

Пусть S_Z — обратимый scale operator для компонент z , а $T_0 > 0$. Тогда

$$\hat{E}_1(z) = \int_0^1 \left\| S_Z^{-1} T_0 \partial_t z(T_0 t) \right\|^2 dt,$$

$$\hat{E}_2(z) = \int_0^1 \left\| S_Z^{-1} T_0^2 \partial_t^2 z(T_0 t) \right\|^2 dt.$$

Для регулярной кривой длины $L_{\text{path}} > 0$:

$$\hat{E}_\kappa = \frac{1}{L_{\text{path}}} \int (\ell_0 \kappa(s))^2 ds.$$

Все три величины безразмерны.

9.4. Скаляризация

Пусть $\mathcal{K}_\lambda$ — конечный набор активных диагностик, $\delta_k = \nu_k(D_k) \in [0, \infty]$ — объявленные нормировки, а $\phi_k : [0, \infty] \rightarrow [0, \infty]$ — объявленные penalty maps. Определим

$$J_\lambda(\theta; y) = w_y \hat{D}_y w_1 \hat{E}_1 w_2 \hat{E}_2 w_\kappa \hat{E}_\kappa \sum_{k \in \mathcal{K}_\lambda} w_k \phi_k(\delta_k) I_{\mathcal{A}_\lambda}(\theta)$$

с $w_\bullet \geq 0$ и суммой весов 1. Здесь $I_{\mathcal{A}_\lambda}$ равно 0 на $\mathcal{A}_\lambda$ и $+\infty$ вне неё. Скаляризация является decision functional, а не универсальным геометрическим инвариантом.

Предложение 9.1 (Инвариантность при смене единиц). *Если каждый член J_λ построен из безразмерных отношений, нормированных мер и согласованно преобразованных ковариаций, то значение J_λ не меняется при согласованной замене единиц.*

Доказательство. Каждое отношение физической величины к эталону той же размерности инвариантно. Mahalanobis residual инвариантен при одновременном преобразовании данных и ковариации. Нормированные меры безразмерны. Следовательно, инвариантна каждая сумма и вся их безразмерная линейная комбинация. $\square$

Определение 9.2 (Вариационный оцениватель).

$$\hat{\theta}_\lambda(y) = \arg \min_{\theta \in \Theta_\lambda} J_\lambda(\theta; y).$$

Значение является множеством, пока единственность не доказана.

10. Существование, неединственность и интерпретация

Теорема 10.1 (Прямой метод). Пусть V — рефлексивное банахово пространство, $\mathcal{A}_\lambda \subset V$ непусто и последовательно слабо замкнуто. Пусть $J_\lambda : V \rightarrow (-\infty, +\infty]$ собственный, ограниченный снизу, коэрцитивный и последовательно слабо ниже полунепрерывный. Тогда

$$\arg \min_V J_\lambda \neq \emptyset.$$

Доказательство. Возьмём минимизирующую последовательность. Коэрцитивность делает её ограниченной. Рефлексивность даёт слабо сходящуюся подпоследовательность. Слабая замкнутость сохраняет предел в $\mathcal{A}_\lambda$, а слабая нижняя полунепрерывность даёт значение не больше $\liminf$ значений последовательности. Предел достигает инфимума. $\square$

Ограничение 10.2. Теорема не даёт единственности, устойчивости, идентификации или физической истинности forward model. Для каждого конкретного пространства необходимо отдельно доказать коэрцитивность и слабую нижнюю полунепрерывность.

Определение 10.3 (Интерпретационная карта). Реконструированная геометрическая структура задаётся объявленной картой

$$\mathcal{G}_\lambda : \Theta_\lambda \rightarrow \text{Geom}_\lambda.$$

Выходом задачи является множество $\mathcal{G}_\lambda(\hat{\theta}_\lambda(y))$, а не автоматически “геометрия пространства”.

11. Восстановимость

Теорема 11.1 (Факторизация скалярной величины). Пусть $O : X \rightarrow Y$ — отображение множеств и $q : X \rightarrow V$. Существует $\bar{q} : O(X) \rightarrow V$ такое, что

$$q = \bar{q} \circ O$$

тогда и только тогда, когда q постоянна на каждом слое $O^{-1}(y)$. Карта $\bar{q}$ единственна на $O(X)$.

Доказательство. Необходимость следует из равенства $O(x) = O(x')$. Для достаточности положим $\bar{q}(y) = q(x)$ для любого $x \in O^{-1}(y)$. Постоянство на слоях обеспечивает корректность определения. $\square$

В стохастическом случае условие $H_b(q(X) | Y) = 0$ означает восстановимость почти наверное только относительно объявленного распределения и не заменяет структурную теорему.

12. Quantity and Protocol Ledger

12.1. Обязательный протокол

Поле	Нормативное содержание
hidden_space	категория X , топология или σ -алгебра, область динамики
system_descriptor	L_X, T_X, M_X, E_X , число степеней свободы и прочие собственные масштабы
reduction	сигнатура $R_\lambda : X \rightarrow X_\lambda$ и параметры разрешения
observation	сигнатура $O_\lambda : X_\lambda \rightarrow Y_\lambda^0$
noise_kernel	$K_\lambda(dy y^0)$, калибровка и ковариация
frame	координаты, базис, camera pose и закон преобразования
quantizer	сетка, ADC, partition или правило missing data
horizon	T_{hor} и temporal resolution δt
estimator	Θ_λ , loss, regularizers, веса и solver
uncertainty	covariance, interval, posterior или risk bound
defects	кодомен каждого D_k и нормировка ν_k

12.2. Ключевые величины

Величина	Размерность d	Тип	Условие
$\ell_0, \ell_{\text{ref}}, \varepsilon_x$	$(1, 0, 0, 0, 0, 0, 0)$	scalar	положительны
$T_0, T_{\text{hor}}, \delta t$	$(0, 0, 1, 0, 0, 0, 0)$	scalar	положительны
κ	$(-1, 0, 0, 0, 0, 0, 0)$	scalar	требует smoothing protocol
Σ_y	квадрат единицы y	covariance tensor	положительно определена или regularized
δ_k	0	loss	нормировка и zero policy обязательны
J_λ	0	loss	decision functional
H_b	0	information	основание b обязательно
N_{bits}	0	information	отображение count $\rightarrow$ bit объявляется

13. Журнал исправлений: v0.1 к v0.2

1. Generic $S_j : M \rightarrow \Sigma_j$ заменён регулярным множеством уровня, включением ι_j и дискретизатором Q_j .

2. Пакет параметров унифицирован как $\theta \in \Theta_\lambda$.
3. Камера перенесена в протокол; неизвестная камера допускается только как оцениваемый параметр.
4. Ghost-mode заменён слоем совместимости или posterior.
5. Континуальная рациональная раскраска заменена конечным конфликтным графом.
6. Пифагоров штраф полностью безразмерен; ортогональный остаток нормирован.
7. Все логарифмы содержат отношение к эталону.
8. Closure defect нормирован разрешением выхода и горизонтом.
9. Риманов “причинный шар” заменён отношением J^- .
10. Информационные пределы не суммируются с энергиями и геометрическими остатками.
11. Невыведенная uncertainty product удалена из теоремного ядра.
12. Функционал состоит только из безразмерных членов.
13. $\arg \min$ оставлен множеством; геометрия получается отдельной интерпретационной картой.
14. Компактная эвристика заменена точной теоремой прямого метода.

14. Claim firewall и критерий эталонности P0

Доказано или определено:

1. типизированная композиция редукции, observation map и noise kernel;
2. корректный тип геометрического сечения;
3. безразмерность всех активных членов J_λ ;
4. существование minimizer при гипотезах прямого метода;
5. критерий структурной факторизации величины;
6. протокол-зависимый характер диагностик.

Не доказано и не заявляется:

1. единственность или устойчивость любой конкретной реконструкции;
2. универсальность Pythagorean, Heronian, chromatic или golden-ratio priors;
3. универсальная uncertainty relation;
4. вывод квантовой механики, гравитации или термодинамики;
5. физическая реальность любой структуры только из факта минимизации.

Документ проходит P0, если машинный manifest подтверждает уникальность символов в каждой области, полные сигнатуры карт, статусы утверждений и отсутствие сложения сырых разнотипных дефектов.

15. Заключение

Функциональный интерфейс теперь имеет строгую роль: он задаёт воспроизводимую задачу наблюдения и оценивания

$$x \xrightarrow{R_\lambda} x_\lambda \xrightarrow{O_\lambda} y_\lambda^0 \xrightarrow{K_\lambda} y \xrightarrow{\hat{\theta}_\lambda} \hat{\theta},$$

а затем отделяет результат оценивания от его геометрической интерпретации. Это минимально необходимая архитектура, на которую можно переносить информационный, метрический, тензорный и прикладные модули без смешения единиц и математических типов.

Список литературы

- [1] J. M. Lee, *Introduction to Smooth Manifolds*, 2nd ed., Springer, 2013.
- [2] V. I. Bogachev, *Measure Theory*, Vols. I-II, Springer, 2007.
- [3] I. Ekeland and R. Temam, *Convex Analysis and Variational Problems*, SIAM, 1999.
- [4] H. W. Engl, M. Hanke, and A. Neubauer, *Regularization of Inverse Problems*, Kluwer, 1996.
- [5] J. Kaipio and E. Somersalo, *Statistical and Computational Inverse Problems*, Springer, 2005.
- [6] B. O'Neill, *Semi-Riemannian Geometry with Applications to Relativity*, Academic Press, 1983.
- [7] R. Landauer, Irreversibility and heat generation in the computing process, *IBM Journal of Research and Development* **5** (1961), 183–191.
- [8] J. D. Bekenstein, Universal upper bound on the entropy-to-energy ratio for bounded systems, *Physical Review D* **23** (1981), 287–298.
- [9] N. Margolus and L. B. Levitin, The maximum speed of dynamical evolution, *Physica D* **120** (1998), 188–195.

Tensorial Observation Geometry in General Relativity

Coordinate covariance, frame/coframe duality, observation maps, and tensor descent

Stassis Research Program

Strict reference revision v0.3 (P0) — 2026

Status. First tensor/GR reference document migrated to GO-QS v0.1. It corrects the tetrad reconstruction formula, declares the unit convention, states exact hypotheses for tensor descent, and separates coordinate, gauge, apparatus, and observation transformations. It does not modify Einstein's equations or claim a new theory of gravity.

Abstract

This note supplies a type-safe tensorial layer for Geometry of Observation. Four operations are separated: passive coordinate re-description, active diffeomorphism, local Lorentz frame gauge, and a possibly non-injective observation channel. Frame coefficients $e_a{}^\mu$ and coframe coefficients $e^a{}_\mu$ are distinguished; the covariant metric is reconstructed from the coframe, while the inverse metric is reconstructed from the frame. The default convention is signature $(-, +, +, +)$ and geometrized units $c = G = 1$, with the SI conversion stated explicitly. For a smooth observation map, the radical of a pulled-back pseudo-Riemannian form is computed exactly, showing why nontrivial kernel is sufficient but not necessary for degeneracy in Lorentzian signature. Smooth scalar descent and descent of covariant tensors along a surjective submersion are stated with their required fiber hypotheses. A local Lorentz gauge change is distinguished from a physical reorientation of an apparatus. A GO-QS quantity and protocol ledger closes the document.

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1 Scope, status, and conventions

Unless stated otherwise, manifolds are smooth, Hausdorff, and second countable. A spacetime (M, g) is a four-dimensional, time-oriented Lorentzian manifold with signature

$$\eta_{ab} = \text{diag}(-1, 1, 1, 1).$$

The curvature convention is

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z.$$

All relativistic formulae use the GO-QS unit context `GR_geometrized`: $c = G = 1$. Hence a proper-time parametrized four-velocity satisfies $g(u, u) = -1$. If proper time is measured in SI seconds and $u_{\text{SI}} = d\gamma / d\tau$, then

$$g(u_{\text{SI}}, u_{\text{SI}}) = -c^2.$$

Boundary 1.1 (Claim boundary). The tensor calculus, frame bundle, Fermi-Walker transport, and submersion results used below are standard. The contribution here is a corrected, typed interface to observation geometry. No new field equation, force, or gravitational degree of freedom is introduced.

2 Four transformations that must not be conflated

2.1 Passive coordinate change

Let $x : U \rightarrow \mathbb{R}^4$ and $x' : U' \rightarrow \mathbb{R}^4$ be charts. On the overlap,

$$x' \circ x^{-1} : x(U \cap U') \rightarrow x'(U \cap U')$$

is a diffeomorphism between coordinate images. It changes components, not the geometric point or tensor. With

$$J^\alpha{}_\mu = \frac{\partial x'^\alpha}{\partial x^\mu}, \quad (J^{-1})^\mu{}_\alpha = \frac{\partial x^\mu}{\partial x'^\alpha},$$

the basis and components transform contragrediently.

2.2 Active diffeomorphism

An active diffeomorphism $\varphi : M \rightarrow M$ maps fields by pullback or pushforward. Whether it represents a gauge redundancy, a physical symmetry, or a change of boundary data depends on the theory and boundary conditions. It is not identified with a passive chart change merely because both use Jacobians.

2.3 Local Lorentz frame gauge

For an orthonormal frame e_a , a local Lorentz transformation is

$$e'_a = \Lambda_a{}^b e_b, \quad \Lambda(x) \in \text{SO}^+(1, 3).$$

It changes the frame representation. If all frame components of physical fields are transformed consistently, gauge-invariant readouts do not change.

2.4 Observation and reduction

GO-QS assigns distinct signatures:

$$T_g : E \rightarrow E, \quad R_\lambda : E \rightarrow E_\lambda, \quad O_\lambda : E_\lambda \rightarrow Y_\lambda^0, \quad K_\lambda(dy \mid y^0).$$

T_g is invertible; R_λ and O_λ need not be. Information loss is therefore a property of a declared channel, not of covariance itself.

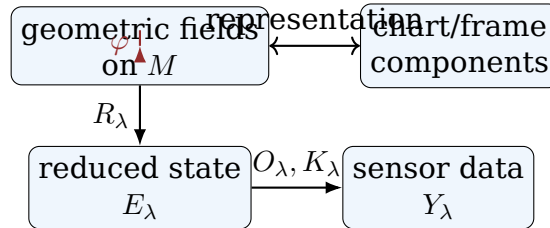


Figure 1: Representation changes preserve the geometric object; an observation pipeline can quotient or randomize it.

3 Tensors and coordinate covariance

Definition 3.1 (Tensor). At $p \in M$, a tensor of type (r, s) is a multilinear map

$$T : (T_p^* M)^r \times (T_p M)^s \rightarrow \mathbb{R}.$$

For $V = V^\mu \partial_\mu$ and $\omega = \omega_\mu dx^\mu$,

$$V'^\alpha = \frac{\partial x'^\alpha}{\partial x^\mu} V^\mu, \quad \omega'_\alpha = \frac{\partial x^\mu}{\partial x'^\alpha} \omega_\mu.$$

Consequently, the contraction $\omega_\mu V^\mu$ is coordinate independent. General tensor components receive one Jacobian per upper index and one inverse Jacobian per lower index.

Remark 3.2 (Coordinate passport). A component array is not a complete GO-QS quantity. It must carry the chart, basis, coordinate dimensions, index variance, and frame/gauge law. If coordinates have different physical dimensions, component dimensions are index dependent.

4 Metric, connection, curvature, and basepoint

The metric is

$$g = g_{\mu\nu} dx^\mu \otimes dx^\nu, \quad ds^2 = g_{\mu\nu} dx^\mu dx^\nu.$$

Under a coordinate change,

$$g'_{\alpha\beta} = \frac{\partial x^\mu}{\partial x'^\alpha} \frac{\partial x^\nu}{\partial x'^\beta} g_{\mu\nu},$$

and the quadratic form ds^2 is unchanged. In Lorentzian signature, this quadratic form is not a metric distance function.

For $p \neq q$, the vector spaces $T_p M$ and $T_q M$ have no canonical identification. A connection supplies a transport rule. The Levi-Civita connection is characterized by

$$\nabla g = 0, \quad \text{Tor}_\nabla = 0,$$

and in coordinates

$$\Gamma^\rho_{\mu\nu} = \frac{1}{2} g^{\rho\sigma} (\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}).$$

The Christoffel symbols are not tensor components. The curvature tensor is

$$R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma}.$$

The Ricci, scalar-curvature, and Einstein tensors are written

$$\text{Ric}_{\sigma\nu} = R^\rho_{\sigma\rho\nu}, \quad \text{Scal}_g = g^{\sigma\nu} \text{Ric}_{\sigma\nu},$$

$$\text{Ein}(g)_{\mu\nu} = \text{Ric}_{\mu\nu} - \frac{1}{2} \text{Scal}_g g_{\mu\nu}.$$

The notation $\text{Ein}(g)$ avoids a collision between the standard symbol $G_{\mu\nu}$ and the GO-QS symmetry-group symbol G . With geometrized stress-energy, Einstein's equation is

$$\text{Ein}(g)_{\mu\nu} = 8\pi T_{\mu\nu}.$$

At the SI boundary it is $\text{Ein}(g)_{\mu\nu} = (8\pi G_N/c^4) T_{\mu\nu}^{\text{SI}}$.

An affinely parametrized geodesic satisfies

$$\nabla_{\dot{\gamma}} \dot{\gamma} = 0.$$

Coordinate acceleration is not invariant; the covariant equation is.

5 Frame/coframe duality and the corrected tetrad formula

Definition 5.1 (Frame and coframe). A local frame and its dual coframe are

$$e_a = e_a^\mu \partial_\mu, \quad \vartheta^a = e^a_\mu dx^\mu,$$

with

$$e^a{}_\mu e_b{}^\mu = \delta^a_b, \quad e^a{}_\mu e_a{}^\nu = \delta_\mu{}^\nu.$$

The frame is orthonormal when

$$g_{\mu\nu} e_a{}^\mu e_b{}^\nu = \eta_{ab}.$$

Proposition 5.2 (Metric reconstruction). *For a Lorentz-orthonormal frame/coframe pair,*

$$g_{\mu\nu} = \eta_{ab} e^a{}_\mu e^b{}_\nu, \quad g^{\mu\nu} = \eta^{ab} e_a{}^\mu e_b{}^\nu.$$

Proof. The abstract identity $g = \eta_{ab} \vartheta^a \otimes \vartheta^b$ gives the first component formula. Inverting the coframe matrix and the metric gives the second. Direct contraction with $e_c{}^\mu e_d{}^\nu$ returns η_{cd} . $\square$

Boundary 5.3 (Correction to v0.2). The expression $\eta_{ab} e_a{}^\mu e_b{}^\nu$ reconstructs $g^{\mu\nu}$, not $g_{\mu\nu}$. The covariant metric requires the dual coframe coefficients $e^a{}_\mu$.

5.1 Example: spherical Minkowski coframe

On a chart with $r > 0$ and $0 < \theta < \pi$,

$$g = -dt^2 + dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2$$

in $c = 1$ units. An orthonormal coframe is

$$\vartheta^0 = dt, \quad \vartheta^1 = dr, \quad \vartheta^2 = r d\theta, \quad \vartheta^3 = r \sin \theta d\phi.$$

Then $g = \eta_{ab} \vartheta^a \otimes \vartheta^b$. Coordinate singularities at the polar axis are not curvature singularities.

6 Observers and transport

A future-directed observer is a timelike worldline

$$\gamma : I \rightarrow M, \quad u = \dot{\gamma}, \quad g(u, u) = -1.$$

Its acceleration is $a = \nabla_u u$, with $g(a, u) = 0$.

Definition 6.1 (Observer tetrad). An observer tetrad along γ is an orthonormal frame (e_0, e_1, e_2, e_3) with $e_0 = u$. The spatial axes span

$$u^\perp = \{X \in T_{\gamma(\tau)} M : g(X, u) = 0\}.$$

Definition 6.2 (Fermi-Walker transport). In the declared $c = 1$ convention, a vector field X along γ is Fermi-Walker transported when

$$\nabla_u X = \left(u \otimes a^\flat - a \otimes u^\flat \right) X.$$

For $a = 0$ this reduces to parallel transport.

In SI variables with $g(u_{\text{SI}}, u_{\text{SI}}) = -c^2$, the right-hand side is divided by c^2 . This conversion boundary is mandatory.

Proposition 6.3 (No canonical spatial triad). *At a point p , after fixing a unit future timelike vector u , the set of oriented orthonormal bases of $u^\perp$ is a torsor for $\text{SO}(3)$. Hence g and u do not select a preferred spatial triad.*

Proof. $u^\perp$ is a three-dimensional Euclidean vector space. $\text{SO}(3)$ acts freely and transitively on its oriented orthonormal bases. $\square$

6.1 Vacuum is not flatness

Einstein vacuum, $R_{\mu\nu} = 0$, does not imply $R^\rho{}_{\sigma\mu\nu} = 0$. For a Jacobi field ξ along a geodesic, the declared curvature convention gives

$$\nabla_u \nabla_u \xi = R(u, \xi)u.$$

Tidal relative acceleration can therefore provide a local relational signal even in vacuum.

7 Frame gauge versus physical apparatus orientation

The oriented, time-oriented orthonormal frame bundle is

$$\pi_{\mathcal{F}} : \mathcal{F}^+(M, g) \rightarrow M,$$

with structure group $\text{SO}^+(1, 3)$. A local Lorentz gauge transformation changes a representative frame. A physical apparatus, however, has material axes, couplings, and a response model. Rotating the apparatus relative to an external field changes its physical configuration; it is not erased by relabelling components.

Model 7.1 (Gauge-covariant sensor response). Let s denote physical fields and F a frame representative. A well-defined scalar readout $\mathcal{R}$ satisfies

$$\mathcal{R}(F\Lambda, \rho(\Lambda)^{-1}s) = \mathcal{R}(F, s)$$

for simultaneous passive frame change. An active apparatus rotation A acts on the apparatus state alone and may produce $\mathcal{R}(A \cdot F, s) \neq \mathcal{R}(F, s)$.

Boundary 7.2. The phrase “frame-dependent readout” is ambiguous unless it states whether the frame is a gauge representative or a material apparatus orientation. GO-QS stores these in different protocol fields.

8 The GO-QS tensor/GR observation package

For a relativistic application, use

$$\mathfrak{D}_{\text{GR}} = \left(E, \Phi, \xi, \Lambda, \{E_\lambda, Y_\lambda^0, Y_\lambda, R_\lambda, O_\lambda, K_\lambda, \hat{q}_\lambda\}_{\lambda \in \Lambda}, G_{\text{diff}}, G_{\text{LL}}, \mathcal{Q}, \mathcal{D} \right).$$

Here E is a state or field configuration space, not spacetime alone. G_{diff} and G_{LL} record their actions separately. A local protocol minimally contains

$$\lambda = (\gamma, u, \mathcal{E}_{\text{app}}, \varepsilon_x, \delta\tau, B_\nu, \Sigma_\eta, Q_{\text{quant}}, \mathcal{C}_{\text{coord}}, \mathcal{C}_{\text{LL}}, T_{\text{hor}}).$$

$\mathcal{E}_{\text{app}}$ is the material apparatus orientation; $\mathcal{C}_{\text{LL}}$ is a gauge convention.

9 Pullback geometry and its radical

Let $O : E \rightarrow B$ be smooth and let h be a nondegenerate pseudo-Riemannian metric on B . Define

$$(O^*h)_e(v, w) = h_{O(e)}(dO_e v, dO_e w).$$

The vertical space is $\mathcal{V}_e = \ker dO_e$.

Theorem 9.1 (Exact radical of a pullback form).

$$\text{rad}((O^*h)_e) = (dO_e)^{-1} \left((\text{im } dO_e)^{\perp_h} \right).$$

In particular, $\ker dO_e \subseteq \text{rad}((O^*h)_e)$.

Proof. A vector v belongs to the radical precisely when $h(dO_e v, dO_e w) = 0$ for every $w \in T_e E$. This is equivalent to $dO_e v$ being orthogonal to $\text{im } dO_e$, which gives the stated preimage. $\square$

Corollary 9.2. If h is Riemannian, then

$$\text{rad}((O^*h)_e) = \ker dO_e.$$

The same equality holds for pseudo-Riemannian h whenever O is a submersion at e .

Proof. For a positive-definite form, $\text{im } dO_e \cap (\text{im } dO_e)^{\perp_h} = \{0\}$. For a submersion, $\text{im } dO_e = T_{O(e)}B$, whose orthogonal complement is zero by nondegeneracy of h . $\square$

Remark 9.3. In Lorentzian signature, injectivity of dO_e does not guarantee nondegeneracy of O^*h : the image can be a degenerate null subspace. Thus “non-immersion implies degeneracy” is correct, but its converse fails.

9.1 No automatic Lorentzian distance

A Lorentzian metric supplies causal cones, proper time on timelike curves, and a time-separation function under suitable hypotheses. It does not supply a positive metric distance analogous to a Riemannian distance. Statistical residuals must use a positive-definite sensor covariance or a declared loss, not the indefinite form h as a norm.

10 Descent and recoverability

Theorem 10.1 (Smooth scalar descent). Let $O : E \rightarrow B$ be a surjective submersion and $q : E \rightarrow \mathbb{R}$ smooth. There is a unique smooth $\bar{q} : B \rightarrow \mathbb{R}$ satisfying

$$q = \bar{q} \circ O$$

if and only if q is constant on every fiber $O^{-1}(b)$.

Proof. Necessity is immediate. For sufficiency, define $\bar{q}(b) = q(e)$ for $e \in O^{-1}(b)$. A submersion admits smooth local sections $\sigma : U \rightarrow E$, and on U , $\bar{q} = q \circ \sigma$, proving smoothness. Surjectivity gives uniqueness. $\square$

Definition 10.2 (Horizontal covariant tensor). A covariant k -tensor T on E is horizontal for O if $T(v_1, \dots, v_k) = 0$ whenever at least one v_i is vertical.

Theorem 10.3 (Descent of covariant tensors). *Let $O : E \rightarrow B$ be a surjective submersion with connected fibers, and let T be a smooth covariant k -tensor on E . Then there is a unique covariant tensor $\bar{T}$ on B with*

$$T = O^*\bar{T}$$

if and only if:

1. T is horizontal; and
2. $\mathcal{L}_V T = 0$ for every smooth vertical vector field V .

Proof sketch. Pullbacks are horizontal and invariant under vertical flows. Conversely, horizontality makes the value of T depend only on projected tangent vectors. Vertical-flow invariance makes that value constant along each connected fiber. Local sections of the submersion define compatible local tensors on B , which glue uniquely. $\square$

Boundary 10.4 (Disconnected fibers and contravariant indices). If fibers are disconnected, vertical Lie invariance does not compare different components; explicit cross-component consistency is also required. The stated criterion is for covariant tensors. Descent of contravariant or mixed tensors requires a separate projectability condition and is not obtained by replacing indices formally.

11 Readouts on the frame bundle

Theorem 11.1 (Failure to descend to spacetime). *Let $\mathcal{R} : \mathcal{F}^+(M, g) \rightarrow Y$. If two frames F_1, F_2 over the same event satisfy*

$$\pi_{\mathcal{F}}(F_1) = \pi_{\mathcal{F}}(F_2), \quad \mathcal{R}(F_1) \neq \mathcal{R}(F_2),$$

then no map $\bar{\mathcal{R}} : M \rightarrow Y$ satisfies $\mathcal{R} = \bar{\mathcal{R}} \circ \pi_{\mathcal{F}}$.

Proof. Such a factorization would assign the same value $\bar{\mathcal{R}}(p)$ to both frames, contradicting the hypothesis. $\square$

This set-theoretic theorem does not decide whether the variation is a gauge artifact or a physical apparatus response. That decision requires the simultaneous transformation law in the protocol.

12 Worked examples

12.1 Riemannian orthographic projection

For

$$\pi : \mathbb{R}^3 \rightarrow \mathbb{R}^2, \quad \pi(x, y, z) = (x, y),$$

and $h = dx^2 + dy^2$,

$$\pi^*h = dx^2 + dy^2, \quad [\pi^*h] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Its radical equals $\ker d\pi = \text{span}\{\partial_z\}$.

12.2 Lorentzian null immersion

Let

$$i : \mathbb{R} \rightarrow \mathbb{R}^{1,1}, \quad i(s) = (s, s), \quad h = -dt^2 + dx^2.$$

di_s is injective, but

$$i^*h = -ds^2 + ds^2 = 0.$$

Thus an immersion can induce a degenerate pullback in indefinite signature. This is the counterexample excluded by an unjustified Riemannian intuition.

13 Quantity and Protocol Ledger

13.1 Coordinate-dimension rule

Store a coordinate passport $[x^\mu]$ for every chart. Since $[ds^2] = L^2$,

$$[g_{\mu\nu}] = \frac{L^2}{[x^\mu][x^\nu]}.$$

If all x^μ have length dimension, $g_{\mu\nu}$ is dimensionless. If $x^0 = t$ is measured in seconds while spatial coordinates are lengths, metric components have different dimensions and cannot be treated as an unannotated numerical matrix.

Object	Abstract dimension	GO-QS kind	Required metadata
x^μ	coordinate-specific	scalar/angle	chart and unit
g	$ds^2 : L^2$	covariant tensor	signature and coordinate passport
u	unitless in $c = 1$ normalization	vector	worldline and parameter
$a = \nabla_u u$	inverse length in geometrized parametrization	vector	unit context
$\Gamma^\rho_{\mu\nu}$	component-specific	connection coefficients	chart; not a tensor
$R^\rho_{\sigma\mu\nu}$	component-specific	tensor	curvature convention
Scal_g	L^{-2}	scalar	metric and sign convention
$e_a{}^\mu$	coordinate-specific	frame coefficients	frame, chart, Lorentz gauge
$e^a{}_\mu$	dual coordinate-specific	coframe coefficients	inverse-pair check
dO_e	output/input units	linear map	domain, codomain, rank threshold
O^*h	inherited from h	covariant tensor	signature and radical

13.2 Minimal relativistic protocol

Field	Required content
unit_context	SI or GR geometrized; explicit conversion boundary
signature	here $(-, +, +, +)$
curvature_convention	definition of $R(X, Y)Z$
coordinates	chart, ranges, units, singular sets
frame	frame and coframe matrices plus inverse-pair validation
lorentz_gauge	representative and transformation law
apparatus_orientation	material axes, distinct from gauge
observer	γ, u, a and transport law
observation	$R_\lambda, O_\lambda, K_\lambda$ with signatures
resolution	spatial, proper-time, spectral, and angular
uncertainty	covariance/posterior/risk and calibration
causal_domain	time orientation and admissible J^- region

14 Correction log: v0.2 to v0.3

1. Replaced the erroneous covariant metric reconstruction by the coframe formula and added the inverse-metric frame formula.
2. Declared signature, curvature convention, and GR_geometrized unit context.
3. Added the SI normalization $g(u_{\text{SI}}, u_{\text{SI}}) = -c^2$ and the $1/c^2$ Fermi-Walker conversion.
4. Separated passive coordinates, active diffeomorphisms, local Lorentz gauge, physical apparatus rotation, reduction, and observation.
5. Replaced the informal basic-tensor equivalence by a theorem for a surjective submersion with connected fibers.
6. Added the disconnected-fiber and mixed-tensor limitations.
7. Computed the exact radical of a pseudo-Riemannian pullback.
8. Added a null-immersion counterexample.
9. Explicitly denied a positive metric-distance interpretation for a Lorentzian form.
10. Added a coordinate-dimension passport and a complete GO-QS protocol ledger.

15 Claim audit

Established statements used:

1. tensor contractions are coordinate independent;
2. different tangent spaces require a transport rule;
3. frame and coframe reconstruct inverse and covariant metrics, respectively;
4. proper acceleration, Fermi-Walker rotation, and curvature are geometrically meaningful after conventions are fixed;
5. the radical formula characterizes local pullback degeneracy;

6. scalar and covariant-tensor descent require fiber compatibility.

Not claimed:

1. coordinate transformations are lossy observation channels;
2. every diffeomorphism is gauge under arbitrary boundary conditions;
3. an apparatus orientation is identical to a Lorentz gauge choice;
4. a Lorentzian metric is a positive distance;
5. every hidden tensor is recoverable;
6. this interface modifies general relativity.

16 Conclusion

The corrected tensorial interface is summarized by

covariance preserves representation-independent content;
a declared observation channel may quotient it.

The statement is now type safe. Coordinate and gauge changes act invertibly on representations; apparatus orientation belongs to the physical protocol; observation maps have explicit domains, codomains, and kernels; Lorentzian pullbacks are treated as indefinite bilinear forms; and tensor recovery is a descent problem with stated global hypotheses.

References

- [1] S. Kobayashi and K. Nomizu, *Foundations of Differential Geometry*, Vol. I, Wiley, 1963.
- [2] J. M. Lee, *Introduction to Smooth Manifolds*, 2nd ed., Springer, 2013.
- [3] J. M. Lee, *Introduction to Riemannian Manifolds*, 2nd ed., Springer, 2018.
- [4] B. O'Neill, *Semi-Riemannian Geometry with Applications to Relativity*, Academic Press, 1983.
- [5] R. M. Wald, *General Relativity*, University of Chicago Press, 1984.
- [6] M. Nakahara, *Geometry, Topology and Physics*, 2nd ed., CRC Press, 2003.
- [7] R. S. Palais, *Foundations of global non-linear analysis*, W. A. Benjamin, 1968.

Information-Theoretic Observation Geometry

Typed information loss, data processing, and recoverable invariants

Stassis Research Program

Strict GO Core reference revision v0.2 — 2026

Status. This revision is the information-theoretic P1 reference module for GO Core v0.2. It fixes the logarithm base, separates raw and normalized defects, records the degenerate-source zero policy, types the full observation protocol, and distinguishes probabilistic information loss from metric covering loss. All theorem statements below are standard finite-alphabet information theory unless explicitly marked as a model or interface proposition.

Abstract

An observation is represented by a reduction, a deterministic readout, and a stochastic channel, rather than by an unspecified map. For finite hidden and observed alphabets and a declared logarithm base $b > 1$, the raw information defect is the conditional entropy $\mathcal{D}_{\lambda, b}^{\text{info}}(P_X) = H_b(X | Y_\lambda)$. It depends on both the channel and the source law. Its normalized form divides by $H_b(X)$ with an explicit zero-source policy and is invariant under a change of information unit. Deterministic channels admit an exact fiber formula; post-processing monotonically decreases retained mutual information and increases the defect; Kullback-Leibler distinguishability contracts. Exact recovery of a finite target $Q = f(X)$ is equivalent to $H_b(Q | Y) = 0$, while Fano's inequality supplies a one-sided obstruction to approximate recovery. Continuous variables are handled through relative entropy, mutual information, and declared quantizers, not through coordinate-dependent differential entropy. The document supplies a typed interface and claim audit; it does not claim new information-theoretic inequalities or a derivation of physical dynamics.

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1 Scope, conventions, and claim boundary

Fix a logarithm base $b > 1$. Unless a standard-Borel extension is stated explicitly, every alphabet is finite and every random variable is defined on a common probability space. Probability laws are denoted by P_X , P_{XY} , and so forth. We use

$$0 \log_b 0 := 0, \quad \log_b \frac{p}{0} := +\infty \quad (p > 0).$$

Information quantities are physically dimensionless but have the semantic type information; their numerical unit is fixed by b (bits for $b = 2$, nats for $b = e$).

Boundary 1.1 (Claim boundary). Shannon entropy is not identified with thermodynamic entropy without a separate physical model. An information channel does not create a hidden state, and information theory alone does not derive spacetime, gravity, mass, charge, or consciousness. The contribution of this note is a type-safe interface between standard information theory and the Geometry of Observation protocol.

1.1 The GO observation protocol

For protocol $\lambda \in \Lambda$, write

$$X \xrightarrow{R_\lambda} X_\lambda \xrightarrow{O_\lambda} U_\lambda \xrightarrow{K_\lambda} Y_\lambda \xrightarrow{\hat{q}_\lambda} \hat{Q}_\lambda.$$

Here R_λ is a model reduction, O_λ is an ideal deterministic readout, and $K_\lambda(y | u)$ is a Markov kernel. The induced hidden-to-data channel is

$$W_\lambda(y | x) = K_\lambda(y | O_\lambda(R_\lambda x)). \quad (1)$$

If the reduction itself is stochastic, it is absorbed into W_λ by kernel composition. A frame or coordinate transformation is instead an invertible representation map and is not counted as information loss.

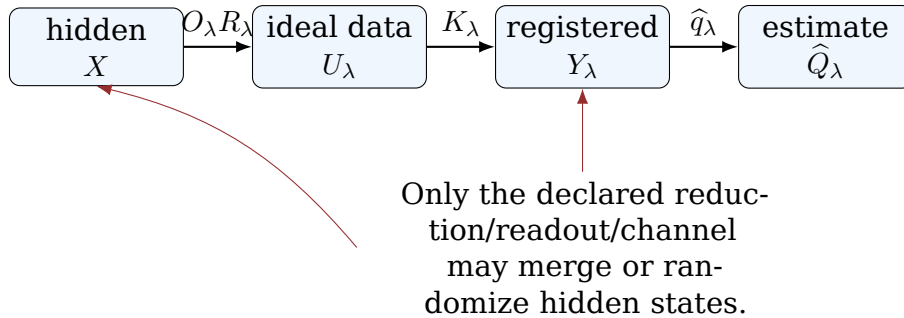


Figure 1: Typed observation chain. A channel is distinct from a change of coordinates or frame.

2 Information quantities and change of unit

Definition 2.1 (Entropy, conditional entropy, and mutual information). For a finite random variable X ,

$$H_b(X) = - \sum_{x \in \mathcal{X}} P_X(x) \log_b P_X(x).$$

For a pair (X, Y) ,

$$H_b(X | Y) = \sum_{y \in \mathcal{Y}} P_Y(y) H_b(X | Y = y),$$

and

$$I_b(X; Y) = H_b(X) - H_b(X | Y) = H_b(Y) - H_b(Y | X).$$

Definition 2.2 (Relative entropy). For laws P, Q on a finite set,

$$D_{\text{KL},b}(P \| Q) = \sum_x P(x) \log_b \frac{P(x)}{Q(x)} \in [0, +\infty].$$

It is finite precisely when $P \ll Q$ in the finite setting.

Theorem 2.3 (Information-unit change). *Let $b, c > 1$. Every entropy, conditional entropy, mutual information, and finite relative entropy in base b is related to its base- c value by*

$$H_b = \log_b(c) H_c, \quad I_b = \log_b(c) I_c, \quad D_{\text{KL},b} = \log_b(c) D_{\text{KL},c}.$$

Consequently, any ratio of two finite positive information quantities computed with the same base is independent of the base.

Proof. The identity $\log_b t = \log_b(c) \log_c t$ holds for $t > 0$. Substitution in each finite sum gives the first three relations. The common positive factor cancels in a ratio. $\square$

Remark 2.4 (Semantic units). The theorem is the information analogue of a coherent physical unit change. Subtracting a bit-valued entropy from a nat-valued entropy without conversion is forbidden, even though both are physically dimensionless.

3 Observation defect and deterministic fibers

Definition 3.1 (Raw information defect). For source law P_X and channel W_λ producing Y_λ , define

$$\mathcal{D}_{\lambda,b}^{\text{info}}(P_X) := H_b(X | Y_\lambda).$$

This is a function of the pair (P_X, W_λ) ; it is not an intrinsic scalar of the channel alone.

Definition 3.2 (Normalized information defect). The normalized defect is

$$\delta_\lambda^{\text{info}}(P_X) = \begin{cases} \frac{H_b(X | Y_\lambda)}{H_b(X)}, & H_b(X) > 0, \\ 0, & H_b(X) = 0. \end{cases}$$

The second line is the declared *degenerate-source zero policy*: a source supported at one state contains no source distinguishability to lose.

Proposition 3.3 (Decomposition, bounds, and unit invariance). *For every finite source and channel,*

$$H_b(X) = I_b(X; Y_\lambda) + \mathcal{D}_{\lambda,b}^{\text{info}}(P_X), \quad 0 \leq \mathcal{D}_{\lambda,b}^{\text{info}}(P_X) \leq H_b(X).$$

Moreover $0 \leq \delta_\lambda^{\text{info}}(P_X) \leq 1$, and $\delta_\lambda^{\text{info}}(P_X)$ is independent of b .

Proof. The identity is the definition of mutual information. Nonnegativity of conditional entropy and mutual information gives the bounds. The normalized claims follow from theorem 2.3 and the zero policy in definition 3.2. $\square$

Theorem 3.4 (Deterministic decomposition and fiber formula). *Let $Y = O(X)$ for a deterministic map $O : \mathcal{X} \rightarrow \mathcal{Y}$. Then*

$$H_b(X) = H_b(Y) + H_b(X | Y), \quad I_b(X; Y) = H_b(Y),$$

and

$$H_b(X | Y) = \sum_{y: P_Y(y) > 0} P_Y(y) \left[- \sum_{x \in O^{-1}(y)} P_{X|Y}(x | y) \log_b P_{X|Y}(x | y) \right]. \quad (2)$$

The defect vanishes if and only if X is a function of Y almost surely. Equivalently, every fiber meeting $\text{supp } P_X$ contains exactly one point of positive conditional probability.

Proof. Since Y is a function of X , $H_b(Y | X) = 0$. Apply the chain rule to $H_b(X, Y)$ in both orders. Expanding the definition of conditional entropy gives (2). A finite conditional entropy is zero exactly when every positive-probability conditional law is a point mass. $\square$

4 Post-processing, preorder, and divergence contraction

Definition 4.1 (GO post-processing preorder). For two channels with common input alphabet, write

$$W_1 \preceq_{\text{GO}} W_2 \iff W_2 = T \circ W_1$$

for some Markov kernel T . Thus W_2 is a garbling of W_1 ; the left channel is at least as informative. This convention agrees with the deterministic relation-refinement order used in the GO core.

Theorem 4.2 (Data processing and defect monotonicity). *If $X \rightarrow Y \rightarrow Z$ is a Markov chain, then*

$$I_b(X; Z) \leq I_b(X; Y), \quad H_b(X | Z) \geq H_b(X | Y).$$

More exactly,

$$H_b(X | Z) - H_b(X | Y) = I_b(X; Y | Z) \geq 0.$$

Proof. The chain rule gives

$$I_b(X; Y, Z) = I_b(X; Z) + I_b(X; Y | Z)$$

and

$$I_b(X; Y, Z) = I_b(X; Y) + I_b(X; Z | Y).$$

The Markov property makes the last conditional mutual information zero. The stated identities follow. $\square$

Theorem 4.3 (Relative-entropy contraction). *For a Markov kernel $W : \mathcal{X} \rightsquigarrow \mathcal{Y}$ and laws P, Q on $\mathcal{X}$,*

$$D_{\text{KL}, b}(WP \| WQ) \leq D_{\text{KL}, b}(P \| Q).$$

Proof. Apply the log-sum inequality to $a_x = P(x)W(y | x)$ and $b_x = Q(x)W(y | x)$ for each y , then sum over y . Extended values are understood when absolute continuity fails. $\square$

Definition 4.4 (Channel-level KL contraction profile). For a fixed finite channel W , define

$$\eta_{\text{KL},b}(W) = \sup_{\substack{P,Q: \\ 0 < D_{\text{KL},b}(P\|Q) < \infty}} \frac{D_{\text{KL},b}(WP\|WQ)}{D_{\text{KL},b}(P\|Q)},$$

with value 0 if the admissible set is empty.

Proposition 4.5. $0 \leq \eta_{\text{KL},b}(W) \leq 1$, and the value is independent of b .

Proof. The bounds follow from theorem 4.3; base independence follows from the common factor in theorem 2.3. $\square$

5 Recoverable targets

Definition 5.1 (Exact recoverability). Let $Q = f(X)$ take values in a finite alphabet $\mathcal{Q}$. It is recoverable from Y if there is a decoder $g : \mathcal{Y} \rightarrow \mathcal{Q}$ such that $Q = g(Y)$ almost surely.

Theorem 5.2 (Entropy criterion for exact recoverability). For finite Q and Y ,

$$Q \text{ is recoverable from } Y \iff H_b(Q | Y) = 0.$$

If $Y = O(X)$ is deterministic, this is equivalent to f being constant on each fiber $O^{-1}(y) \cap \text{supp } P_X$, up to null states.

Proof. The conditional entropy vanishes precisely when each conditional law $P_{Q|Y=y}$ with $P_Y(y) > 0$ is a point mass. Selecting its unique atom defines $g(y)$. The deterministic fiber statement is the same condition written in terms of O . $\square$

Corollary 5.3 (Target recovery is weaker than state recovery). It is possible that

$$H_b(X | Y) > 0 \quad \text{but} \quad H_b(f(X) | Y) = 0.$$

Thus recovery claims must name the target; recovery of one invariant does not imply recovery of the hidden state.

Theorem 5.4 (Fano inequality). Let $|\mathcal{Q}| = M \geq 2$, let $\hat{Q} = g(Y)$ be any decoder, and put $P_e = P(\hat{Q} \neq Q)$. Then

$$H_b(Q | Y) \leq h_b(P_e) + P_e \log_b(M - 1),$$

where

$$h_b(p) = -p \log_b p - (1 - p) \log_b(1 - p).$$

Remark 5.5 (One-sided use). Fano's inequality obstructs very small error when the unresolved conditional entropy is large. Its converse use requires additional hypotheses or sharper reverse bounds. No claim that "small conditional entropy automatically supplies a stable physical inverse" is made here.

5.1 Target-normalized defect

For nonconstant $Q = f(X)$, define

$$\delta_{Q,\lambda}^{\text{info}}(P_X) = \frac{H_b(Q | Y_\lambda)}{H_b(Q)}.$$

If $H_b(Q) = 0$, the same degenerate-target zero policy assigns value 0. This target-specific defect is base invariant and lies in $[0, 1]$.

6 Calibrating examples

6.1 Finite parity aliasing

Let X be uniform on $\{0, 1, \dots, 2m - 1\}$ and let $Y = X \bmod 2$. Then

$$H_b(X) = \log_b(2m), \quad H_b(Y) = \log_b 2, \quad H_b(X | Y) = \log_b m.$$

The channel retains parity and loses the index within each parity fiber. For $b = 2$, the retained information is exactly one bit.

6.2 Binary symmetric readout

Let $X \sim \text{Bernoulli}(1/2)$ and $Y = X \oplus N$, where $N \sim \text{Bernoulli}(\alpha)$ is independent and $0 \leq \alpha \leq 1/2$. Then

$$H_2(X | Y) = h_2(\alpha), \quad I_2(X; Y) = 1 - h_2(\alpha), \quad \delta^{\text{info}} = h_2(\alpha).$$

This example separates deterministic fiber merging from stochastic sensor noise.

6.3 Quantized hidden circle

Let a hidden variable be uniform on a circle of circumference $2\pi r$. Choose a declared arc-length quantizer whose cells have diameter at most $\varepsilon_\ell > 0$. With

$$m_\varepsilon = \left\lceil \frac{2\pi r}{\varepsilon_\ell} \right\rceil$$

equal-probability arc cells, the unresolved discrete fiber entropy grows as

$$\log_b m_\varepsilon = \log_b \frac{r}{\varepsilon_\ell} + O(1).$$

The logarithm contains the dimensionless ratio r/ε_ℓ . This is a quantized probabilistic statement, not an entropy assigned directly to an unquantized continuum.

6.4 Conditional chromatic multiplicity

Let P be an observed cell and C a hidden color, phase, or label. Observation of P alone leaves defect

$$H_b(P, C | P) = H_b(C | P).$$

If $C = c(P)$ is a deterministic field, this quantity is exactly zero. Nonzero chromatic information loss requires several conditionally possible labels at at least one observed cell; a deterministic graph lift by itself does not create hidden entropy.

7 Continuous variables and declared quantizers

For random elements in standard Borel spaces, define mutual information by

$$I_b(X; Y) = D_{\text{KL}, b}(P_{XY} \| P_X \otimes P_Y),$$

possibly $+\infty$. This definition is invariant under measurable bijections of the alphabets. By contrast, if p_X is a density relative to a dimensional coordinate measure dx , a dimensionally admissible coordinate differential entropy requires a positive reference

volume v_* with the same units as dx :

$$h_{b,v_*}(X) = - \int p_X(x) \log_b(p_X(x)v_*) dx.$$

It depends on v_* and changes under coordinates; it can also be negative. The unreferenceed shorthand $-\int p \log p dx$ is prohibited by the GO logarithm rule. Differential entropy is therefore not used as a coordinate-free GO defect.

Let $q_\varepsilon : \mathcal{X} \rightarrow \{1, \dots, N_\varepsilon\}$ be an explicitly declared finite quantizer or measurable partition. The finite observable

$$X_\varepsilon = q_\varepsilon(X)$$

has the well-typed defect

$$H_b(X_\varepsilon | Y).$$

Every limit $\varepsilon \downarrow 0$ must state the family of quantizers, the metric and reference scale, and whether the limit is finite, divergent, or renormalized. Covering entropy constrains how many geometric cells are available, but it is not generally equal to a distribution-dependent conditional entropy. That distinction is developed in the companion Metric Entropy v0.2 module.

8 Observation design

Model 8.1 (Information bottleneck interface). Let $Q = f(X)$ be the target and require the Markov condition $Q - X - Y$. For a declared channel class $\mathcal{W}$ and information budget R_b in base- b units, one may study

$$\sup_{W \in \mathcal{W}} I_b(Q; Y) \quad \text{subject to} \quad I_b(X; Y) \leq R_b.$$

The channel class, source law, alphabet restrictions, and existence of an optimizer are part of the model, not automatic consequences of the notation.

Model 8.2 (Typed rate-distortion interface). Let d_Q have physical dimension $\mathbf{d}$ and choose a strictly positive reference distortion d_{ref} of the same dimension. The normalized loss

$$\bar{d}_Q(q, \hat{q}) = \frac{d_Q(q, \hat{q})}{d_{\text{ref}}}$$

is dimensionless. A rate-distortion problem may then constrain $\mathbb{E}[\bar{d}_Q] \leq D$ with dimensionless D . The source, admissible reproduction alphabet, and channel class must be declared.

9 GO Core quantity and protocol ledger

9.1 Map register

Map	Domain	Codomain	GO kind and loss status
R_λ	hidden state X	reduced state X_λ	reduction; loss possible
O_λ	X_λ	ideal datum U_λ	deterministic observation; loss possible
K_λ	U_λ	law on Y_λ	stochastic channel; loss possible
T	Y	law on Z	post-processing channel; loss possible
$\hat{q}_\lambda$	Y_λ	target estimate	estimator; loss is not a map attribute

9.2 Quantity register

Quantity	Physical dimension	Semantic type	Required protocol fields
$P_X(x), W(y x)$	0	probability	alphabet, normalization, source/channel law
$H_b, I_b, D_{\text{KL},b}$	0	information in base b	logarithm base and finiteness policy
$\mathcal{D}_{\lambda,b}^{\text{info}}$	0	raw information defect	source law, channel, target (if any)
$\delta_\lambda^{\text{info}}$	0	normalized loss	positive denominator or declared zero policy
d_Q	declared d	distortion	target space, metric/loss, reference distortion
$\bar{d}_Q$	0	normalized loss	$d_{\text{ref}} > 0$ with the same dimension

The complete protocol ledger declares:

1. hidden alphabet and source law;
2. system descriptor and protocol index λ ;
3. reduction, ideal observation, noise channel, and observed alphabet;
4. reference frame (or the explicit statement not applicable);
5. quantizer/partition and spatial/temporal resolution, including not used where appropriate;
6. observation horizon;
7. estimator or direct-readout declaration;
8. exactness, uncertainty, and missing-data policy;
9. neutral/SI unit context and information base;
10. raw defect, normalizer, and zero policy.

10 Claim audit and conclusion

Established under stated hypotheses	Not established by this module
Conditional entropy measures unresolved finite-source uncertainty for a declared law and channel.	An intrinsic, prior-free scalar loss for every observation device.
Post-processing decreases mutual information and contracts KL divergence.	Post-processing can reconstruct information absent from its input.
$H_b(Q Y) = 0$ characterizes exact finite-target recovery almost surely.	Recovery of one invariant implies recovery of the full hidden state.
Quantized continuous variables admit finite information defects.	Differential entropy is a coordinate-invariant hidden-state defect.
The normalized defect is invariant under information-unit changes.	Information defect is automatically thermodynamic entropy or physical energy.

The module closes a probabilistic interface:

$$\boxed{\text{declared source} + \text{declared channel} \mapsto (I_b(X; Y), H_b(X | Y), \eta_{\text{KL}}(W))}.$$

These are typed, noninterchangeable objects. The results are mathematically standard; the framework contribution is their integration into a reproducible observation protocol with explicit units, targets, normalizations, and claim status.

References

- [1] C. E. Shannon, A mathematical theory of communication, *Bell System Technical Journal* **27** (1948), 379–423 and 623–656.
- [2] D. Blackwell, Equivalent comparisons of experiments, *Annals of Mathematical Statistics* **24** (1953), 265–272, doi:10.1214/aoms/1177729032.
- [3] I. Csiszar, Information-type measures of difference of probability distributions and indirect observations, *Studia Scientiarum Mathematicarum Hungarica* **2** (1967), 299–318.
- [4] R. M. Fano, *Transmission of Information: A Statistical Theory of Communications*, MIT Press, Cambridge, MA, 1961.
- [5] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, second edition, Wiley, Hoboken, 2006.
- [6] I. Csiszar and J. Korner, *Information Theory: Coding Theorems for Discrete Memoryless Systems*, second edition, Cambridge University Press, Cambridge, 2011.
- [7] R. M. Gray, *Entropy and Information Theory*, second edition, Springer, New York, 2011.
- [8] N. Tishby, F. C. Pereira, and W. Bialek, The information bottleneck method, in *Proceedings of the 37th Annual Allerton Conference on Communication, Control, and Computing*, 1999, 368–377.
- [9] S. Amari, *Information Geometry and Its Applications*, Springer, Tokyo, 2016.

Metric Entropy and Observational Entropy Defect

Matched resolutions, hidden fibers, curvature, and heat-trace complexity

Stassis Research Program

Strict GO Core reference revision v0.2 — 2026

Status. This revision is the metric-scale P1 reference module for GO Core v0.2. It replaces dimensional logarithms by reference-scale ratios, uses separate input and output resolutions, types the Lipschitz constant, assigns the heat parameter geometric dimension L^2 before nondimensionalization, and removes the untyped curvature/gradient/entropy sum of v0.1.

Abstract

Covering numbers quantify the number of metric cells needed to resolve a set at a declared scale. Their logarithms are information-valued because their arguments are counts, but all scale limits must use dimensionless ratios. For a Lipschitz observation $O : (X, d_X) \rightarrow (Y, d_Y)$, the physically correct interface uses two resolutions, ε_X and ε_Y , satisfying $L_O \varepsilon_X \leq \varepsilon_Y$. The matched-scale defect

$$\mathcal{D}_{O,b}^{\text{met}}(A; \varepsilon_X, \varepsilon_Y) = \log_b N_{\varepsilon_X}(A, d_X) - \log_b N_{\varepsilon_Y}(O(A), d_Y)$$

is then nonnegative. Its normalized version has an explicit zero policy and is invariant under changes of length unit and information unit. Defects telescope exactly under composable matched observations. Product fibers, a hidden circle, finite parity, and chromatic graph lifts are treated with stated metric hypotheses. Curvature enters only through dimensionless small-ball asymptotics. The heat-trace logarithm is typed as a smooth spectral complexity proxy, not as Shannon or thermodynamic entropy. The module is a strict interface built from standard metric, Riemannian, and spectral facts; no new gravity law is claimed.

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1 Scope and terminology

This document uses *metric entropy* in the Kolmogorov-Tikhomirov covering sense

$$\mathcal{H}_{b,\varepsilon}(A, d) = \log_b N_\varepsilon(A, d),$$

not in the Kolmogorov-Sinai dynamical sense. The logarithm base $b > 1$ is always declared. A physical metric can be length-valued; an abstract metric may carry another declared scale type. Every set placed inside a logarithm is assumed nonempty. The empty-set covering number is not assigned an entropy in this module.

Boundary 1.1 (Claim boundary). Covering entropy is neither thermodynamic entropy nor physical energy. Curvature influences local volume and hence covering estimates, but this does not derive Einstein's equations. A hidden metric fiber is a mathematical non-identifiability mechanism, not automatically an extra physical dimension.

2 Covering, packing, and scale passports

Definition 2.1 (Internal covering number). Let (X, d) be a metric space, $A \subseteq X$, and $\varepsilon > 0$ with the same scale type as d . Using closed balls $\overline{B}_d(x, \varepsilon)$, define

$$N_\varepsilon(A, d) = \min \left\{ |S| : S \subseteq A, A \subseteq \bigcup_{s \in S} \overline{B}_d(s, \varepsilon) \right\},$$

and set $N_\varepsilon(A, d) = +\infty$ when no finite cover exists.

Definition 2.2 (Packing number). For $\varepsilon > 0$, define

$$P_\varepsilon(A, d) = \sup \{ |S| : S \subseteq A, d(x, x') > \varepsilon \text{ for } x \neq x' \}.$$

For a totally bounded set this number is finite; for compact A the supremum is attained.

Proposition 2.3 (Covering-packing sandwich). For every totally bounded A and $\varepsilon > 0$,

$$P_{2\varepsilon}(A, d) \leq N_\varepsilon(A, d) \leq P_\varepsilon(A, d).$$

Proof. An ε -ball contains at most one point of a 2ε -separated set, which gives the first inequality. A maximal ε -separated set is an ε -net, giving the second. $\square$

Definition 2.4 (Covering entropy). For $N_\varepsilon(A, d) < \infty$,

$$\mathcal{H}_{b,\varepsilon}(A, d) := \log_b N_\varepsilon(A, d).$$

Its physical dimension is **0**, its semantic type is information, and its information unit is fixed by b .

2.1 Coherent changes of length and information unit

Theorem 2.5 (Metric-unit covariance). *For every scale factor $a > 0$,*

$$N_{a\varepsilon}(A, ad) = N_\varepsilon(A, d), \quad \mathcal{H}_{b,a\varepsilon}(A, ad) = \mathcal{H}_{b,\varepsilon}(A, d).$$

Proof. $\overline{B}_{ad}(x, a\varepsilon) = \overline{B}_d(x, \varepsilon)$ as subsets of X , so the admissible covers are identical. $\square$

Theorem 2.6 (Information-unit covariance). *For $b, c > 1$,*

$$\mathcal{H}_{b,\varepsilon}(A, d) = \log_b(c) \mathcal{H}_{c,\varepsilon}(A, d).$$

Therefore ratios of finite positive covering entropies computed at the same base are base independent.

3 Dimension without dimensional logarithms

Choose a positive reference scale ℓ_* with the same physical dimension as d , and write

$$\bar{d} = \frac{d}{\ell_*}, \quad \rho = \frac{\varepsilon}{\ell_*}.$$

Both are dimensionless. The lower and upper box dimensions are

$$\underline{\dim}_B(A, d) = \liminf_{\varepsilon \downarrow 0} \frac{\mathcal{H}_{b,\varepsilon}(A, d)}{\log_b(\ell_*/\varepsilon)}, \quad (1)$$

$$\overline{\dim}_B(A, d) = \limsup_{\varepsilon \downarrow 0} \frac{\mathcal{H}_{b,\varepsilon}(A, d)}{\log_b(\ell_*/\varepsilon)}. \quad (2)$$

When they agree, their common value is $\dim_B(A, d)$.

Proposition 3.1 (Convention independence). *The lower and upper box dimensions are independent of the logarithm base, the coherent length unit, and the choice of fixed positive reference length ℓ_* .*

Proof. Changing the logarithm base multiplies numerator and denominator by the same positive constant. A coherent unit change leaves the ratios d/ℓ_* and ε/ℓ_* unchanged by theorem 2.5. Replacing ℓ_* by $a\ell_*$ adds the constant $\log_b a$ to the denominator; this does not alter the limiting quotient. $\square$

Remark 3.2 (Prohibited shorthand). The expression $\log(1/\varepsilon)$ is not physically typed when ε is a length. It is permitted only after an explicit nondimensionalization, as $\log(1/\rho)$ or $\log(\ell_*/\varepsilon)$.

4 Lipschitz observations at matched resolutions

Let d_X and d_Y have declared scale types U_X and U_Y . A Lipschitz constant has type U_Y/U_X :

$$d_Y(Ox, Ox') \leq L_O d_X(x, x').$$

When both metrics are lengths, L_O is physically dimensionless.

Definition 4.1 (Matched scale pair). A pair $(\varepsilon_X, \varepsilon_Y)$ is admissible for O if

$$\varepsilon_X > 0, \quad \varepsilon_Y > 0, \quad L_O \varepsilon_X \leq \varepsilon_Y.$$

Theorem 4.2 (Matched-scale covering contraction). *If $O : (X, d_X) \rightarrow (Y, d_Y)$ is L_O -Lipschitz and $(\varepsilon_X, \varepsilon_Y)$ is matched, then for every totally bounded $A \subseteq X$,*

$$N_{\varepsilon_Y}(O(A), d_Y) \leq N_{\varepsilon_X}(A, d_X),$$

and hence

$$\mathcal{H}_{b, \varepsilon_Y}(O(A), d_Y) \leq \mathcal{H}_{b, \varepsilon_X}(A, d_X).$$

Proof. The image of every d_X -ball of radius ε_X is contained in a d_Y -ball of radius $L_O \varepsilon_X \leq \varepsilon_Y$. Map the centers of an optimal finite input cover through O . $\square$

Definition 4.3 (Matched-scale observational defect). For an admissible scale pair, define

$$\mathcal{D}_{O,b}^{\text{met}}(A; \varepsilon_X, \varepsilon_Y) := \mathcal{H}_{b, \varepsilon_X}(A, d_X) - \mathcal{H}_{b, \varepsilon_Y}(O(A), d_Y) \geq 0.$$

This is an information-valued scale profile, not a physical length or energy.

Definition 4.4 (Normalized metric defect). Set

$$\delta_O^{\text{met}}(A; \varepsilon_X, \varepsilon_Y) = \begin{cases} \frac{\mathcal{D}_{O,b}^{\text{met}}(A; \varepsilon_X, \varepsilon_Y)}{\mathcal{H}_{b, \varepsilon_X}(A, d_X)}, & \mathcal{H}_{b, \varepsilon_X}(A, d_X) > 0, \\ 0, & \mathcal{H}_{b, \varepsilon_X}(A, d_X) = 0. \end{cases}$$

The second line is the declared one-cell zero policy.

Proposition 4.5 (Invariances and bounds). $0 \leq \delta_O^{\text{met}} \leq 1$. *It is invariant under information-base changes and under coherent rescaling of each metric together with its own resolution.*

Proof. Nonnegativity follows from theorem 4.2; the upper bound uses nonnegativity of the image entropy. Base factors cancel by theorem 2.6, and coherent metric rescaling leaves both covering numbers unchanged by theorem 2.5. $\square$

Theorem 4.6 (Exact composition law). *Let $O_1 : X \rightarrow Y$ and $O_2 : Y \rightarrow Z$ be Lipschitz. If*

$$L_1 \varepsilon_X \leq \varepsilon_Y, \quad L_2 \varepsilon_Y \leq \varepsilon_Z,$$

then

$$\begin{aligned} \mathcal{D}_{O_2 \circ O_1, b}^{\text{met}}(A; \varepsilon_X, \varepsilon_Z) &= \mathcal{D}_{O_1, b}^{\text{met}}(A; \varepsilon_X, \varepsilon_Y) \\ &\quad + \mathcal{D}_{O_2, b}^{\text{met}}(O_1(A); \varepsilon_Y, \varepsilon_Z). \end{aligned}$$

Proof. Expand the two terms on the right. The intermediate covering entropy $\mathcal{H}_{b, \varepsilon_Y}(O_1(A), d_Y)$ cancels. Each term is nonnegative by theorem 4.2. $\square$

Remark 4.7. The uncorrected same-symbol difference $\mathcal{H}_{b, \varepsilon}(A, d_X) - \mathcal{H}_{b, \varepsilon}(O(A), d_Y)$ has no invariant meaning until the input and output scale types and unit conversion are declared.

5 Hidden product fibers

Proposition 5.1 (Product covering bounds). *Let B, F be nonempty totally bounded metric spaces and equip $B \times F$ with*

$$d_\infty((b, f), (b', f')) = \max \{d_B(b, b'), d_F(f, f')\}.$$

Then

$$N_\varepsilon(B, d_B) \leq N_\varepsilon(B \times F, d_\infty) \leq N_\varepsilon(B, d_B) N_\varepsilon(F, d_F).$$

For the projection $\pi : B \times F \rightarrow B$,

$$0 \leq \mathcal{D}_{\pi, b}^{\text{met}}(B \times F; \varepsilon, \varepsilon) \leq \mathcal{H}_{b, \varepsilon}(F, d_F).$$

Proof. An ε -cover of the product restricts to an ε -cover of the isometric copy $B \times \{f_0\}$, giving the lower bound. The Cartesian product of input covers gives the upper bound. The entropy inequalities follow by taking logarithms. $\square$

Theorem 5.2 (Compact manifold fiber asymptotics). *Let B and F be compact Riemannian manifolds of dimensions m and n , and let a common reference length $\ell_* > 0$ nondimensionalize their distance metrics. For the max product metric,*

$$\mathcal{D}_{\pi, b}^{\text{met}}(B \times F; \varepsilon, \varepsilon) = n \log_b \frac{\ell_*}{\varepsilon} + O(1) \quad (\varepsilon \downarrow 0).$$

Proof. For every compact smooth k -manifold, local Euclidean comparison and a finite atlas yield constants $0 < c < C < \infty$ such that

$$c \left(\frac{\ell_*}{\varepsilon} \right)^k \leq N_\varepsilon(M) \leq C \left(\frac{\ell_*}{\varepsilon} \right)^k$$

for sufficiently small ε . Apply this to $B \times F$ and B and subtract the logarithms. $\square$

Corollary 5.3 (Hidden circle). *For the intrinsic geodesic metric on a circle S_R^1 of radius R ,*

$$N_\varepsilon(S_R^1) = \left\lceil \frac{\pi R}{\varepsilon} \right\rceil, \quad 0 < \varepsilon < \pi R.$$

Hence a forgotten circle contributes

$$\log_b \frac{R}{\varepsilon} + O(1)$$

to the small-scale projection defect.

Proof. A closed geodesic ball of radius ε covers an arc of length 2ε . The circle has circumference $2\pi R$; equally spaced centers attain the stated ceiling. $\square$

Corollary 5.4 (Finite parity fiber). *Let $F = \mathbb{Z}/2\mathbb{Z}$ with point separation $a > 0$. Then*

$$\mathcal{H}_{b, \varepsilon}(F) = \begin{cases} \log_b 2, & 0 < \varepsilon < a, \\ 0, & \varepsilon \geq a. \end{cases}$$

The finite defect therefore depends explicitly on whether the protocol resolves the two points.

6 Operational bridge to probabilistic information

The packing number is the largest codebook size at a declared minimum separation. A uniform random variable X_S on a finite ε -separated codebook S has

$$H_b(X_S) = \log_b |S| \leq \log_b P_\varepsilon(A, d).$$

Conversely, a chosen ε -cover produces a deterministic finite quantizer with at most $N_\varepsilon(A, d)$ labels, so its output entropy is at most $\mathcal{H}_{b,\varepsilon}(A, d)$ for every source law. Together with proposition 2.3, this gives an operational packing/covering bracket on finite-resolution distinguishability.

Boundary 6.1 (No automatic equality). The metric defect depends on worst-case covering capacity, while $H_b(X | Y)$ depends on a source law and conditional probabilities. They can agree in calibrated finite symmetric examples, but there is no general identity between them.

7 Chromatic graph lifts

Let (X, d_X) be length-valued and let (C, d_C) be a declared feature space. Choose a scale converter $\lambda > 0$ of type $[d_X]/[d_C]$ and define

$$D_\lambda((x, a), (y, c))^2 = d_X(x, y)^2 + \lambda^2 d_C(a, c)^2.$$

For a field $c : X \rightarrow C$, its graph is

$$\Gamma_c = \{(x, c(x)) : x \in X\}.$$

Proposition 7.1 (Typed chromatic entropy sandwich). *If c is L_c -Lipschitz, then λL_c is dimensionless and*

$$\mathcal{H}_{b,\varepsilon}(X, d_X) \leq \mathcal{H}_{b,\varepsilon}(\Gamma_c, D_\lambda) \leq \mathcal{H}_{b,\varepsilon/\sqrt{1+(\lambda L_c)^2}}(X, d_X).$$

Proof. Projection $\Gamma_c \rightarrow X$ is 1-Lipschitz. The graph map $x \mapsto (x, c(x))$ is $\sqrt{1 + (\lambda L_c)^2}$ -Lipschitz. Apply theorem 4.2 to both maps. $\square$

The chromatic excess

$$\mathcal{L}_{b,\varepsilon}^{\text{ch}}(X, c; \lambda) = \mathcal{H}_{b,\varepsilon}(\Gamma_c, D_\lambda) - \mathcal{H}_{b,\varepsilon}(X, d_X)$$

is therefore nonnegative. It measures graph-metric complexity, not hidden probabilistic label entropy. If labels are random conditional on position, the separate quantity $H_b(C | X)$ belongs to the information-theoretic module.

8 Local volume, curvature, and a typed pressure model

Let (M, g) be a closed Riemannian n -manifold. Fix $\ell_* > 0$ and define

$$\bar{d} = \frac{dg}{\ell_*}, \quad d\bar{V} = \frac{dV_g}{\ell_*^n}, \quad \overline{\text{Scal}} = \ell_*^2 \text{Scal}_g, \quad \bar{r} = \frac{r}{\ell_*}.$$

Every barred quantity in this display is dimensionless.

Proposition 8.1 (Dimensionless small-ball expansion). *As $\bar{r} \downarrow 0$,*

$$\bar{V}(B_{\bar{d}}(x, \bar{r})) = \omega_n \bar{r}^n \left[1 - \frac{\overline{\text{Scal}}(x)}{6(n+2)} \bar{r}^2 + O(\bar{r}^4) \right].$$

Remark 8.2. The statement is local and asymptotic. The sign of scalar curvature controls the leading relative volume correction; it does not by itself determine a finite-scale covering number or a gravitational field equation.

Choose a positive reference information h_* in base- b units and a positive dimensionless reference volume $\bar{v}_*$. A typed local complexity density is

$$\hat{\rho}_{b,\varepsilon,r}(x) = \frac{\mathcal{H}_{b,\varepsilon}(B_g(x,r), d_g)}{h_*} \frac{\bar{v}_*}{\bar{V}(B_d(x, \bar{r}))}.$$

Both ratios are dimensionless and both denominators have strict positivity policies.

Model 8.3 (Dimensionless entropy-pressure functional). Let $\Phi : [0, \infty) \rightarrow [0, \infty)$ be declared and let $d\bar{\mu}_g = d\bar{V}/\bar{V}(M)$. Then

$$\mathcal{P}_{b,\varepsilon,r}[g] = \int_M \Phi(\hat{\rho}_{b,\varepsilon,r}(x)) d\bar{\mu}_g(x)$$

is dimensionless. Any combination with curvature, gradient, or physical energy terms must first normalize each component separately:

$$\mathcal{J} = \sum_{k=1}^N w_k \phi_k(\nu_k), \quad w_k \geq 0, \quad \sum_k w_k = 1,$$

where every ν_k is a declared nonnegative dimensionless quantity with zero and infinity policies. This is a decision functional, not a universal invariant.

9 Heat trace with a geometric scale

Let $\Delta_g \geq 0$ be the nonnegative Laplace-Beltrami operator on a closed Riemannian manifold. Its eigenvalues have physical dimension L^{-2} . Define

$$\bar{\Delta} = \ell_*^2 \Delta_g, \quad \bar{\lambda}_j = \ell_*^2 \lambda_j, \quad \tau = \frac{s}{\ell_*^2} > 0,$$

where the geometric heat parameter s has dimension L^2 . Then

$$\tau \bar{\lambda}_j = s \lambda_j$$

is dimensionless.

Definition 9.1 (Heat-trace complexity).

$$Z_g(\tau) = \text{Tr}(e^{-\tau \bar{\Delta}}) = \sum_{j=0}^{\infty} e^{-\tau \bar{\lambda}_j}, \quad \mathcal{C}_{b,\tau}^{\text{heat}}(M, g) = \log_b Z_g(\tau).$$

Boundary 9.2 (Semantic status). $\log_b Z_g(\tau)$ is a log heat trace, analogous to a log partition function. It is not Shannon entropy, von Neumann entropy, or thermodynamic entropy unless a separate normalized state and physical model are supplied.

Theorem 9.3 (Small- τ heat-trace expansion). *For a closed smooth n -manifold,*

$$Z_g(\tau) \sim (4\pi\tau)^{-n/2} \left[\bar{V}(M) + \frac{\tau}{6} \int_M \overline{\text{Scal}} d\bar{V} + O(\tau^2) \right] \quad (\tau \downarrow 0).$$

All terms in the bracket are dimensionless.

If t_{phys} is a diffusion time and κ has dimension $L^2 T^{-1}$, then

$$\tau = \frac{\kappa t_{\text{phys}}}{\ell_*^2}$$

is the required dimensionless parameter. Writing $e^{-t_{\text{phys}}\Delta_g}$ without a diffusivity is dimensionally incorrect when Δ_g has dimension L^{-2} .

10 Shell graphs and a conditional cylindrical example

For a finite configuration $P = \{p_1, \dots, p_N\}$, define a shell graph using a radius $r > 0$ and tolerance $\sigma \geq 0$ of the same dimension:

$$E_{r,\sigma}(P) = \{\{i, j\} : 1 \leq i < j \leq N, |d(p_i, p_j) - r| < \sigma\}.$$

Under coherent rescaling $(d, r, \sigma) \mapsto (ad, ar, a\sigma)$, the graph is unchanged. Exact unit-distance incidence corresponds to a declared Euclidean reference length and $\sigma = 0$; it is not inferred from covering numbers alone.

Model 10.1 (Cylindrical parameter fiber). Suppose a nondimensionalized compact parameter space has the product form

$$\mathcal{P} = S^2 \times D \times S^1 \times SO(2),$$

where D is a finite set of q separated diameter states. With a regular product metric,

$$\mathcal{H}_{b,\rho}(\mathcal{P}) = 4 \log_b(1/\rho) + \log_b q + O(1).$$

If a Lipschitz observation has bounded image in a one-dimensional interval, then its image covering entropy is at most $\log_b(1/\rho) + O(1)$ at matched dimensionless scales. Thus the defect is at least

$$3 \log_b(1/\rho) + \log_b q + O(1)$$

under these stated hypotheses. No such rate is asserted for an arbitrary non-Lipschitz readout or an undeclared product metric.

11 GO Core quantity and protocol ledger

Quantity	Physical dimension	Semantic type	Required protocol fields
$d_X, \varepsilon_X, \ell_{*,X}$	U_X	metric scale	metric, ball convention, input resolution, reference scale
$d_Y, \varepsilon_Y, \ell_{*,Y}$	U_Y	metric scale	output metric, output resolution, reference scale
L_O	U_Y/U_X	Lipschitz scale converter	domain, codomain, certified or assumed bound
$N_\varepsilon, P_\varepsilon$	0	count	internal/external centers, closed/open balls, separation convention
$\mathcal{H}_{b,\varepsilon}$	0	information in base b	logarithm base and finite/infinite policy
$\mathcal{D}_{O,b}^{\text{met}}$	0	scale-profile information defect	matched pair $(\varepsilon_X, \varepsilon_Y)$
δ_O^{met}	0	normalized loss	positive entropy denominator or one-cell zero policy
s	L^2	geometric heat parameter	ℓ_* or physical diffusivity map
$\tau = s/\ell_*^2$	0	normalized heat scale	strictly positive ℓ_*
λ	$[d_X]/[d_C]$	feature scale converter	feature metric and chromatic protocol

The protocol ledger declares the hidden metric space, system descriptor, reduction (or identity), deterministic observation, stochastic channel (or noiseless statement), observed metric space, reference frame or not applicable, quantizer/partition, both spatial resolutions, temporal resolution, observation horizon, estimator/direct readout, uncertainty/exactness status, SI/neutral unit contexts, and defect normalizations.

12 Claim audit and conclusion

Established under stated hypotheses	Not established by this module
Matched Lipschitz observations contract covering numbers.	A same-numerical-scale entropy difference is invariant across arbitrary metric units.
Product fibers contribute their box dimension to logarithmic defect growth.	Every hidden mathematical fiber is a physical compact dimension.
The normalized defect is invariant under coherent length and information unit changes.	Covering entropy is automatically equal to conditional Shannon entropy.
Scalar curvature enters the leading small-ball volume correction.	An entropy-pressure ansatz derives a gravitational field equation.
The nondimensionalized heat trace has standard spectral asymptotics.	The heat-trace logarithm is thermodynamic entropy without a state model.

The strict observable is the scale profile

$$(\varepsilon_X, \varepsilon_Y) \mapsto \mathcal{D}_{O,b}^{\text{met}}(A; \varepsilon_X, \varepsilon_Y), \quad L_O \varepsilon_X \leq \varepsilon_Y.$$

It is not collapsed into a single universal defect unless a protocol chooses a scale distribution or a decision functional. The principal addition of v0.2 is type safety and

compositional closure; the covering, manifold, and heat-kernel results themselves are standard.

References

- [1] A. N. Kolmogorov and V. M. Tikhomirov, ε -entropy and ε -capacity of sets in function spaces, *American Mathematical Society Translations, Series 2* **17** (1961), 277–364, doi:10.1090/trans2/017/10.
- [2] R. M. Dudley, *Real Analysis and Probability*, Cambridge University Press, Cambridge, 2002.
- [3] D. Burago, Y. Burago, and S. Ivanov, *A Course in Metric Geometry*, American Mathematical Society, Providence, 2001.
- [4] K. Falconer, *Fractal Geometry: Mathematical Foundations and Applications*, third edition, Wiley, Chichester, 2014.
- [5] P. Petersen, *Riemannian Geometry*, third edition, Springer, Cham, 2016.
- [6] A. Gray, *Tubes*, second edition, Birkhauser, Basel, 2004.
- [7] S. Minakshisundaram and A. Pleijel, Some properties of the eigenfunctions of the Laplace-operator on Riemannian manifolds, *Canadian Journal of Mathematics* **1** (1949), 242–256, doi:10.4153/CJM-1949-021-5.
- [8] P. B. Gilkey, *Invariance Theory, the Heat Equation, and the Atiyah–Singer Index Theorem*, second edition, CRC Press, Boca Raton, 1995.
- [9] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, second edition, Wiley, Hoboken, 2006.

Distance and Scale Interface under Observation Maps

Global pseudometrics, quotient geometry, matched resolutions, and calibrated shells

Stassis Research Program

Strict GO Core reference revision v0.2 — 2026

Status. This document replaces *Unit Distances under Observation Maps v0.1* as the distance-scale reference module for GO Core v0.2. A numerical value “1” is never compared with a physical distance without a declared reference radius. The global observed pseudometric is separated from the local pullback tensor, all resolution maps are typed, and exact-shell statements are separated from tolerant or noisy decisions.

Abstract

Let $O : (X, d_X) \rightarrow (Y, d_Y)$ be an observation map between metric spaces. The globally observed distance on hidden states is the pseudometric $d_O(x, x') = d_Y(Ox, Ox')$. Its zero classes are the observation fibers, and the metric quotient is canonically isometric to the image $O(X)$. This global construction is distinct from the local pullback quadratic form O^*g_Y ; the corresponding path pseudometric can exceed the ambient image distance unless a minimizing image geodesic is liftable. A distance shell is defined relative to a positive, typed radius r_* , while finite resolution and measurement error require a tolerance τ and an explicit decision rule. Under coherent changes of unit, d_Y, r_*, τ , and the output resolution rescale together, leaving all normalized predicates invariant. Lipschitz observations propagate input resolution to output resolution and compose multiplicatively. The orthogonal-projection cylinder, fiber multiplicities in observed unit-distance graphs, and periodic half-step aliasing are derived as applications. The module is an interface built from standard metric and differential geometry; it does not solve the classical Erdos unit-distance problem.

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1 Scope, objects, and claim boundary

The purpose of this module is to make every use of “distance” and “scale” in Geometry of Observation type-checkable. It supplies a common interface for metric entropy, ruler traces, mechanics, and later observation modules. It does not impose a physical ontology on the hidden space.

Boundary 1.1 (Claim boundary). An observation-induced pseudometric is a standard pullback construction. Fiber multiplicity can create quadratically many hidden-state pairs, but this is not a new lower bound for the classical planar unit-distance function. A degenerate pullback tensor does not by itself imply a physical extra dimension.

Definition 1.2 (Typed metric datum). A typed metric datum is a tuple

$$\mathfrak{M}_X = (X, d_X, U_X),$$

where $d_X : X \times X \rightarrow \mathbb{R}_{\geq 0}U_X$ is a metric and U_X names its scale type. For a physical spatial metric, U_X may be metres; for a normalized abstract metric, $U_X = 1$.

The notation $\mathbb{R}_{\geq 0}U_X$ is a bookkeeping device: it records that d_X is length-valued (or more generally scale-valued), not an untyped real number. Ratios of two positive quantities with the same scale type are dimensionless.

Definition 1.3 (System and protocol passports). The intrinsic descriptor and the observation protocol are distinct:

$$\xi_X = (X, d_X, \text{diam } X, \text{intrinsic regularity data}, \dots),$$

$$\lambda = (O, d_Y, r_*, \varepsilon_X, \varepsilon_Y, \tau, Q, E, U, \dots).$$

Here $r_* > 0$ is a declared output-space reference radius, $\varepsilon_X, \varepsilon_Y > 0$ are input and output resolutions, $\tau \geq 0$ is a decision tolerance, Q is a quantizer or partition, E is an estimator or direct readout, and U is an uncertainty model.

An intrinsic diameter is not an instrument resolution. A shell radius is not a regression reference length. The same physical dimension therefore does not imply the same semantic role.

2 The global observed pseudometric

Definition 2.1 (Observed pseudometric). Let (Y, d_Y) be a metric space and let $O : X \rightarrow Y$ be any map. Define

$$d_O(x, x') := d_Y(O(x), O(x')).$$

Theorem 2.2 (Pseudometric and separation criterion). *The function d_O is a pseudometric on X . It is a metric if and only if O is injective.*

Proof. Nonnegativity and symmetry follow from d_Y . For $x, x', x'' \in X$,

$$d_O(x, x'') = d_Y(Ox, Ox'') \leq d_Y(Ox, Ox') + d_Y(Ox', Ox'') = d_O(x, x') + d_O(x', x'').$$

Thus the triangle inequality holds. Moreover, $d_O(x, x') = 0$ if and only if $O(x) = O(x')$. Hence separation is equivalent to injectivity. $\square$

Definition 2.3 (Observation equivalence). Write

$$x \sim_O x' \iff O(x) = O(x').$$

Let $X/\sim_O$ denote the set of observation fibers.

Theorem 2.4 (Canonical quotient metric). *The formula*

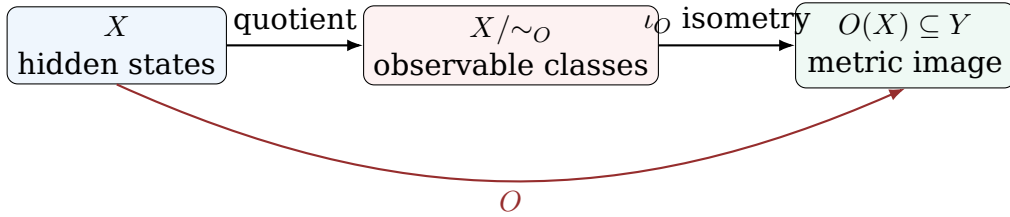
$$\bar{d}_O([x], [x']) := d_Y(Ox, Ox')$$

is well-defined and is a metric on $X/\sim_O$. The map

$$\iota_O : X/\sim_O \longrightarrow O(X), \quad [x] \mapsto O(x),$$

is an isometry onto the metric subspace $O(X) \subseteq Y$.

Proof. Changing representatives leaves both images unchanged, so the formula is well-defined. The pseudometric axioms descend from theorem 2.2; zero distance now implies equality of classes. The map ι_O is a bijection and preserves the displayed distance. $\square$



3 Local pullback form versus global distance

Suppose now that X and Y are smooth manifolds, Y has Riemannian metric g_Y , and $O : X \rightarrow Y$ is smooth.

Definition 3.1 (Pullback quadratic form).

$$g_O := O^* g_Y, \quad (g_O)_x(v, w) = (g_Y)_{O(x)}(dO_x v, dO_x w).$$

Proposition 3.2 (Degeneracy directions). g_O is positive semidefinite and

$$\ker dO_x \subseteq \ker (g_O)_x.$$

If g_Y is positive definite, equality holds.

Proof. Positive semidefiniteness follows from $g_O(v, v) = g_Y(dO v, dO v) \geq 0$. If $dO v = 0$, then $g_O(v, w) = 0$ for every w . Conversely, if $g_O(v, v) = 0$, positive definiteness of g_Y gives $dO v = 0$. $\square$

For every piecewise C^1 curve γ in X ,

$$L_{g_O}(\gamma) = L_{g_Y}(O \circ \gamma).$$

Define the extended path pseudometric

$$\rho_O(x, x') := \inf_{\gamma: x \rightsquigarrow x'} L_{g_O}(\gamma),$$

with value $+\infty$ if there is no admissible path.

Proposition 3.3 (Ambient versus intrinsic image distance). *For all path-connected pairs,*

$$d_O(x, x') \leq \rho_O(x, x').$$

Equality holds if a d_Y -minimizing geodesic from $O(x)$ to $O(x')$ lies in $O(X)$ and has a lift joining x to x' .

Proof. Every image curve joining $O(x)$ to $O(x')$ has length at least their ambient Riemannian distance. Taking the infimum gives the inequality. A lift of a minimizing geodesic realizes equality. $\square$

Remark 3.4. The local tensor g_O , the path pseudometric ρ_O , and the global pullback pseudometric $d_O = d_Y \circ (O \times O)$ are therefore not interchangeable without the hypotheses of proposition 3.3. This corrects the ambiguity in v0.1.

4 Reference shells and calibrated decisions

Definition 4.1 (Exact and tolerant shells). Let $r_* > 0$ and $\tau \geq 0$ have the same scale type as d_Y . For $x \in X$, define

$$S_O(x; r_*) := \{x' \in X : d_O(x, x') = r_*\},$$

$$S_O^\tau(x; r_*) := \{x' \in X : |d_O(x, x') - r_*| \leq \tau\}.$$

The exact shell is the case $\tau = 0$.

Definition 4.2 (Observed distance graph). For a finite set $A = \{x_1, \dots, x_n\} \subseteq X$, define

$$G_{O, r_*, \tau}(A) := (A, E_{O, r_*, \tau}),$$

$$\{x_i, x_j\} \in E_{O, r_*, \tau} \iff |d_O(x_i, x_j) - r_*| \leq \tau.$$

Boundary 4.3 (Exact equality and measured data). For continuously noisy data, exact numerical equality is generally not a stable event. The mathematical shell $\tau = 0$ remains valid, but an empirical classifier must declare a tolerance or a hypothesis test.

Proposition 4.4 (Deterministic error guarantee). *Suppose an estimated distance $\hat{d}_{ij}$ obeys*

$$|\hat{d}_{ij} - d_O(x_i, x_j)| \leq u$$

with a declared bound $u \geq 0$. The rule

$$|\hat{d}_{ij} - r_*| \leq \tau$$

has the following guarantees:

- (i) every exact-shell pair is accepted if $\tau \geq u$;
- (ii) every accepted pair satisfies $|d_O(x_i, x_j) - r_*| \leq \tau + u$.

Proof. Both statements are direct applications of the triangle inequality. $\square$

If u is probabilistic rather than deterministic, the confidence level, dependence structure, and multiple-testing policy belong to the protocol.

5 Coherent unit covariance

Let $a > 0$ be a change of numerical length unit. Define

$$d'_Y = a d_Y, \quad r'_* = a r_*, \quad \tau' = a \tau, \quad \varepsilon'_Y = a \varepsilon_Y.$$

Theorem 5.1 (Unit covariance of shells and graphs).

$$S_{O, d'_Y}^{\tau'}(x; r'_*) = S_{O, d_Y}^{\tau}(x; r_*),$$

and therefore

$$G_{O, r'_*, \tau'}(A; d'_Y) = G_{O, r_*, \tau}(A; d_Y).$$

Proof.

$$|a d_O(x, x') - a r_*| \leq a \tau \iff |d_O(x, x') - r_*| \leq \tau.$$

$\square$

The normalized quantities

$$\bar{d}_O = \frac{d_O}{r_*}, \quad \bar{\tau} = \frac{\tau}{r_*}, \quad \bar{\varepsilon}_Y = \frac{\varepsilon_Y}{r_*}$$

are invariant under the same coherent rescaling. A formula $d_O(x, x') = 1$ is admissible only after the convention $r_* = 1 U_Y$ has been declared. The typed formula is $d_O(x, x') = r_*$.

Definition 5.2 (Dimensionless log-scale coordinate). For a positive reference scale ℓ_* and resolution $\varepsilon > 0$,

$$s_b(\varepsilon; \ell_*) := \log_b \frac{\ell_*}{\varepsilon}, \quad b > 1.$$

No logarithm of a bare physical length is admissible.

6 Matched resolutions under Lipschitz observation

Definition 6.1 (Typed Lipschitz converter). An observation $O : (X, d_X) \rightarrow (Y, d_Y)$ is L_O -Lipschitz if

$$d_Y(Ox, Ox') \leq L_O d_X(x, x')$$

for all x, x' . Semantically,

$$[L_O] = U_Y/U_X.$$

Proposition 6.2 (Resolution propagation). *If an input cell has d_X -diameter at most ε_X , its image has d_Y -diameter at most $L_O \varepsilon_X$. Hence a conservative matched output resolution satisfies*

$$L_O \varepsilon_X \leq \varepsilon_Y.$$

Proof. Apply the Lipschitz inequality to every pair in the input cell. $\square$

Theorem 6.3 (Composition law). *If $O_1 : (X, d_X) \rightarrow (Y, d_Y)$ is L_1 -Lipschitz and $O_2 : (Y, d_Y) \rightarrow (Z, d_Z)$ is L_2 -Lipschitz, then $O_2 \circ O_1$ is $L_2 L_1$ -Lipschitz. Matched scales propagate as*

$$\varepsilon_Y \geq L_1 \varepsilon_X, \quad \varepsilon_Z \geq L_2 \varepsilon_Y \implies \varepsilon_Z \geq L_2 L_1 \varepsilon_X.$$

Proof. Substitute the first Lipschitz inequality into the second. The scale statement follows by transitivity. $\square$

This is the distance-scale bridge to the matched covering defect in the Metric Entropy v0.2 module.

7 Orthogonal projection: cylinder and quotient

Let

$$\pi : \mathbb{R}^3 \rightarrow \mathbb{R}^2, \quad \pi(x, y, z) = (x, y),$$

with Euclidean metrics.

Proposition 7.1 (Observed shell under projection). *For $X_0 = (x_0, y_0, z_0)$ and $r_* > 0$,*

$$S_\pi(X_0; r_*) = \{(x, y, z) : (x - x_0)^2 + (y - y_0)^2 = r_*^2\}.$$

Thus the exact observed shell is a circular cylinder of radius r_ .*

Proof. The condition $d_\pi(X, X_0) = \|\pi(X) - \pi(X_0)\|_2 = r_*$ is precisely the displayed equation; z is free. $\square$

Corollary 7.2 (Hidden distance profile). *If $d_\pi(X, X') = r_*$, then*

$$\|X - X'\|_2^2 = r_*^2 + (z - z')^2.$$

One observed shell radius therefore corresponds to a continuum of hidden Euclidean distances greater than or equal to r_ .*

The quotient $\mathbb{R}^3 / \sim_\pi$ is isometric to $\mathbb{R}^2$; the cylinder is the preimage of a planar circle, not a new type of planar unit-distance configuration.

8 Fiber multiplicity and counting rules

Two non-equivalent counts must be declared:

- (a) *visible count*: replace A by the set of distinct images $O(A)$ and count image pairs;
- (b) *hidden-multiplicity count*: count pairs of distinct hidden states whose images satisfy the shell predicate.

Theorem 8.1 (Complete bipartite fiber lift). *Let $F_a = O^{-1}(a)$ and $F_b = O^{-1}(b)$, with $d_Y(a, b) = r_*$. If a finite sample contains m states in F_a and k states in F_b , its hidden-multiplicity graph contains $K_{m,k}$ and at least mk shell edges.*

Proof. Every cross-fiber pair has observed distance $d_Y(a, b) = r_*$. $\square$

Proposition 8.2 (Reduction to the visible problem). *If O is injective on a finite sample A and the count is taken over distinct images, then*

$$|E_{O, r_*, 0}(A)| = \#\{\{a, b\} \subset O(A) : d_Y(a, b) = r_*\}.$$

Proof. The restriction $O|_A$ is a vertex bijection preserving the edge predicate. $\square$

9 Dimensionally valid periodic and half-step readout

Let a hidden diameter ladder be

$$d_n = d_0 + n\Delta d, \quad n \in \mathbb{Z},$$

where d_0 and Δd are lengths. Let $P > 0$ be a physical period and ϕ_0 an angle. Define

$$h(n) = \cos \left(2\pi \frac{d_n - d_0}{P} + \phi_0 \right).$$

The trigonometric argument is an angle because $(d_n - d_0)/P$ is dimensionless.

Theorem 9.1 (Rational-step factorization). *If*

$$\frac{\Delta d}{P} = \frac{p}{q}, \quad p \in \mathbb{Z}, \quad q \in \mathbb{N}, \quad \gcd(p, q) = 1,$$

then h factors through $\mathbb{Z}/q\mathbb{Z}$.

Proof. For every n ,

$$h(n+q) = \cos \left(2\pi \frac{p(n+q)}{q} + \phi_0 \right) = \cos \left(2\pi \frac{pn}{q} + 2\pi p + \phi_0 \right) = h(n).$$

□

The theorem states factorization, not injectivity on the q residue classes; cosine symmetry may identify additional classes.

Corollary 9.2 (Half-step parity). *If $\Delta d = P/2$ and $\phi_0 = 0$, then*

$$h(n) = \cos(\pi n) = (-1)^n,$$

so the readout is exactly the parity quotient $\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$.

This replaces the dimensionally invalid expression $d_n = d_0 + n/2$ unless a length unit for the half-step has first been declared.

10 Unit-distance graphs as an application

For $P \subset \mathbb{R}^2$ finite and $r_* > 0$, write

$$U_{r_*}(P) := \# \{ \{p, q\} \subset P : \|p - q\|_2 = r_* \}.$$

Euclidean similarity $p \mapsto p/r_*$ gives

$$\max_{|P|=n} U_{r_*}(P) = \max_{|P|=n} U_1(P).$$

Thus the classical extremal function is independent of the chosen positive reference radius after coherent rescaling.

For an observation map $O : X \rightarrow \mathbb{R}^2$, define instead

$$U_{O, r_*}^{\text{hidden}}(A) := \# E_{O, r_*, 0}(A).$$

This quantity can be quadratic because of theorem 8.1. It is a different problem: multiplicity inside observation fibers is part of the data.

Boundary 10.1 (Erdos firewall). The present interface proves no new asymptotic upper or lower bound for the classical planar unit-distance function. It only states the data needed to define observed-distance variants without dimensional or counting ambiguity.

11 Normative protocol ledger

For a static theorem with exact metric data, the stochastic-channel field is recorded as *explicit-noiseless*, while temporal resolution and observation horizon are recorded as *not-applicable*. They become active fields for time-dependent geometry. This module defines no scalar defect, so *defect-normalizations* is recorded as *not-applicable*; a downstream distortion or decision loss must provide its own typed normalization.

Field	Type	Required declaration
Hidden space	set, manifold, or state space	admissible points and equivalence before observation
Hidden metric d_X	U_X	metric convention and domain
Observation O	map $X \rightarrow Y$	deterministic, stochastic, or composed channel role
Noise/channel policy	kernel or exact flag	stochastic law or explicit noiseless statement
Observed metric d_Y	U_Y	ambient or intrinsic convention
Observed pseudometric d_O	U_Y	global formula $d_Y(Ox, Ox')$
Reference radius r_*	U_Y , positive	shell or incidence target
Input resolution ε_X	U_X , positive	quantizer or cover convention
Output resolution ε_Y	U_Y , positive	matched-scale rule
Temporal resolution	time or not applicable	required for time-dependent geometry
Observation horizon	time or not applicable	required for a dynamic trace
Tolerance τ	U_Y , nonnegative	exact, bounded-error, or hypothesis-test policy
Uncertainty u or law	U_Y or distribution	confidence/dependence statement
Counting rule	discrete	visible images or hidden multiplicities
Reference scale ℓ_*	same type as scale variable	normalization and zero policy
Logarithm base b	dimensionless, $b > 1$	every logarithmic trace
Estimator	declared map	output, window, and uncertainty summary
Defect normalization	typed ratio or not applicable	required before any scalar loss aggregation

12 Claim audit

Statement	Status	Necessary hypotheses
$d_O = d_Y \circ (O \times O)$ is a pseudometric	theorem	d_Y metric, O a map
$X/\sim_O$ is isometric to $O(X)$	theorem	quotient by equal observations
$g_O = O^*g_Y$ determines the same global distance as d_O	not generally true	requires a suitable lift of minimizing image paths
Observed exact shells are unit-covariant	theorem	coherent rescaling of metric, radius, and tolerance
Fiber multiplicity produces $K_{m,k}$	theorem	two finite sampled fibers at the target separation
Half-step readout recovers the hidden state	false	parity readout is many-to-one
Observation-map graphs solve the classical unit-distance problem	not claimed	the classical problem is only the injective visible special case

13 Conclusion

The common distance-scale object is not a bare number. It is

$$(X, d_X, O, Y, d_Y, d_O, r_*, \varepsilon_X, \varepsilon_Y, \tau, \mathbf{U}, \text{counting rule}).$$

The global pseudometric identifies exactly the observation fibers; its quotient is the metric image. Physical comparisons use typed radii and tolerances, logarithms use reference ratios, and resolutions propagate through declared Lipschitz converters. These rules are the dependency needed by scale-trace modules such as Mandelbrot rulers.

References

- [1] D. Burago, Y. Burago, and S. Ivanov, *A Course in Metric Geometry*, American Mathematical Society, 2001.
- [2] J. Heinonen, *Lectures on Analysis on Metric Spaces*, Springer, 2001.
- [3] J. M. Lee, *Riemannian Manifolds: An Introduction to Curvature*, Springer, 1997.
- [4] P. Erdos, "On sets of distances of n points," *American Mathematical Monthly* 53 (1946), 248–250.

Planck-to-Cosmos Observation Rulers

Typed logarithmic charts, scale passports, and cosmological landmarks

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Strict GO Core reference revision v1.1 — 2026

Status. This document supersedes *Planck-to-Cosmos Observation Rulers v1.0*. It is a reference module for GO Core v0.2 and the Distance-Scale Interface v0.2. The revision separates intrinsic system scales from instrument resolutions, treats Planck units as normalizations rather than cutoffs, and treats cosmic entries as model-, epoch-, and convention-dependent landmarks. The logarithmic chart is standard mathematics; no new fundamental law is claimed.

Abstract

For a positive length, invariant mass, and duration, a declared reference triple $\mathbf{a} = (a_L, a_M, a_T)$ and base $b > 1$ define the logarithmic chart

$$\chi_{b,\mathbf{a}}(L, M, T) = \left(\log_b \frac{L}{a_L}, \log_b \frac{M}{a_M}, \log_b \frac{T}{a_T} \right).$$

This map is a global smooth chart from the positive scale cone $(\mathbb{R}_{>0})^3$ to $\mathbb{R}^3$. Coherent unit changes leave it invariant, anchor changes translate it, and base changes act by a scalar linear map. Coordinate differences support a family of weighted log-scale metrics; there is no canonical physical reason to assign equal weight to one decade of length, mass, and time. Intrinsic scale catalogues belong to the system descriptor ξ , whereas spatial, mass-equivalent, and temporal resolutions belong to an observation protocol λ . Their anchor-independent comparison is the resolution margin $g_i = \log_b(q_i/\varepsilon_i)$. Monomial physical relations become linear constraint surfaces in log coordinates. In Planck normalization, the causal condition $L \leq cT$ gives $\chi_L \leq \chi_T$, the reduced Compton relation gives $\chi_L = -\chi_M$, and the Schwarzschild relation gives $\chi_L = \chi_M + \log_b 2$. A reproducible cosmological ledger then shows why a bare “mass of the observable Universe” is not a typed quantity: baryonic, total-matter, and total-energy mass equivalents differ by more than a decade under the same rounded horizon convention.

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1 Scope and claim boundary

The original ruler note correctly normalized physical quantities before taking logarithms. Its unresolved issue was semantic rather than algebraic: object sizes, instrument resolutions, and cosmological landmarks were placed in the same coordinates without distinct passports. The notation

$$Y_X(s, \mu, \tau) = O_{s, \mu, \tau}(X)$$

also treated every point of $\mathbb{R}^3$ as if it automatically specified an observation channel. It does not. A channel requires a declared measurement model, frame, calibration, noise law, estimator, and applicable axes.

Boundary 1.1 (Claim firewall). The Planck length and Planck time are not asserted to be experimentally proved minimum scales. The present comoving particle horizon is not the diameter of the entire Universe. Similar numerical spans of roughly sixty decades do not imply a new physical law. A camera, clock, or spectrometer need not measure all of length, invariant mass, and time.

The strict claim is limited to the following:

Positive physical scales admit useful typed logarithmic coordinates. Systems generally determine labelled sets or regions of such coordinates, while observation protocols determine separate resolution coordinates and response traces.

2 Typed scale data and the two-passport rule

Definition 2.1 (Scale item). A scale item is a record

$$\sigma = (\text{id}, \text{axis}, q, \text{role}, \text{frame}, \text{convention}, \text{U}),$$

where $\text{axis} \in \{L, M, T\}$, $q > 0$ has the corresponding physical dimension, role identifies the measurand, and U contains exactness or uncertainty information.

The role is indispensable. A particle radius, a wavelength, a correlation length, and a detector pixel pitch all have dimension L , but they are not interchangeable. Likewise, invariant rest mass is not relativistic energy divided by an undeclared constant, and proper time is not a coordinate duration without a clock congruence.

Definition 2.2 (Intrinsic scale catalogue). For a system X , let

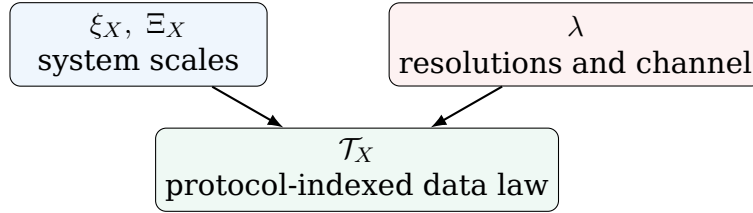
$$\Xi_X = \{\sigma_\alpha : \alpha \in A_X\}$$

be a labelled, possibly partial and multivalued catalogue of intrinsic or model-derived scales. It is a component of the system descriptor ξ_X . A missing axis is recorded as not-applicable; it is not encoded by 0 or $+\infty$.

Definition 2.3 (Observation protocol passport). An observation protocol is a tuple

$$\lambda = (R_\lambda, O_\lambda, K_\lambda, Q_\lambda, \varepsilon_\lambda, \mathbf{r}_\lambda^*, \mathbf{F}_\lambda, \hat{\theta}_\lambda, \mathbf{U}_\lambda).$$

Here R_λ is a reduction, O_λ is an ideal deterministic observation, K_λ is a stochastic kernel or an explicit noiseless flag, Q_λ is a quantizer or partition, ε_λ contains only applicable resolutions, $\mathbf{r}_\lambda^*$ contains their reference scales, $\mathbf{F}_\lambda$ records frames and conventions, $\hat{\theta}_\lambda$ is an estimator or declared direct readout, and $\mathbf{U}_\lambda$ records uncertainty.



Remark 2.4. A mass axis in λ exists only if the channel measures mass, invariant mass, or a declared mass-equivalent. If the actual threshold is an energy ε_E and the model uses $E = Mc^2$, then

$$\varepsilon_M = \frac{\varepsilon_E}{c^2}$$

is a typed converter. Without such a converter, “mass resolution” is not available.

3 The logarithmic scale chart

Definition 3.1 (Positive scale cone and chart). Let

$$\mathcal{S}_{LMT} := \mathbb{R}_{>0}^{(L)} \times \mathbb{R}_{>0}^{(M)} \times \mathbb{R}_{>0}^{(T)}.$$

For a positive reference triple $\mathbf{a} = (a_L, a_M, a_T)$ and $b > 1$, define

$$\chi_{b,\mathbf{a}}(L, M, T) = \begin{pmatrix} \chi_L \\ \chi_M \\ \chi_T \end{pmatrix} := \begin{pmatrix} \log_b(L/a_L) \\ \log_b(M/a_M) \\ \log_b(T/a_T) \end{pmatrix}.$$

Theorem 3.2 (Global chart). *The map $\chi_{b,\mathbf{a}} : \mathcal{S}_{LMT} \rightarrow \mathbb{R}^3$ is a smooth diffeomorphism with inverse*

$$\chi_{b,\mathbf{a}}^{-1}(x_L, x_M, x_T) = (a_L b^{x_L}, a_M b^{x_M}, a_T b^{x_T}).$$

Proof. Each one-dimensional map $q \mapsto \log_b(q/a)$ is smooth, strictly increasing, and bijective from $\mathbb{R}_{>0}$ to $\mathbb{R}$, with the displayed smooth inverse. The product map has the same properties. $\square$

The chart is an exact reparameterization, not a reduction. Information loss enters when a scale item is selected from a system, rounded, thresholded, or passed through an observation channel.

Definition 3.3 (Scale region). For a catalogue Ξ_X , a declared extraction rule E_r selects compatible triples or partial tuples. Its logarithmic image is

$$\mathcal{R}_{X,r} = \chi_{b,\mathbf{a}}(E_r(\Xi_X)).$$

An extended or multiscale object is represented by this labelled set or region, not by a forced single point.

Boundary 3.4 (No canonical triple). An atom has several length scales and several characteristic times. A galaxy has luminous, half-mass, virial, and halo radii. Selecting one value on each axis is a protocol or modelling decision. The phrase “the object occupies the point (χ_L, χ_M, χ_T) ” is valid only after the extraction rule has been declared.

4 Unit, anchor, and base transformation laws

Theorem 4.1 (Coordinate covariance). *Let $q, a > 0$ have the same physical dimension.*

- (i) *Under a coherent unit conversion $(q, a) \mapsto (\alpha q, \alpha a)$ with $\alpha > 0$, $\log_b(q/a)$ is unchanged.*
- (ii) *If the reference changes from a to $a' > 0$, then*

$$\chi_{b,a'}(q) = \chi_{b,a}(q) - \log_b \frac{a'}{a}.$$

- (iii) *If the base changes from b to $b' > 1$, then*

$$\chi_{b',a}(q) = \frac{\ln b}{\ln b'} \chi_{b,a}(q).$$

Proof. The first statement follows from cancellation of α . The second is the logarithm of $(q/a)/(a'/a)$. The third follows from the change-of-base formula. $\square$

Thus an anchor change translates log-space. Coordinate differences, axis orderings, and any construction depending only on differences are anchor independent. A change from decades to octaves or nats rescales the chart.

Definition 4.2 (Weighted log-scale distance). For a symmetric positive-definite dimensionless matrix $W \in \mathbb{R}^{3 \times 3}$, define

$$d_{W,b}(\mathbf{q}, \mathbf{q}') := \sqrt{(\chi_{b,a}(\mathbf{q}) - \chi_{b,a}(\mathbf{q}'))^\top W (\chi_{b,a}(\mathbf{q}) - \chi_{b,a}(\mathbf{q}'))}.$$

Theorem 4.3 (Metric and transformation law). *The function $d_{W,b}$ is a metric on $\mathcal{S}_{LMT}$, independent of the anchor a and invariant under coherent unit changes. Under a base change $b \mapsto b'$, it remains numerically invariant if*

$$W_{b'} = \left(\frac{\ln b'}{\ln b} \right)^2 W_b.$$

Proof. The chart in theorem 3.2 is injective, and the positive-definite quadratic form defines a norm on coordinate differences. Anchor translations cancel in those differences. Unit invariance follows from theorem 4.1. If $c_{bb'} = \ln b / \ln b'$, then $\Delta \chi_{b'} = c_{bb'} \Delta \chi_b$; the displayed transformation of W cancels $c_{bb'}^2$. $\square$

Boundary 4.4 (Metric convention). $W = I$ declares that one decade of length, mass, and time contributes equally. This may be useful for visualization, but it is not a physical law. A scientifically interpretable W must come from a decision problem, tolerances, or a covariance model.

5 Descriptor coordinates versus resolution coordinates

The Distance-Scale Interface uses a resolution coordinate whose orientation is opposite to the descriptor coordinate.

Definition 5.1 (Protocol resolution depth). For an applicable protocol axis i with resolution $\varepsilon_i > 0$ and reference $r_i^* > 0$, define

$$\eta_i(\lambda) := \log_b \frac{r_i^*}{\varepsilon_i}.$$

Finer resolution increases η_i . If $r_i^* = a_i$, then

$$\eta_i = -\chi_{b,a_i}(\varepsilon_i).$$

Definition 5.2 (Resolution margin). For a system scale $q_i > 0$ and protocol resolution $\varepsilon_i > 0$ on the same typed axis, define

$$g_i(q_i, \lambda) := \log_b \frac{q_i}{\varepsilon_i}.$$

Proposition 5.3 (Anchor-independent comparison). *For any common positive anchor a_i ,*

$$g_i = \chi_{b,a_i}(q_i) - \chi_{b,a_i}(\varepsilon_i).$$

Hence g_i is invariant under coherent unit changes and common anchor changes.

Proof. Subtract the two logarithms and combine their arguments. □

In a simple scale-separation model, $g_i > 0$ means that the intrinsic scale exceeds the nominal resolution. It does not by itself specify contrast, signal-to-noise ratio, inverse-problem conditioning, or a detection probability.

Definition 5.4 (Protocol-indexed observation trace). For a declared family Λ of protocols, a deterministic trace is

$$\mathcal{T}_X^{\text{det}} = \{(\eta(\lambda), O_\lambda(X)) : \lambda \in \Lambda\}.$$

For a stochastic channel it is

$$\mathcal{T}_X^{\text{stoch}} = \{(\eta(\lambda), \text{Law}(Y_\lambda | X, \lambda)) : \lambda \in \Lambda\}.$$

The shorthand O_η is admissible only after the map $\eta \mapsto \lambda(\eta)$ and all nuisance parameters have been defined.

6 Monomial algebra and constraint surfaces

Theorem 6.1 (Log-linearization of positive monomial laws). *Let positive inputs $q = (q_1, \dots, q_n)$ and positive outputs r_i satisfy*

$$r_i = C_i \prod_{j=1}^n q_j^{A_{ij}}, \quad r_i^* = C_i \prod_{j=1}^n (q_j^*)^{A_{ij}},$$

where the references are dimensionally matched. Then

$$\log_b \frac{r_i}{r_i^*} = \sum_{j=1}^n A_{ij} \log_b \frac{q_j}{q_j^*}.$$

In vector form, $\chi_r = A\chi_q$.

Proof. Divide the two monomials, cancel C_i , and apply the logarithm product and power laws. $\square$

This theorem is the precise structural reason that dimensional monomials become hyperplanes in logarithmic coordinates. It is a coordinate simplification, not a claim that arbitrary nonlinear physics becomes linear.

6.1 Planck identities

Use the standard definitions

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}, \quad m_P = \sqrt{\frac{\hbar c}{G}}, \quad t_P = \sqrt{\frac{\hbar G}{c^5}}.$$

Direct substitution gives

$$t_P = \frac{\ell_P}{c}, \quad \ell_P m_P = \frac{\hbar}{c}, \quad m_P c^2 = \frac{\hbar}{t_P}.$$

The three Planck anchors are therefore built from common constants and are not statistically or conceptually independent axes.

6.2 Causal, quantum, and gravitational examples

Corollary 6.2 (Causal half-space). *If L and T refer to the same causal process and $L \leq cT$, then in Planck normalization*

$$\chi_L \leq \chi_T.$$

Proof. Because $\ell_P = ct_P$,

$$\frac{L/\ell_P}{T/t_P} = \frac{L}{cT} \leq 1.$$

Apply the increasing logarithm. $\square$

If L and T belong to unrelated catalogue entries, the hypothesis of corollary 6.2 fails and no such inequality follows.

Corollary 6.3 (Mass-energy, density, Compton, and Schwarzschild planes). *In Planck normalization:*

$$\begin{aligned} E = Mc^2, \quad E_P = m_P c^2 &\implies \chi_E = \chi_M, \\ \rho = M/L^3, \quad \rho_P = m_P/\ell_P^3 &\implies \chi_\rho = \chi_M - 3\chi_L, \\ \bar{\lambda}_C = \hbar/(Mc) &\implies \chi_{\bar{\lambda}_C} = -\chi_M, \\ r_s = 2GM/c^2 &\implies \chi_{r_s} = \chi_M + \log_b 2. \end{aligned}$$

Proof. The first two identities follow from theorem 6.1. For the reduced Compton length, use $\ell_P m_P = \hbar/c$:

$$\frac{\bar{\lambda}_C}{\ell_P} = \frac{m_P}{M}.$$

For the Schwarzschild radius, $Gm_P/c^2 = \ell_P$, hence

$$\frac{r_s}{\ell_P} = 2 \frac{M}{m_P}.$$

Taking logarithms proves the result. $\square$

Boundary 6.4 (Physical-domain boundary). The lines in corollary 6.3 are relations among specifically coupled measurands. They do not turn the whole scale catalogue into one three-dimensional state space, and they do not by themselves establish a quantum-gravity model.

7 Planck anchors and coordinate uncertainty

Anchor	Definition	2022 CODATA value	Status
Planck length ℓ_P	$\sqrt{\hbar G/c^3}$	$1.61625 \times 10^{-35} \text{ m}$	normalization, not a proved minimum length
Planck mass m_P	$\sqrt{\hbar c/G}$	$2.17643 \times 10^{-8} \text{ kg}$	normalization, not a minimum mass
Planck time t_P	$\sqrt{\hbar G/c^5}$	$5.39125 \times 10^{-44} \text{ s}$	normalization, not a proved minimum duration

The 2022 CODATA values inherit uncertainty primarily through G ; c and h are exact defining constants in the revised SI. Consequently, the Planck anchors are correlated. Treating their listed standard uncertainties as independent can give an incorrect covariance for derived scale coordinates.

Proposition 7.1 (First-order uncertainty of a log ratio). *Let*

$$\chi = \log_b(q/a)$$

with $q, a > 0$. To first order,

$$u^2(\chi) = \frac{1}{(\ln b)^2} \left[\frac{u^2(q)}{q^2} + \frac{u^2(a)}{a^2} - 2 \frac{\text{Cov}(q, a)}{qa} \right].$$

For several correlated inputs,

$$\Sigma_\chi = J \Sigma_{q,a} J^\top,$$

where J is the Jacobian of the logarithmic map.

Proof. Differentiate $\chi = (\ln q - \ln a)/\ln b$ and apply the first-order covariance propagation formula. $\square$

The formula is a local approximation. Large, asymmetric, or model-posterior uncertainties should be propagated by the full distribution rather than by a symmetric linearized error bar.

8 Cosmological landmarks are typed model outputs

The phrase “observable-universe diameter” must name a cosmological distance. The legacy 93 Gly entry is retained only as a rounded *present comoving particle-horizon diameter* in a declared $a_0 = 1$ convention. It is not a light-travel distance and not the size of the entire Universe. Cosmological distance definitions are distinct even when they share the SI dimension of length.

The mass axis has a larger ambiguity. For a flat-volume illustrative ledger, let

$$\rho_{c0} = \frac{3H_0^2}{8\pi G}, \quad V_{\text{flat}} = \frac{4\pi}{3} R_{\text{ph},0}^3.$$

Then three different mass-equivalent landmarks are

$$M_b = \Omega_b \rho_{c0} V_{\text{flat}}, \quad M_m = \Omega_m \rho_{c0} V_{\text{flat}}, \quad M_E = \rho_{c0} V_{\text{flat}}.$$

M_b counts the baryonic component, M_m all pressureless matter in the adopted parameterization, and M_E the total critical mass-equivalent. Dark energy is not a collection of rest-mass particles; the last quantity is explicitly an energy-equivalent bookkeeping scale.

Landmark	Value in SI	Coordinate	Convention
Present comoving particle-horizon diameter	$8.79848 \times 10^{26} \text{ m}$	61.7359	rounded 93 Gly landmark
FLRW cosmic age	$4.35400 \times 10^{17} \text{ s}$	60.9072	Planck 2018 baseline, $13.797 \pm 0.023 \text{ Gyr}$
Baryonic mass-equivalent	$1.50025 \times 10^{53} \text{ kg}$	60.8384	$\Omega_b \rho_{c0} V$
All-matter mass-equivalent	$9.58576 \times 10^{53} \text{ kg}$	61.6439	$\Omega_m \rho_{c0} V$
Total-energy mass-equivalent	$3.04310 \times 10^{54} \text{ kg}$	62.1456	$\rho_{c0} V$

The table uses a rounded 93 Gly diameter and a Planck 2018 flat base- Λ CDM parameter ledger: $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.315$, and $\Omega_b = 0.0493$. Because the rounded horizon entry and the full parameter covariance are not supplied, no combined confidence interval is fabricated. The values are reproducible landmarks, not a precision cosmological analysis.

Proposition 8.1 (The legacy mass anchor is ambiguous). *Under this single illustrative ledger,*

$$M_b \approx 1.50 \times 10^{53} \text{ kg}, \quad M_m \approx 9.59 \times 10^{53} \text{ kg}, \quad M_E \approx 3.04 \times 10^{54} \text{ kg}.$$

Their Planck-mass coordinates are approximately 60.84, 61.64, and 62.15, respectively. Therefore an unqualified 10^{53} kg entry cannot serve as a unique cosmic mass anchor.

Proof. Substitute the declared H_0 , density fractions, and flat-volume convention into the preceding definitions, then divide by m_P and take $\log_{10}$. $\square$

Boundary 8.2 (Epoch and model dependence). Changing the cosmological model, horizon definition, spatial curvature, epoch, or matter component changes these landmarks. They are catalogue records, not constants defining the logarithmic coordinate system.

9 Selected reproducible catalogue

The following table is generated from the machine-readable input ledger. Entries marked “order of magnitude” or “declared reference” are visual landmarks rather than measurements. Every coordinate uses the corresponding 2022 CODATA Planck anchor and base 10.

Entry	Axis	SI value	Coordinate	Status
planck length	length	$1.6163 \times 10^{-35} \text{ m}$	0.000	CODATA recommended
atomic order	length	$1.0000 \times 10^{-10} \text{ m}$	24.791	order of magnitude marker
human length reference	length	$1.0000 \times 10^0 \text{ m}$	34.791	declared not population estimate
earth diameter	length	$1.2742 \times 10^7 \text{ m}$	41.897	rounded mean diameter
astronomical unit	length	$1.4960 \times 10^{11} \text{ m}$	45.966	exact IAU definition
observable particle horizon diameter	length	$8.7985 \times 10^{26} \text{ m}$	61.736	rounded model dependent
electron mass	mass	$9.1094 \times 10^{-31} \text{ kg}$	-22.378	CODATA recommended
proton mass	mass	$1.6726 \times 10^{-27} \text{ kg}$	-19.114	CODATA recommended
planck mass	mass	$2.1764 \times 10^{-8} \text{ kg}$	0.000	CODATA recommended
human mass reference	mass	$7.0000 \times 10^1 \text{ kg}$	9.507	declared not population estimate
earth mass	mass	$5.9722 \times 10^{24} \text{ kg}$	32.438	rounded astronomical value
solar mass	mass	$1.9885 \times 10^{30} \text{ kg}$	37.961	rounded astronomical value
planck time	time	$5.3912 \times 10^{-44} \text{ s}$	0.000	CODATA recommended
femtosecond	time	$1.0000 \times 10^{-15} \text{ s}$	28.268	SI prefix exact
second	time	$1.0000 \times 10^0 \text{ s}$	43.268	SI base unit
day	time	$8.6400 \times 10^4 \text{ s}$	48.205	conventional exact duration
Julian year	time	$3.1558 \times 10^7 \text{ s}$	50.767	conventional exact duration
universe age	time	$4.3540 \times 10^{17} \text{ s}$	60.907	model inferred parameter

The equality of the light-year length coordinate and the Julian-year time coordinate to the displayed precision is not mysterious: $1 \text{ ly} = c \times 1 \text{ Julian year}$ and $\ell_P = ct_P$. It is an exact consequence of matched definitions.

10 Normative reporting contract

For a static catalogue, stochastic-channel, quantizer, temporal resolution, and observation-horizon fields may be not-applicable; the field must still be present. For an actual measurement trace, applicable fields require values rather than prose placeholders.

Field	Type	Required declaration
Hidden system X	state space or object	domain and admissible states
System descriptor ξ_X	typed record	intrinsic model data and scale catalogue Ξ_X
Scale item	typed positive quantity	axis, role, frame, convention, value or interval, uncertainty

Field	Type	Required declaration
Extraction rule	reduction	which labelled items form a tuple or region
Reference triple a	positive typed tuple	Planck or alternative anchors, with source
Logarithm base b	dimensionless, $b > 1$	decades, octaves, or other declared base
Observation protocol λ	channel passport	reduction, observation, noise/exactness, quantizer, estimator
Applicable axes	subset of $\{L, M, T\}$	missing axes recorded as not applicable
Spatial resolution	positive length or not applicable	frame and calibration
Mass resolution	positive mass or not applicable	direct channel or explicit energy/mass converter
Temporal resolution	positive duration or not applicable	clock and frame convention
Observation horizon	duration or not applicable	acquisition interval
Reference scales r^*	positive typed tuple	normalization of resolution depth
Resolution margin	dimensionless	matched measurand and resolution
Frame passport	metadata	proper, coordinate, rest, comoving, or other declared convention
Cosmology passport	model metadata or not applicable	model, epoch, distance, volume, and component definitions
Uncertainty	covariance, law, or exact flag	dependence and propagation semantics
Weight matrix W	positive definite or not applicable	rationale and base-transformation policy
Defect normalization	typed ratio or not applicable	required before scalar loss aggregation

11 Claim audit

Statement	Status	Necessary hypotheses
$\chi_{b,a}$ is a global chart on $(\mathbb{R}_{>0})^3$	theorem	positive scales, positive anchors, $b > 1$
Log coordinates are invariant under a unit change	theorem	quantity and anchor converted coherently
Anchor changes preserve coordinate differences	theorem	common typed anchor change
$d_{W,b}$ is a scale-space metric	theorem	W symmetric positive definite
$W = I$ is the unique physical metric	false	no such canonical choice is proved
Every system has one canonical (L, M, T) point	false	requires a declared extraction rule

Statement	Status	Necessary hypotheses
Every protocol measures all three axes	false	missing axes are allowed
$g_i > 0$ guarantees detection	false	only a simple scale-separation diagnostic
Planck units are minimum physical scales	not claimed	they are used as normalizations
The causal inequality gives $\chi_L \leq \chi_T$	corollary	paired length and time of the same causal process
Compton and Schwarzschild relations become affine lines	corollary	declared standard formulas and Planck normalization
93 Gly is the diameter of the entire Universe	false	rounded present comoving particle-horizon convention only
10^{53} kg is the unique mass of the observable Universe	false	component, volume, epoch, and model are required
Three spans near sixty decades imply a new law	not claimed	numerical similarity depends on selected landmarks

12 Conclusion

The strict Planck-to-cosmos ruler object is not a bare point in $\mathbb{R}^3$. It is the pair

$$(\xi_X, \Xi_X; \lambda, \varepsilon_\lambda, \mathbf{r}_\lambda^*),$$

together with a positive reference tuple, logarithm base, frame and cosmology conventions, uncertainty model, and a protocol-indexed observation trace. The descriptor chart

$$\chi_{b,a}(\mathbf{q}) = \log_b(\mathbf{q}/\mathbf{a})$$

is unit invariant and turns positive monomial relations into linear constraints. The resolution margin $\log_b(q_i/\varepsilon_i)$ compares a typed system scale with a typed protocol resolution without conflating the two. Cosmic quantities are optional model outputs. This separation closes the central blocker in v1.0 and makes the module compatible with the GO Core distance-scale interface.

References

- [1] P. J. Mohr, D. B. Newell, B. N. Taylor, and E. Tiesinga, “CODATA Recommended Values of the Fundamental Physical Constants: 2022,” *Reviews of Modern Physics* 97 (2025), 025002.
- [2] National Institute of Standards and Technology, *Reference on Constants, Units, and Uncertainty: 2022 CODATA recommended values*, accessed 2026.

- [3] Bureau International des Poids et Mesures, *The International System of Units (SI Brochure)*, 9th ed., version 4.01, 2026.
- [4] Joint Committee for Guides in Metrology, *Evaluation of Measurement Data — Guide to the Expression of Uncertainty in Measurement*, JCGM 100:2008.
- [5] Planck Collaboration, “Planck 2018 results. I. Overview and the cosmological legacy of Planck,” *Astronomy & Astrophysics* 641 (2020), A1.
- [6] Planck Collaboration, “Planck 2018 results. VI. Cosmological parameters,” *Astronomy & Astrophysics* 641 (2020), A6.
- [7] D. W. Hogg, “Distance measures in cosmology,” arXiv:astro-ph/9905116 (1999).
- [8] D. Burago, Y. Burago, and S. Ivanov, *A Course in Metric Geometry*, American Mathematical Society, 2001.

Mandelbrot Rulers as Observation-Scale Geometry

Dimensionless traces, protocol equivalence, finite-window exponents, and uncertainty

Stassis Research Program

Strict GO Core reference revision v1.1 — 2026

Status. This revision supersedes v1.0. Every physical scale is normalized by a positive reference length before entering a logarithm. Intrinsic object scales are separated from instrument resolutions. Finite-scale exponents require a declared scale window, weighting rule, residual diagnostic, and uncertainty semantics. A statistical confidence interval is forbidden unless a stochastic error model is supplied.

Abstract

Mandelbrot's coastline problem is naturally expressed as a family of scale-indexed observation protocols rather than as a single instrument-free length. For a bounded set A in a metric space, a protocol records a positive reference length ℓ_* , physical resolutions ε , a covering, divider, or dilation rule, and an error model. The dimensionless coordinate $t = \log_b(\ell_*/\varepsilon)$ and the trace $y(t) = \log_b N_\varepsilon(A)$ are invariant under coherent changes of length unit. Lower and upper box dimensions are independent of both ℓ_* and the logarithm base. Lipschitz observations cannot increase upper box dimension at matched scales, while bi-Lipschitz observations preserve it. Protocols whose counts are comparable up to fixed multiplicative and scale constants have the same asymptotic exponent, although finite-scale amplitudes and residuals may differ. For finite data, the weighted slope estimator is defined together with its window; a deterministic stability interval is distinguished from a probabilistic confidence interval. The Koch curve supplies an exact divider benchmark. Natural coastlines need not possess a unique universal exponent outside a declared finite scale regime.

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1 Scope and claim firewall

This module uses Mandelbrot’s ruler argument as a strict test of the distance-scale interface. It does not replace Hausdorff measure, Minkowski content, or geometric measure theory. It distinguishes three levels:

$$\text{set and metric} \longrightarrow \text{scale-indexed protocol} \longrightarrow \begin{array}{l} \text{asymptotic or} \\ \text{finite-window summary} \end{array}$$

Boundary 1.1 (Claim boundary). Scale-dependent measurements do not imply that geometry is subjective. The set, metric, protocol, and scale are mathematical data. A log-log line over a short range is not proof of exact self-similarity. Different protocols may share an exponent while differing in prefactor, oscillation, lacunarity, bias, and uncertainty.

2 Typed scale protocol

Let (X, d) be a metric space whose distance has scale type U_L , and let $A \subseteq X$ be nonempty and bounded.

Definition 2.1 (Intrinsic descriptor). The intrinsic descriptor is

$$\xi_A = (A, d, \text{diam}_d A, \text{regularity and topology data}, \dots).$$

It contains properties of the mathematical object and metric, not the instrument resolution.

Definition 2.2 (Scale-observation protocol). A finite scale protocol is a tuple

$$\mathcal{P} = (\ell_*, b, I, \{\varepsilon_i\}_{i=1}^m, \mathcal{M}, \mathbf{A}, \vartheta, \mathcal{W}, \mathbf{E}),$$

where

- (i) $\ell_* > 0$ is a reference length with type U_L ;
- (ii) $b > 1$ is the logarithm base;
- (iii) $I = [\varepsilon_{\min}, \varepsilon_{\max}]$ is the physical scale range;

- (iv) each $\varepsilon_i \in I$ is a positive instrument resolution;
- (v) $\mathcal{M}$ is the measurement family (cover, divider, dilation, or another declared operator);
- (vi) A is the anchoring, grid, stopping, or boundary convention;
- (vii) ϑ collects nuisance settings such as grid translation, rotation, and sampling phase;
- (viii) $\mathcal{W} \subseteq \{1, \dots, m\}$ is a fit window;
- (ix) E is an exactness, deterministic-error, or stochastic error model.

The scale coordinate is

$$t_i := \log_b \frac{\ell_*}{\varepsilon_i}.$$

Its argument is dimensionless. A bare expression $\log(1/\varepsilon)$ is prohibited when ε is a physical length.

Definition 2.3 (Scale trace). For a protocol output $Y_{\mathcal{P}}(A; \varepsilon)$, the finite trace is

$$T_{\mathcal{P}}(A) := \{(t_i, Y_{\mathcal{P}}(A; \varepsilon_i)) : 1 \leq i \leq m\}.$$

An asymptotic trace additionally specifies the directed limit $\varepsilon \downarrow 0$ and the admissible protocol family.

3 Covering trace and box dimensions

Use internal closed-ball covers:

$$N_{\varepsilon}(A, d) = \min \left\{ |S| : S \subseteq A, A \subseteq \bigcup_{s \in S} \overline{B}_d(s, \varepsilon) \right\},$$

with value $+\infty$ if no finite cover exists. For totally bounded A , the count is finite for every $\varepsilon > 0$.

Definition 3.1 (Covering entropy). For $b > 1$,

$$\mathcal{H}_{b, \varepsilon}(A, d) := \log_b N_{\varepsilon}(A, d).$$

This is a covering-complexity quantity with information unit fixed by b ; it is not thermodynamic entropy.

Proposition 3.2 (Scale monotonicity). *If $0 < \varepsilon_1 \leq \varepsilon_2$, then*

$$N_{\varepsilon_1}(A, d) \geq N_{\varepsilon_2}(A, d).$$

Consequently $t \mapsto \mathcal{H}_{b, \ell_ b^{-t}}(A, d)$ is nondecreasing.*

Proof. Every ε_1 -ball lies in the ball with the same center and radius ε_2 , so any ε_1 -cover can be enlarged without increasing its cardinality. $\square$

Definition 3.3 (Lower and upper box dimensions).

$$\underline{\dim}_B A := \liminf_{\varepsilon \downarrow 0} \frac{\log_b N_{\varepsilon}(A, d)}{\log_b(\ell_*/\varepsilon)},$$

$$\overline{\dim}_B A := \limsup_{\varepsilon \downarrow 0} \frac{\log_b N_\varepsilon(A, d)}{\log_b(\ell_*/\varepsilon)}.$$

If they agree, their common value is $\dim_B A$.

Theorem 3.4 (Reference and base independence). *The lower and upper dimensions in definition 3.3 are independent of the positive reference length ℓ_* and of the logarithm base $b > 1$.*

Proof. Changing the base multiplies numerator and denominator by the same positive constant. Replacing ℓ_* by $c\ell_*$ adds the constant $\log_b c$ to the denominator. Since $\log_b(\ell_*/\varepsilon) \rightarrow +\infty$, this constant does not alter the liminf or limsup quotient. $\square$

Remark 3.5. Reference independence is an asymptotic statement. A finite-window intercept depends on ℓ_* , while a fitted slope with an intercept does not.

4 Observation maps and dimension

Theorem 4.1 (Matched Lipschitz cover). *If $O : (X, d_X) \rightarrow (Y, d_Y)$ is L_O -Lipschitz with $L_O > 0$, then*

$$N_{L_O\varepsilon}(O(A), d_Y) \leq N_\varepsilon(A, d_X).$$

Hence

$$\overline{\dim}_B O(A) \leq \overline{\dim}_B A, \quad \underline{\dim}_B O(A) \leq \underline{\dim}_B A$$

whenever the displayed dimensions are defined by compatible reference scales.

Proof. The image of every d_X -ball of radius ε is contained in a d_Y -ball of radius $L_O\varepsilon$. Map a finite input cover into the output. For the dimension statement, take logarithms and note that the fixed scale factor L_O shifts the logarithmic denominator by a constant. $\square$

Corollary 4.2 (Bi-Lipschitz invariance). *If, for some $0 < m_O \leq L_O < \infty$,*

$$m_O d_X(x, x') \leq d_Y(Ox, Ox') \leq L_O d_X(x, x')$$

on A , then lower and upper box dimensions are preserved.

Proof. Apply theorem 4.1 to O and to the inverse $O^{-1} : O(A) \rightarrow A$, which is m_O^{-1} -Lipschitz. $\square$

Thus the exponent is an invariant of the bi-Lipschitz observation class, not of arbitrary lossy observations.

5 When different rulers have the same exponent

Finite-scale traces depend on the protocol. The following criterion separates protocol dependence from asymptotic exponent dependence.

Definition 5.1 (Exponent-equivalent protocols). Two positive count traces $N_\varepsilon^{(1)}$ and $N_\varepsilon^{(2)}$ are exponent-equivalent if there exist constants $a_1, a_2, c_1, c_2 > 0$ and $\varepsilon_0 > 0$ such that, for $0 < \varepsilon < \varepsilon_0$,

$$c_1 N_{a_1\varepsilon}^{(1)} \leq N_\varepsilon^{(2)} \leq c_2 N_{a_2\varepsilon}^{(1)}.$$

Theorem 5.2 (Protocol-comparison criterion). *Exponent-equivalent protocols have the same lower and upper logarithmic scaling exponents.*

Proof. Take logarithms. The constants c_j add bounded terms to the numerators, and the constants a_j add bounded terms to the logarithmic scale coordinates. Division by $\log(\ell_*/\varepsilon) \rightarrow +\infty$ removes all such bounded terms. Apply the two inequalities to obtain matching liminf and limsup bounds. $\square$

Remark 5.3. Theorem 5.2 does not say that the traces are equal. Prefactors, log-periodic residuals, crossovers, and finite-window biases remain protocol-dependent.

For Euclidean subsets, standard axis-aligned cube covers, ball covers, and comparable grids satisfy relations of this type, so they recover the same box exponent even though their finite counts differ.

6 Divider, covering-length, and dilation protocols

6.1 Divider or ruler walk

A ruler walk along an ordered curve requires a stopping and anchoring rule. There is no protocol-free integer $N_\varepsilon^{\text{ruler}}$. Once a rule $\mathbf{A}$ is declared, define

$$L_\varepsilon^{\text{ruler}}(A; \mathbf{A}) := \varepsilon N_\varepsilon^{\text{ruler}}(A; \mathbf{A}).$$

The product has physical dimension L .

If, for some D and dimensionless $C_{\mathbf{A}} > 0$,

$$N_\varepsilon^{\text{ruler}}(A; \mathbf{A}) = C_{\mathbf{A}} \left(\frac{\ell_*}{\varepsilon} \right)^D (1 + o(1)),$$

then the normalized measured length obeys

$$\frac{L_\varepsilon^{\text{ruler}}}{\ell_*} = C_{\mathbf{A}} \left(\frac{\varepsilon}{\ell_*} \right)^{1-D} (1 + o(1)).$$

For $D > 1$ this diverges as $\varepsilon \downarrow 0$; for $D = 1$ it can converge to a finite protocol-dependent prefactor.

Boundary 6.1 (No universal divider). For self-intersections, inlets narrower than ε , disconnected sets, or digitized curves, a greedy divider may skip different pieces depending on traversal and anchoring. The algorithm is therefore part of the observation map.

6.2 Covering-length proxy

For a curve-like set one may define the distinct proxy

$$L_\varepsilon^{\text{cov}}(A) := \varepsilon N_\varepsilon(A, d).$$

This is not automatically the arclength of a curve. It is a typed scale-dependent proxy derived from covering complexity.

6.3 Minkowski sausage

For bounded $A \subset \mathbb{R}^m$, let

$$A_\varepsilon := \{x \in \mathbb{R}^m : \text{dist}(x, A) \leq \varepsilon\}, \quad V_A(\varepsilon) := \text{Vol}_m(A_\varepsilon).$$

The normalized volume trace is

$$\bar{V}_A(\varepsilon) := \frac{V_A(\varepsilon)}{\ell_*^m}.$$

If the limit exists, a Minkowski exponent can be written dimensionally as

$$D_M = m - \lim_{\varepsilon \downarrow 0} \frac{\log_b \bar{V}_A(\varepsilon)}{\log_b(\varepsilon/\ell_*)}.$$

Equivalently, finite nonzero Minkowski content corresponds to

$$\bar{V}_A(\varepsilon) = C_M \left(\frac{\varepsilon}{\ell_*} \right)^{m-D_M} (1 + o(1)),$$

with dimensionless $C_M > 0$.

7 Finite-window exponent estimation

An asymptotic dimension is a limit. Experimental and numerical data are finite. Let

$$t_i = \log_b \frac{\ell_*}{\varepsilon_i}, \quad y_i = \log_b N_{\varepsilon_i}(A, d), \quad i \in \mathcal{W},$$

where $\mathcal{W}$ contains at least two distinct values of t_i .

Definition 7.1 (Weighted finite-window slope). For declared weights $w_i > 0$, set

$$\begin{aligned} \bar{t}_w &= \frac{\sum_{i \in \mathcal{W}} w_i t_i}{\sum_{i \in \mathcal{W}} w_i}, & \bar{y}_w &= \frac{\sum_{i \in \mathcal{W}} w_i y_i}{\sum_{i \in \mathcal{W}} w_i}, \\ \hat{D}_{\mathcal{W},w} &= \frac{\sum_{i \in \mathcal{W}} w_i (t_i - \bar{t}_w)(y_i - \bar{y}_w)}{\sum_{i \in \mathcal{W}} w_i (t_i - \bar{t}_w)^2}. \end{aligned}$$

The fitted intercept is $\hat{\alpha} = \bar{y}_w - \hat{D}_{\mathcal{W},w} \bar{t}_w$.

Theorem 7.2 (Unit, reference, and base covariance of the slope). *The slope in definition 7.1 is unchanged by:*

- (i) *coherent changes of length unit* $(\ell_*, \varepsilon_i) \mapsto (a\ell_*, a\varepsilon_i)$;
- (ii) *replacing ℓ_* by $c\ell_*$ while retaining an intercept*;
- (iii) *changing the logarithm base simultaneously in t_i and y_i .*

Proof. In (i), every ratio ℓ_*/ε_i is unchanged. In (ii), all t_i receive the same additive constant, which cancels in $t_i - \bar{t}_w$. In (iii), all t_i and y_i are multiplied by the same positive constant, so numerator and denominator receive the same squared factor. $\square$

Definition 7.3 (Residual diagnostics). Let

$$r_i = y_i - \hat{\alpha} - \hat{D}_{\mathcal{W},w} t_i.$$

A minimum deterministic report contains

$$\left(\mathcal{W}, \{\varepsilon_i\}, \{w_i\}, \hat{D}_{\mathcal{W},w}, \max_{i \in \mathcal{W}} |r_i|, \sqrt{\frac{\sum w_i r_i^2}{\sum w_i}} \right).$$

Definition 7.4 (Deterministic pairwise-slope range).

$$I_{\mathcal{W}}^{\text{pair}} := \left[\min_{\substack{i < j \\ i, j \in \mathcal{W}}} \frac{y_j - y_i}{t_j - t_i}, \max_{\substack{i < j \\ i, j \in \mathcal{W}}} \frac{y_j - y_i}{t_j - t_i} \right].$$

This interval measures finite-window slope variation. It is not a confidence interval and carries no coverage probability.

7.1 When a confidence interval is legitimate

Suppose repeated measurements or a declared stochastic model justify

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\eta}, \quad \boldsymbol{\beta} = (\alpha, D)^\top, \quad \mathbb{E}[\boldsymbol{\eta}] = 0, \quad \text{Cov}(\boldsymbol{\eta}) = \boldsymbol{\Sigma},$$

where $\mathbf{X}$ has rows $(1, t_i)$ and $\boldsymbol{\Sigma}$ is positive definite. The generalized least-squares estimator is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \boldsymbol{\Sigma}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\Sigma}^{-1} \mathbf{y}.$$

If $\boldsymbol{\eta}$ is multivariate Gaussian and $\boldsymbol{\Sigma}$ is known, then

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^\top \boldsymbol{\Sigma}^{-1} \mathbf{X})^{-1},$$

and a normal-theory interval for D is available. If variance parameters are estimated, the corresponding degrees of freedom and estimator must be declared.

Boundary 7.5 (Correlated scales). Counts at nested resolutions are usually correlated because they are computed from the same set. Treating the log-counts as independent homoscedastic observations is not a default law. Without an error model, report deterministic residual and window stability only.

8 Grid phase and nuisance-parameter uncertainty

For box counts in $\mathbb{R}^m$, let ϑ encode grid translation and, when relevant, rotation. The finite count is then

$$N_{\varepsilon}^{\text{grid}}(A; \vartheta).$$

The protocol must choose one of the following:

- (a) a fixed, reproducible ϑ ;
- (b) an extremal count over a declared nuisance set;
- (c) an average or quantile under a declared probability law for ϑ .

A spread over arbitrarily chosen grid phases is a sensitivity analysis, not automatically a sampling confidence interval. Pixelization, segmentation, map projection, and boundary extraction add further protocol layers and must not be absorbed into the exponent without a record.

9 Exact benchmark: the Koch curve

Let the initial segment have length $\ell_0 > 0$. At construction stage n , the generator-aligned segment ruler has

$$\varepsilon_n = \ell_0 3^{-n}, \quad N_n^{\text{seg}} = 4^n.$$

Therefore

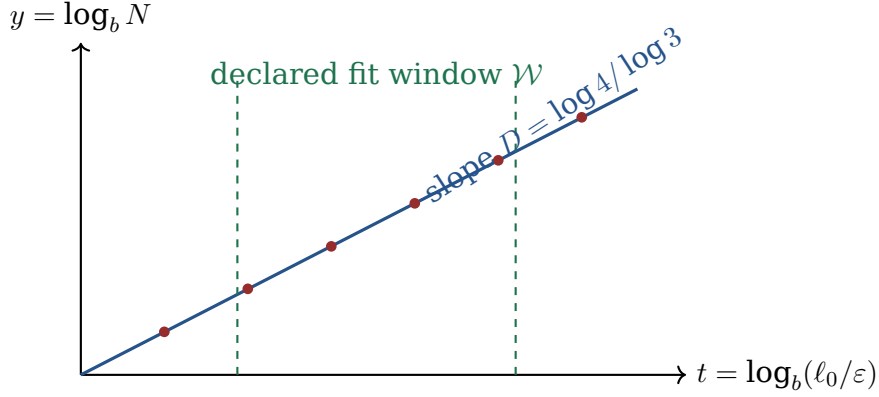
$$t_n = \log_b \frac{\ell_0}{\varepsilon_n} = n \log_b 3, \quad y_n = \log_b N_n^{\text{seg}} = n \log_b 4,$$

and every nontrivial exact stage window gives

$$\hat{D} = \frac{\log_b 4}{\log_b 3} = \frac{\log 4}{\log 3}.$$

The measured divider length satisfies

$$\frac{L_n}{\ell_0} = \frac{N_n^{\text{seg}} \varepsilon_n}{\ell_0} = \left(\frac{4}{3}\right)^n = \left(\frac{\varepsilon_n}{\ell_0}\right)^{1-D}.$$



Remark 9.1. The equality $N_n^{\text{seg}} = 4^n$ belongs to the generator-aligned divider protocol. It is not asserted as the exact count for every axis-aligned box grid. Standard covering protocols recover the same asymptotic dimension through comparison estimates.

The zero residual of this exact synthetic benchmark is not evidence that a natural coastline follows the same law.

10 Natural boundaries and finite scaling regimes

For empirical boundaries there may be lower and upper cutoffs:

$$\varepsilon_{\text{pixel}} \lesssim \varepsilon \lesssim \varepsilon_{\text{domain}}.$$

Below the first cutoff, segmentation or instrument noise dominates. Above the second, the finite domain and large-scale geometry dominate. Intermediate intervals can contain several crossovers rather than one power law.

A reproducible empirical statement has the form:

Under protocol $\mathcal{P}$, on the scale window $[\varepsilon_a, \varepsilon_b]$, the declared estimator gives $\hat{D}$, with the listed residual diagnostic and uncertainty model.

It must not be shortened to “the object has dimension $\hat{D}$ ” unless the limiting hypotheses are independently justified.

11 Connection to GO metric entropy

The metric-entropy module supplies the trace

$$\mathcal{H}_{b,\varepsilon}(A, d) = \log_b N_\varepsilon(A, d).$$

The present module adds:

- (i) the common distance-scale passport;
- (ii) the dimensionless coordinate $t = \log_b(\ell_*/\varepsilon)$;
- (iii) observation-map transformation laws;
- (iv) finite-window estimation and uncertainty semantics;
- (v) protocol-equivalence criteria.

When a differentiable interpolation is justified, an effective local exponent may be written

$$D_{\text{eff}}(t) = \frac{d\mathcal{H}(t)}{dt}.$$

For stepwise exact counts, this derivative is not canonical; a finite difference or smoothing rule is part of the estimator.

12 Normative reporting contract

For exact synthetic sets, the channel is explicitly noiseless. Temporal resolution and observation horizon are not-applicable for a static boundary; they must be supplied for an evolving boundary or time series. This module reports traces and exponent diagnostics rather than a scalar GO defect, so defect-normalizations is not-applicable until a downstream decision loss is introduced.

Field	Type	Required content
Object A and metric d	set and U_L -valued metric	domain, boundedness, total boundedness, metric convention
Intrinsic descriptor ξ_A	system data	diameter, regularity, and known cutoffs
Measurement family $\mathcal{M}$	operator	cover, divider, dilation, or explicit alternative
Noise/channel policy	kernel or exact flag	stochastic model or explicit noiseless statement
Resolution ε	U_L , positive	sampled values and ordering
Temporal resolution/horizon	time or not applicable	mandatory for time-dependent traces
Reference length ℓ_*	U_L , positive	normalization and unit context
Logarithm base b	dimensionless, $b > 1$	information unit
Anchoring A	protocol metadata	grid, traversal, stopping, boundary convention
Nuisance parameter ϑ	protocol metadata	fixed, extremized, or probabilistically averaged
Fit window $\mathcal{W}$	index/scale interval	predeclared or selection rule
Weights/covariance	positive weights or matrix	deterministic fit or stochastic model
Estimator	map	WLS, GLS, pairwise range, or alternative
Residual summary	dimensionless	norm and subwindow stability

Field	Type	Required content
Uncertainty statement	semantics	deterministic sensitivity or probabilistic coverage
Defect normalization	typed ratio or not applicable	required before any scalar loss aggregation
Limit claim	theorem or diagnostic	liminf, limsup, limit, or finite-window only

13 Claim audit

Statement	Status	Required hypotheses
Covering counts are monotone in resolution	theorem	fixed metric and cover convention
Box exponents are unit, reference, and base independent	theorem	positive reference and asymptotic limit
Lipschitz observation cannot increase upper box dimension	theorem	matched output scale
Bi-Lipschitz observation preserves box dimensions	theorem	two-sided distance bounds
All reasonable rulers give identical finite traces	false	only asymptotic comparison may hold
Finite log-log slope is a fractal dimension	diagnostic only	becomes a dimension only under a justified limit
Residual slope range is a confidence interval	false	no coverage probability without a stochastic model
Every natural coastline has one universal dimension	not claimed	finite regimes and crossovers are allowed
Koch divider length diverges	exact benchmark	generator-aligned construction scales
Mandelbrot's ruler argument proves an S-model	not claimed	no such implication is used

14 Conclusion

The mathematically stable object is not one measured length. It is the typed protocol trace

$$\left\{ \left(\log_b \frac{\ell_*}{\varepsilon}, Y_{\mathcal{P}}(A; \varepsilon) \right) : \varepsilon \in I \right\},$$

together with its transformation law, asymptotic exponent when it exists, finite-window residuals, and uncertainty semantics. The exponent is robust only under a specified class of observation maps or exponent-equivalent protocols. This is the strict content of the Mandelbrot ruler layer in Geometry of Observation.

References

- [1] B. B. Mandelbrot, "How Long Is the Coast of Britain? Statistical Self-Similarity and Fractional Dimension," *Science* 156 (1967), 636–638.
- [2] L. F. Richardson, "The problem of contiguity: an appendix to statistics of deadly quarrels," *General Systems Yearbook* 6 (1961), 139–187.
- [3] K. Falconer, *Fractal Geometry: Mathematical Foundations and Applications*, 3rd ed., Wiley, 2014.
- [4] P. Mattila, *Geometry of Sets and Measures in Euclidean Spaces*, Cambridge University Press, 1995.
- [5] C. Tricot, *Curves and Fractal Dimension*, Springer, 1995.
- [6] D. Burago, Y. Burago, and S. Ivanov, *A Course in Metric Geometry*, American Mathematical Society, 2001.

Regular Polyhedra under Typed Observation Filters

Abstract incidence, realization symmetry, Petrials, and identifiability

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Strict GO Core reference revision v1.1 — 28 July 2026

Status. This document supersedes the bilingual working note v1.0 and is an application-layer reference migrated to GO Core v0.2. It does not introduce a new polyhedron, classification, realization theorem, or symmetry algorithm. Its contribution is a typed separation of abstract incidence, geometric realization, admissibility, information loss, exact symmetry, and finite-resolution inference.

Abstract

The phrase “regular polyhedron” changes meaning when convex polyhedra, star polyhedra, abstract maps, or discrete skeletal realizations are placed in the same discussion. This reference fixes the object class before counting. An abstract polyhedron is a ranked poset satisfying the diamond condition and strong flag connectivity; abstract regularity means transitivity of $\text{Aut}(\mathcal{P})$ on flags. A geometric realization β has a generally smaller Euclidean symmetry group G_β , so an abstractly regular object need not be geometrically regular in a chosen realization. Observation specifications are defined on full subgroupoids of realized objects. Admissibility enlargement and informational coarsening are two different preorders: $\mathcal{C}_1 \subseteq \mathcal{C}_2$ enlarges the admitted domain, whereas $Q_{\text{coarse}} = \kappa \circ Q_{\text{fine}}$ loses distinctions. They cannot be compressed into an undefined inclusion of tuples. A cube and its Petrial give an exact non-injectivity control: the vertex-edge skeleton is unchanged, while six square faces are replaced by four Petrie hexagons and the Euler characteristic changes from 2 to 0. Under bounded noise, approximate symmetry residuals and a half-separation criterion give conditional identifiability, not proof of exact symmetry. Counts such as five Platonic solids or $48 = 18 + 30$ discrete regular skeletal polyhedra in $\mathbb{R}^3$ are therefore convention-scoped results, not values of one universal observation-independent enumeration.

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1 Scope and the category of objects

Definition 1.1 (Abstract polyhedron). A rank-3 abstract polyhedron $\mathcal{P}$ is a partially ordered set with a rank map to $\{-1, 0, 1, 2, 3\}$, unique least and greatest faces, and the following properties.

- (i) Every maximal chain, or *flag*, contains exactly one face of each rank.
- (ii) If $F < G$ have ranks $i - 1$ and $i + 1$, exactly two rank- i faces lie strictly between them (diamond condition).
- (iii) Every section is flag-connected, equivalently the usual strong flag-connectivity condition holds.

The proper faces of ranks 0, 1, 2 are vertices, edges, and faces. A reduced flag is the incident triple (v, e, f) .

The tuple (V, E, F, I) from the legacy note is only bookkeeping; by itself it does not impose thinness, connectedness, rank, or section axioms. Those axioms are required before standard results about polyhedra and flags are used.

Definition 1.2 (Realized-polyhedron groupoid). Let $\mathbf{RPol}_3$ be the groupoid whose objects are

$$X = (\mathcal{P}, \beta, \mathsf{T}),$$

where $\mathcal{P}$ is an abstract polyhedron, β is a declared realization in $\mathbb{R}^3$, and T records the realization convention: straight or other edges, planar, skew, star, zig-zag, or helical faces, discreteness, local finiteness, and faithfulness. An arrow $X \rightarrow X'$ is a rank-preserving incidence isomorphism $\varphi : \mathcal{P} \rightarrow \mathcal{P}'$ together with an ambient isometry a intertwining the realizations and all declared edge and face data.

The convention T is part of the type. A convex face-lattice realization, a regular map, and a skeletal apeirohedron are not silently treated as objects of one unqualified category.

2 Flags, abstract regularity, and string C-groups

By the diamond condition, every flag Φ has a unique i -adjacent flag Φ^i , differing only in its rank- i face, for $i = 0, 1, 2$. The resulting edge-coloured flag graph is connected.

Definition 2.1 (Abstract regularity). The automorphism group $\Gamma(\mathcal{P}) = \text{Aut}(\mathcal{P})$ consists of all rank-preserving order automorphisms. The abstract polyhedron is regular when $\Gamma(\mathcal{P})$ acts transitively on $\text{Flag}(\mathcal{P})$.

Proposition 2.2 (Simple flag transitivity). *For an abstract polyhedron, an automorphism fixing one flag is the identity. Hence a regular abstract polyhedron has a simply transitive flag action. If it is finite, then*

$$|\Gamma(\mathcal{P})| = |\text{Flag}(\mathcal{P})| = 4f_1, \tag{1}$$

where $f_1 = |E|$.

Proof. Rank preservation fixes every member of the chosen flag. The diamond condition then fixes each adjacent flag, and strong flag connectivity propagates this to all flags and faces. Thus a flag stabilizer is trivial. Each edge has two incident vertices and two incident faces, so it belongs to four reduced flags. $\square$

Fix a base flag. For a regular $\mathcal{P}$, let ρ_i be the unique automorphism sending it to its i -adjacent flag. For type $\{p, q\}$,

$$\rho_i^2 = 1, \quad (\rho_0 \rho_2)^2 = 1, \quad \text{ord}(\rho_0 \rho_1) = p, \quad \text{ord}(\rho_1 \rho_2) = q, \quad (2)$$

and the intersection property is

$$\langle \rho_i : i \in I \rangle \cap \langle \rho_i : i \in J \rangle = \langle \rho_i : i \in I \cap J \rangle. \quad (3)$$

These data form a rank-3 string C-group. The orders p, q are exact, and additional quotient relations may be present. Therefore $\{p, q\}$ is local type data, not a complete global identifier.

3 Abstract symmetry versus realization symmetry

For a faithful nondegenerate realization β , define

$$G_\beta = \{a \in \text{Isom}(\mathbb{R}^3) : a \text{ preserves the realized vertices, edges, and declared faces}\}.$$

Its action induces a subgroup $\iota_\beta(G_\beta) \leq \text{Aut}(\mathcal{P})$.

Definition 3.1 (Geometric regularity). The realization $(\mathcal{P}, \beta, \mathsf{T})$ is geometrically regular when G_β is transitive on its geometric flags.

Proposition 3.2 (One-way implication). *A faithful geometrically regular realization has an abstractly regular incidence polyhedron. The converse fails.*

Proof. The induced subgroup of $\text{Aut}(\mathcal{P})$ is already flag-transitive, so the full automorphism group is flag-transitive. Conversely, give the abstract cube three pairwise distinct edge lengths in orthogonal directions. Its incidence automorphism group still has order 48 and is flag-transitive, but the rectangular cuboid has only the eight independent coordinate sign changes. Edge directions cannot be permuted, leaving six geometric flag orbits. $\square$

Thus the legacy symbol $G(\mathcal{P})$, described as “geometric or combinatorial depending on the setting”, is ill-typed. The two groups and the homomorphism between them must be named.

4 Observation specifications and two preorders

Definition 4.1 (Typed observation specification). An observation specification is

$$\mathcal{O} = (\mathcal{C}_\mathcal{O}, Y_\mathcal{O}, Q_\mathcal{O}),$$

where $\mathcal{C}_\mathcal{O}$ is a full subgroupoid of $\mathbf{RPol}_3$, stable under isomorphism, $Y_\mathcal{O}$ is a declared data or label space, and

$$Q_\mathcal{O} : \text{Ob}(\mathcal{C}_\mathcal{O}) \longrightarrow Y_\mathcal{O}$$

is constant on the isomorphisms the protocol treats as nuisance transformations.

There are two independent comparisons.

$$\mathcal{O}_1 \preceq_{\text{adm}} \mathcal{O}_2 \iff \mathcal{C}_{\mathcal{O}_1} \text{ is a full subgroupoid of } \mathcal{C}_{\mathcal{O}_2}, \quad (4)$$

$$\mathcal{O}_{\text{fine}} \succeq_{\text{info}} \mathcal{O}_{\text{coarse}} \iff Q_{\text{coarse}} = \kappa \circ Q_{\text{fine}} \text{ on a common domain} \quad (5)$$

for some deterministic κ . Then $\text{Ker } Q_{\text{fine}} \subseteq \text{Ker } Q_{\text{coarse}}$: the coarse channel identifies at least as many objects.

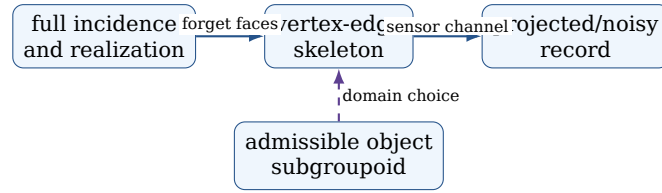
Theorem 4.2 (Scoped filter monotonicity). *Let R be one fixed predicate, such as abstract regularity or geometric regularity under one declared convention. If $\mathcal{O}_1 \preceq_{\text{adm}} \mathcal{O}_2$, then*

$$\{X \in \text{Ob}(\mathcal{C}_{\mathcal{O}_1}) : R(X)\} \subseteq \{X \in \text{Ob}(\mathcal{C}_{\mathcal{O}_2}) : R(X)\}.$$

No ordering of information kernels follows without equation (5), and no count comparison follows if different quotient equivalences are used.

Proof. The set inclusion is immediate from inclusion of the object domains while the predicate is held fixed. The remaining claims concern different data and therefore require a factorization map κ ; domain inclusion alone supplies none. $\square$

Boundary 4.3 (The historical family is not a chain). Removing finiteness while retaining planarity and allowing self-intersection while retaining finiteness relax different predicates. The resulting admissible classes may be incomparable. The old “filter ladder” is therefore replaced by a filter poset.



The horizontal arrows are lossy channels. The vertical arrow is a choice of domain. Calling both operations a “filter” without types caused the legacy order error.

5 Classical spherical count and its hypotheses

Let a finite equivelar polyhedral map of type $\{p, q\}$ have f -vector (V, E, F) . Double counting incidences gives

$$pF = 2E, \quad qV = 2E. \quad (6)$$

If the underlying surface is the sphere,

$$V - E + F = 2, \quad (7)$$

and therefore

$$\frac{1}{p} + \frac{1}{q} - \frac{1}{2} = \frac{1}{E} > 0. \quad (8)$$

For integers $p, q \geq 3$, this leaves precisely

$$(p, q) \in \{(3, 3), (4, 3), (3, 4), (5, 3), (3, 5)\}.$$

This proves the list of possible local types under the stated spherical, finite, thin, equivelar hypotheses. Existence and uniqueness up to similarity of the corresponding convex regular realizations are the separate classical realization theorem. Equation (8) is not a

Table 1: Exact finite controls for the five convex regular types.

Object	p	q	V	E	F	$ \text{Flag} = 4E$
Tetrahedron	3	3	4	6	4	24
Cube	4	3	8	12	6	48
Octahedron	3	4	6	12	8	48
Dodecahedron	5	3	20	30	12	120
Icosahedron	3	5	12	30	20	120

proof classifying star polyhedra, higher-genus maps, infinite tilings, or skeletal apeirohedra.

Duality reverses the face order:

$$\mathcal{P} \mapsto \mathcal{P}^*, \quad (V, E, F) \mapsto (F, E, V), \quad \{p, q\} \mapsto \{q, p\},$$

and preserves the automorphism-group order. Duality is an exact combinatorial operation, not an optical viewpoint change.

6 Skeleton non-identifiability and the Petrial control

For a regular map with distinguished generators (ρ_0, ρ_1, ρ_2) , the Petrie operation replaces the generating triple by

$$(\rho_0\rho_2, \rho_1, \rho_2). \tag{9}$$

Because ρ_0 and ρ_2 commute and are involutions, $\rho_0\rho_2$ is an involution. On the map, the operation keeps vertices and edges and selects Petrie polygons as faces.

Proposition 6.1 (Explicit skeleton non-injectivity). *Let $Q_{\text{skel}}(\mathcal{P}) = (V, E)$. The cube $\mathcal{P}_{\square}$ and its Petrial $\mathcal{P}_{\square}^{\pi}$ satisfy*

$$Q_{\text{skel}}(\mathcal{P}_{\square}) = Q_{\text{skel}}(\mathcal{P}_{\square}^{\pi}), \quad \mathcal{P}_{\square} \not\cong \mathcal{P}_{\square}^{\pi}.$$

The first has six square faces and Euler characteristic 2; the second has four Petrie hexagons and Euler characteristic 0.

Proof. Both structures use the eight cube vertices and twelve cube edges. The six square cycles give $4 \cdot 6 = 24 = 2E$ edge-face incidences. The four length-six Petrie cycles also give $6 \cdot 4 = 24 = 2E$. Their f -vectors are respectively $(8, 12, 6)$ and $(8, 12, 4)$, so they cannot be isomorphic as ranked posets and have Euler characteristics 2 and 0. $\square$

Thus “same skeleton, different face observation” is now an exact counterexample, not a metaphor. A two-dimensional rendering is even coarser: projection can merge vertices, create apparent crossings, or hide depth order. Those effects belong to the sensor channel, not to the incidence definition.

7 Finite-resolution symmetry and identifiability

Let observed vertices be

$$y_i = x_i + \eta_i, \quad \|\eta_i\| \leq \epsilon,$$

after a declared registration, labeling, and incidence-matching protocol. For a candidate isometry g and allowed incidence permutations Π , define the RMS residual

$$r_Y(g) = \min_{\pi \in \Pi} \left(\frac{1}{n} \sum_{i=1}^n \|gy_i - y_{\pi(i)}\|^2 \right)^{1/2}, \quad \delta_Y(g) = \frac{r_Y(g)}{\ell_{\text{ref}}}, \quad (10)$$

where $\ell_{\text{ref}} > 0$ is declared.

Proposition 7.1 (Exact symmetry under bounded noise). *If $gx_i = x_{\pi(i)}$ for one allowed π , then*

$$r_Y(g) \leq 2\epsilon.$$

The converse is false: a small residual certifies only threshold-dependent approximate symmetry.

Proof. Isometries preserve noise norms, so each residual vector is $g\eta_i - \eta_{\pi(i)}$ and has norm at most 2ϵ . The RMS bound follows. Arbitrarily small asymmetric perturbations give nonzero but arbitrarily small residuals. $\square$

Let d_Q be a declared metric on observations modulo nuisance isometries and relabelings, and let $\{M_1, \dots, M_K\}$ be a finite candidate library. Define

$$\Delta_i = \min_{j \neq i} d_Q(M_i, M_j).$$

Theorem 7.2 (Half-separation certificate). *If $\Delta_i > 0$ and $d_Q(Y, M_i) < \Delta_i/2$, then M_i is the unique nearest candidate.*

Proof. For $j \neq i$, the triangle inequality gives

$$d_Q(Y, M_j) \geq d_Q(M_i, M_j) - d_Q(Y, M_i) > \Delta_i/2 > d_Q(Y, M_i).$$

$\square$

This is conditional identifiability inside a declared finite library. It does not show that the true object belongs to that library.

For a finite flag set and a declared group G , let its flag orbits have weights w_j . The diagnostic

$$H_{\text{orb}}^{(2)}(G) = - \sum_j w_j \log_2 w_j \quad (11)$$

vanishes exactly when there is one orbit. It is a combinatorial Shannon diagnostic, not thermodynamic entropy, and it is undefined for an infinite flag set without a probability measure.

8 Convention-scoped taxonomy and claim firewall

Boundary 8.1 (Claim firewall). This reference does not claim a new classification, a new regular polyhedron, completeness beyond a cited convention, or recovery of incidence from a picture. Its strict internal results are the typed filter preorders, the separation of abstract and geometric symmetry, the explicit cube-Petrial non-injectivity control, and the finite-resolution identification bounds.

9 Protocol minimum and benchmark audit

A reproducible polyhedral observation claim must declare:

Table 2: Counts are meaningful only after the object convention is fixed.

Count	Object convention	Permitted interpretation
5	finite convex regular polyhedra in $\mathbb{R}^3$, up to similarity	Classical Platonic classification.
4	classical Kepler-Poinsot regular star polyhedra	Star-face and self-intersection conventions are part of the type.
$48 = 18 + 30$	Schulte’s discrete geometrically regular skeletal polyhedra in $\mathbb{R}^3$: 18 finite and 30 infinite	Published classification in that specific skeletal convention.
Not 48	all abstract regular rank-3 polytopes or all combinatorially regular realizations	The abstract and realization universes are much larger.

1. the abstract object axioms and rank;
2. the realization convention T , ambient space, and faithfulness;
3. whether regularity uses $\text{Aut}(\mathcal{P})$ or G_β ;
4. the admissible subgroupoid and the equivalence used for counting;
5. the observation map, projection, labeling, and face-selection rule;
6. spatial resolution, noise bound or law, registration group, estimator, threshold, and uncertainty;
7. for infinite structures, discreteness, local finiteness, observation window, and boundary treatment.

The executable companion checks all five Platonic incidence ledgers, the spherical type inequality, dual pairs, all cube square and Petrie cycles, the 48-element cube action, the 8-element generic cuboid action, flag-orbit counts, bounded-noise residuals, and half-separation certificates. Passing these controls verifies the claims encoded here; it is not an independent classification proof.

References

- [1] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Encyclopedia of Mathematics and its Applications 92, Cambridge University Press, 2002. doi:10.1017/CBO9780511546686.
- [2] P. McMullen and E. Schulte, “Regular Polytopes in Ordinary Space”, *Discrete & Computational Geometry* 17 (1997), 449–478. doi:10.1007/PL00009304.
- [3] E. Schulte, “Regular Incidence Complexes, Polytopes, and C-Groups”, 2017. arXiv:1711.02297.
- [4] E. Schulte, “Polyhedra, Complexes, Nets and Symmetry”, *Acta Crystallographica A* 70 (2014), 203–216. doi:10.1107/S2053273314000217.
- [5] B. Grünbaum, “Regular Polyhedra—Old and New”, *Aequationes Mathematicae* 16 (1977), 1–20. doi:10.1007/BF01836414.
- [6] A. W. M. Dress, “A Combinatorial Theory of Grünbaum’s New Regular Polyhedra, Part I”, *Aequationes Mathematicae* 23 (1981), 252–265. doi:10.1007/BF02188039.
- [7] A. W. M. Dress, “A Combinatorial Theory of Grünbaum’s New Regular Polyhedra, Part II: Complete Enumeration”, *Aequationes Mathematicae* 29 (1985), 222–243. doi:10.1007/BF02189831.

Frames, Forces, Constraints, and Dissipation

A typed mechanics interface under observation maps

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GO Core mechanics interface v0.1 — 28 July 2026

Status. Normative infrastructure for GO Core v0.2. This interface fixes frame direction, point-vector separation, constraint signs, force/power duality, and the nonnegative convention for dissipation. It supplies a common contract for the Foucault, bobsleigh, and roller-coaster modules. It does not propose a new law of mechanics.

Abstract

Mechanical observation models become ambiguous when positions, displacements, forces, generalized forces, and accelerations are added without a common frame or when “dissipated power” is assigned both signs in the same document. We define a typed interface on a configuration manifold Q . A frame edge $A_{F \leftarrow G}(t) \in \text{SO}(3)$ maps free-vector components from G to F , while a point map additionally carries an origin. Differentiating the affine point map yields the translational, Coriolis, Euler, and centrifugal terms; these are representation terms, not new interactions. Physical forces are covectors through virtual power, and generalized forces belong to T_q^*Q . Equality constraints, unilateral contact, and Coulomb-type tangential reactions are given separate signatures. Dissipation is defined by

$$P_{\text{diss}} := -\langle Q_d, \dot{q} \rangle \geq 0,$$

so a quadratic Rayleigh function produces $P_{\text{diss}} = 2\mathcal{R}$. The Lagrangian energy identity is

$$\dot{E}_L = P_{\text{act}} - P_{\text{diss}} + P_{\text{con}} - \partial_t L.$$

For an ideal scleronomous constraint $P_{\text{con}} = 0$; for a moving constraint it need not vanish. The interface also separates physical frame changes, deterministic observation maps, stochastic measurement channels, and estimators. Its statements are standard mechanics recast as a machine-checkable contract.

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1	Scope and typed objects	2
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7	Observation protocol and transformation classes	6

1 Scope and typed objects

Let Q be a finite-dimensional configuration manifold and let F be a set of declared Euclidean frames. The interface distinguishes the following objects.

Object	Mathematical type	Admissible operation
Point p	affine-space element	affine frame map, point difference
Free vector v	$V_F \simeq \mathbb{R}^3$	linear frame map, norm, dot product
Configuration velocity	$\dot{q} \in T_q Q$	pairing with a covector
Generalized force	$Q_f \in T_q^* Q$	virtual power $\langle Q_f, \dot{q} \rangle$
Spatial force	$F^F \in V_F$	dot product with a coframed velocity
Acceleration	$a^F \in V_F$	addition only in one declared frame
Observation	map or Markov kernel	may be non-injective and lossy

Boundary 1.1 (Type firewall). A frame transform is an invertible change of representation. An observation map may be many-to-one. Neither is a force law. A generalized force is a covector and cannot be added directly to a spatial acceleration. Multiplication by a mass metric or an explicit pullback is required.

2 Frame graph and affine point maps

Definition 2.1 (Directed frame edge). For frames F, G , the matrix

$$A_{F \leftarrow G}(t) \in \text{SO}(3)$$

maps components of a free vector from G to F : $v^F = A_{F \leftarrow G} v^G$. It obeys

$$A_{F \leftarrow G}^{-1} = A_{G \leftarrow F} = A_{F \leftarrow G}^T, \quad A_{F \leftarrow H} = A_{F \leftarrow G} A_{G \leftarrow H}.$$

A point needs an origin in addition to a rotation. Fix an inertial frame I , let $c_F^I(t)$ be the origin of F represented in I , and abbreviate $A(t) = A_{I \leftarrow F}(t)$. Then

$$x^I = c_F^I + A x^F. \quad (1)$$

The translation is forbidden for a free vector: $v^I = A v^F$, not $c_F^I + A v^F$.

Definition 2.2 (Angular velocity convention). The F -components of angular velocity are defined by

$$A^T \dot{A} = [\Omega^F]_{\times}, \quad [\Omega]_{\times} u = \Omega \times u.$$

Thus $\dot{A} = A [\Omega^F]_{\times}$.

Theorem 2.3 (Kinematics of a moving frame). *If $x^F(t)$ is twice differentiable, then*

$$v^I = \dot{c}_F^I + A(v^F + \Omega^F \times x^F), \quad (2)$$

$$a^I = \ddot{c}_F^I + A(a^F + 2\Omega^F \times v^F + \dot{\Omega}^F \times x^F + \Omega^F \times (\Omega^F \times x^F)). \quad (3)$$

Proof. Differentiate equation (1), substitute $\dot{A} = A[\Omega^F]_\times$, and differentiate once more. The product rule gives two equal cross terms $\Omega^F \times v^F$, producing the coefficient 2. $\square$

For a particle of mass m with physical force $F^F = A^\top F^I$, Newton's equation in F is

$$\begin{aligned} ma^F = F^F - mA^\top \ddot{c}_F^I - 2m\Omega^F \times v^F \\ - m\dot{\Omega}^F \times x^F - m\Omega^F \times (\Omega^F \times x^F). \end{aligned} \quad (4)$$

The last four terms are respectively translational, Coriolis, Euler, and centrifugal effective forces.

Boundary 2.4 (Interpretation). The effective terms in equation (4) are produced by the time-dependent representation. They must not be inserted in the inertial equation and then transformed again.

3 Configuration embeddings and constraints

Let a mechanical realization be a possibly time-dependent embedding

$$\Phi_t : Q \longrightarrow (\mathbb{R}^3)^n, \quad q \longmapsto (x_1(t, q), \dots, x_n(t, q)),$$

with masses $m_a > 0$. For a time-independent embedding, the pullback mass form is

$$G_{ij}(q) = \sum_{a=1}^n m_a \partial_i x_a \cdot \partial_j x_a. \quad (5)$$

Proposition 3.1 (Mass-form rank). *The matrix $G(q)$ in equation (5) is positive semidefinite. It is positive definite exactly when $d\Phi_q : T_q Q \rightarrow (\mathbb{R}^3)^n$ is injective.*

Proof. For $\xi \in T_q Q$,

$$\xi^\top G \xi = \sum_a m_a \|d\Phi_q(\xi)_a\|^2 \geq 0.$$

It vanishes exactly on $\ker d\Phi_q$. $\square$

Equality constraints are written

$$h(t, q) = 0, \quad A_h(t, q) = \partial_q h.$$

For a regular ideal constraint with multiplier λ , the reaction covector is

$$Q_h = A_h^\top \lambda.$$

An admissible velocity satisfies

$$A_h \dot{q} + \partial_t h = 0. \quad (6)$$

Therefore its reaction power is

$$P_h = \langle Q_h, \dot{q} \rangle = -\lambda^\top \partial_t h. \quad (7)$$

It vanishes for a scleronomous constraint $\partial_t h = 0$, but not in general for a moving support.

For unilateral contact we fix the sign convention

$$g_j(t, q) \geq 0, \quad N_j \geq 0, \quad N_j g_j = 0, \quad (8)$$

where $g_j > 0$ means separation and N_j is compressive. Impacts, restitution, and measure-valued impulses require a separate nonsmooth extension and are outside the smooth interface.

4 Forces, pullback, and virtual power

Suppose the spatial velocity of application point a is

$$v_a = J_a(t, q)\dot{q} + v_{a,0}(t, q).$$

A spatial force F_a pulls back to

$$Q_a = J_a^\top F_a \in T_q^*Q.$$

For the virtual part,

$$\langle Q_a, \delta q \rangle = F_a \cdot J_a \delta q.$$

The actual spatial power contains the affine contribution:

$$F_a \cdot v_a = \langle Q_a, \dot{q} \rangle + F_a \cdot v_{a,0}.$$

Consequently a time-dependent embedding cannot be treated as a time-independent Jacobian without losing support power.

Proposition 4.1 (Orthogonal covariance of power). *For $A \in \text{SO}(3)$, if $F^I = AF^F$ and $v^I = Av^F$, then*

$$F^I \cdot v^I = F^F \cdot v^F.$$

Proof. $(AF^F)^\top (Av^F) = (F^F)^\top A^\top Av^F$. □

This proposition applies to coframed physical vectors. It does not authorize pairing a force in one frame with a velocity in another.

5 Dissipation and contact friction

Definition 5.1 (Dissipative generalized force). A generalized force $Q_d(t, q, \dot{q})$ is dissipative on an admissible domain if

$$\langle Q_d, \dot{q} \rangle \leq 0.$$

Its nonnegative dissipated power is

$$P_{\text{diss}} := -\langle Q_d, \dot{q} \rangle \geq 0. \quad (9)$$

For a Rayleigh function

$$\mathcal{R}(t, q, \dot{q}) = \frac{1}{2} \dot{q}^\top C(t, q) \dot{q}, \quad C = C^\top \succeq 0,$$

the convention is

$$Q_d = -\partial_{\dot{q}} \mathcal{R} = -C \dot{q}, \quad P_{\text{diss}} = \dot{q}^\top C \dot{q} = 2\mathcal{R}.$$

At contact j , let n_j be the unit normal and $P_{T,j} = I - n_j n_j^\top$. Write the force on the modeled body as

$$F_j = N_j n_j + T_j, \quad T_j \cdot n_j = 0,$$

and the tangential relative velocity as

$$v_{T,j} = P_{T,j}(v_{\text{body},j} - v_{\text{support},j}).$$

A Coulomb-type admissible reaction obeys

$$\|T_j\| \leq \mu_j N_j, \quad T_j \cdot v_{T,j} \leq 0. \quad (10)$$

Thus

$$P_{\text{fric},j} = -T_j \cdot v_{T,j} \geq 0. \quad (11)$$

For sliding with $v_{T,j} \neq 0$, the maximum-dissipation selection is

$$T_j = -\mu_j N_j \frac{v_{T,j}}{\|v_{T,j}\|}.$$

For sticking, $v_{T,j} = 0$, the friction force is set-valued inside the cone and its instantaneous power is zero.

Definition 5.2 (Guarded cone utilization). On an admissible contact state define

$$u_j = \begin{cases} \|T_j\| / (\mu_j N_j), & \mu_j N_j > 0, \\ 0, & \mu_j N_j = 0 \text{ and } T_j = 0. \end{cases}$$

The cone law excludes the missing case $\mu_j N_j = 0$, $T_j \neq 0$. For a finite contact set,

$$S_{\text{slip}} = 1 - \max_j u_j \in [0, 1],$$

with $\max \emptyset := 0$.

6 Lagrange-d'Alembert energy identity

Let $L(t, q, \dot{q})$ be regular on the active smooth regime and set

$$E_L = \langle \partial_{\dot{q}} L, \dot{q} \rangle - L.$$

The forced equations are

$$\frac{d}{dt} \partial_{\dot{q}} L - \partial_q L = Q_{\text{act}} + Q_d + Q_{\text{con}}. \quad (12)$$

Theorem 6.1 (Energy balance). *Every C^2 solution of equation (12) satisfies*

$$\dot{E}_L = P_{\text{act}} - P_{\text{diss}} + P_{\text{con}} - \partial_t L, \quad (13)$$

where

$$P_{\text{act}} = \langle Q_{\text{act}}, \dot{q} \rangle, \quad P_{\text{con}} = \langle Q_{\text{con}}, \dot{q} \rangle.$$

Proof. Direct differentiation gives

$$\dot{E}_L = \left\langle \frac{d}{dt} \partial_{\dot{q}} L - \partial_q L, \dot{q} \right\rangle - \partial_t L.$$

Insert equation (12) and use equation (9). □

Corollary 6.2 (Autonomous ideal case). *If $\partial_t L = 0$, $Q_{\text{act}} = 0$, $Q_d = 0$, and the constraints are ideal and scleronomous, then $\dot{E}_L = 0$.*

Boundary 6.3 (No-double-counting rule). A moving track may be represented either by an ambient formulation with explicit reaction/support power or by reduction to a time-dependent Lagrangian. The same support work must not be added in both places.

7 Observation protocol and transformation classes

A complete protocol is the typed chain

$$X_t \xrightarrow{\mathcal{R}_{\text{red}}} \tilde{X}_t \xrightarrow{\mathcal{O}_\lambda} Z_t \xrightarrow{\mathcal{K}_\lambda} \text{Law}(Y_t | Z_t) \xrightarrow{\hat{\theta}_\lambda} \hat{\theta}_t.$$

The protocol record λ includes:

- (i) frame graph, clock, coordinates, and unit context;
- (ii) system descriptor and constraint convention;
- (iii) spatial and temporal resolution and observation horizon;
- (iv) deterministic readout and stochastic/noiseless declaration;
- (v) calibration, estimator, uncertainty, and missing-data policy;
- (vi) a declared transformation group for every invariant claim.

Boundary 7.1 (Invariant scope). Rigid-motion invariance, covariance under a time-dependent frame, Galilean invariance, label-swap invariance, and invariance under a lossy observation map are different statements. The word “invariant” is rejected unless the acting group and the transformed objects are named.

8 Machine gates

The mechanics extension admits a reference document only if the following gates are satisfied.

Gate	Required declaration	Rejected pattern
Frame closure	every vector sum has one frame; every point map has an origin	$r^I + q^B$ without $A_{I \leftarrow B}$
Force type	spatial force or generalized covector, with pullback map	adding F directly to a
Constraint sign	gap side, reaction sign, complementarity, active set	unsigned g, N convention
Dissipation sign	$P_{\text{diss}} \geq 0$ and force-on-system convention	$\dot{E} = -\sum T \cdot v_{\text{rel}}$ with $T \cdot v_{\text{rel}} \leq 0$

Gate	Required declaration	Rejected pattern
Moving support	explicit $\partial_t h$, support power, or reduced time-dependent L	declaring every ideal constraint workless
Friction guard	zero-load branch for $\ T\ /(\mu N)$	division by $\mu N = 0$
Observation split	frame map, observation, channel, estimator kept separate	calling a camera projection a frame transform
Invariant scope	named group and hypotheses	unqualified “observation invariant”

9 Claim boundary

The interface is a consistency layer. It does not establish:

- existence or uniqueness for arbitrary nonsmooth contact;
- a constitutive law for ice, wheel-rail, or air resistance;
- structural safety, fatigue, or human-tolerance limits;
- a precision geophysical pendulum model;
- scientific novelty of the underlying standard identities.

Any application must add its own regularity, constitutive, parameter, and uncertainty hypotheses.

References

- [1] A. M. Bloch, *Nonholonomic Mechanics and Control*, 2nd ed., Springer, 2015. doi:10.1007/978-1-4939-3017-3.
- [2] B. Brogliato, *Nonsmooth Mechanics: Models, Dynamics and Control*, 3rd ed., Springer, 2016. doi:10.1007/978-3-319-28664-8.
- [3] J. E. Marsden and T. S. Ratiu, *Introduction to Mechanics and Symmetry*, 2nd ed., Springer, 1999. doi:10.1007/978-0-387-21792-5.
- [4] V. I. Arnold, *Mathematical Methods of Classical Mechanics*, 2nd ed., Springer, 1989. doi:10.1007/978-1-4757-2063-1.

Foucault Networks on Celestial Bodies

Frame-consistent local dynamics and nested observation maps

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Strict GO Core reference revision v1.1 — 28 July 2026

Status. This document supersedes the bilingual working draft v1.0. It is migrated to GO Core v0.2 and the Frames-Forces-Dissipation Interface v0.1. The local pendulum is a small-amplitude rotating-frame model; the celestial hierarchy is a finite kinematic laboratory, not an ephemeris or a precision geophysical model.

Abstract

We formulate a network of idealized Foucault pendula on a finite hierarchy of rotating bodies without mixing body, local, inertial, and observer-frame components. A body edge in the hierarchy is transformed through its parent's attitude, a local endpoint displacement is transformed together with the site position, and a moving observer is represented by a full affine frame map rather than an origin subtraction alone. In the body-fixed tangent plane the declared small-amplitude model is

$$\ddot{u} + 2\beta\dot{u} + 2\Omega_n J\dot{u} + \omega_0^2 u = 0, \quad \Omega_n = \Omega^B \cdot n^B.$$

For $\beta = 0$, the substitution $u_C = e^{-i\Omega_n t} u$ proves that the oscillation plane precesses at signed rate $-\Omega_n$ in the chosen orientation. The Coriolis term performs no work, while damping dissipates $P_{\text{diss}} = 2m\beta \|\dot{u}\|^2 \geq 0$. A nested quasiperiodic hierarchy is represented as a continuous readout of a torus flow. Its closure is the image of the subtorus determined by integer frequency relations. Frequency support is preserved under a static Euclidean change of observer and a time-origin shift, but not under an arbitrary time-dependent rotating observer or a lossy camera projection. The result is a typed observation model with explicit claim boundaries, not a new theory of celestial mechanics.

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1 Scope and observation protocol

The experiment is a finite mathematical laboratory: choose bodies b , surface sites ℓ , local pendula, an observer frame O , and a sampling/camera protocol λ . The typed chain is

$$X_t \xrightarrow{\mathcal{R}} \tilde{X}_t \xrightarrow{\mathcal{O}_\lambda} Z_t \xrightarrow{\mathcal{K}_\lambda} \text{Law}(Y_t | Z_t) \xrightarrow{\hat{\theta}_\lambda} \hat{\theta}_t.$$

The reduction $\mathcal{R}$ selects a finite hierarchy and local model, $\mathcal{O}_\lambda$ produces ideal framed endpoints or projected pixels, $\mathcal{K}_\lambda$ represents sampling and noise, and $\hat{\theta}_\lambda$ estimates rates or spectral relations. These maps are not identified.

Boundary 1.1 (Claim firewall). The model excludes tidal elasticity, polar motion, atmospheric loading, suspension anisotropy, finite-amplitude precession, general relativity, and precision ephemerides. Parameters called “planetary” or “galactic” are labels unless supplied by a dated external model with uncertainty.

2 Frame-consistent celestial hierarchy

Fix an inertial reference I . Each body b has a center $R_b^I(t)$ and attitude

$$A_{I \leftarrow B_b}(t) \in \text{SO}(3),$$

where B_b is its body frame. If $p(b)$ is the parent and the relative center vector is stored in the parent frame, recursion is

$$R_b^I = R_{p(b)}^I + A_{I \leftarrow B_{p(b)}} r_{b|p(b)}^{B_{p(b)}}. \quad (1)$$

Thus every addition in equation (1) occurs in I .

At site ℓ , let $r_{b\ell}^{B_b}$ be the body-fixed site point, $n_{b\ell}^{B_b}$ the outward unit normal, and

$$E_{B_b \leftarrow L_{b\ell}} = \begin{pmatrix} e_1^{B_b} & e_2^{B_b} & n^{B_b} \end{pmatrix} \in \text{SO}(3)$$

the local frame. A horizontal pendulum displacement is $u^{L_{b\ell}} = (u_1, u_2, 0)^\top$.

Definition 2.1 (Endpoint point map). The endpoint in the inertial frame is

$$p_{b\ell}^I = R_b^I + A_{I \leftarrow B_b} \left(r_{b\ell}^{B_b} + E_{B_b \leftarrow L_{b\ell}} u^{L_{b\ell}} \right). \quad (2)$$

For a moving observer with origin $c_O^I(t)$ and attitude $A_{I \leftarrow O}(t)$, the observer components are

$$p_{b\ell}^O = A_{O \leftarrow I} (p_{b\ell}^I - c_O^I). \quad (3)$$

A camera map Π_λ then gives $z_{b\ell} = \Pi_\lambda(p_{b\ell}^O)$. In general Π_λ is not invertible.

Remark 2.2 (Correction to v1.0). The expression $R_b^I + A_{I \leftarrow B_b} r_{b\ell}^{B_b} + u_{b\ell}^{B_b} - c_O^I$ mixed the body displacement $u_{b\ell}^{B_b}$ with inertial vectors. Equation equation (2) transforms the site and local displacement before their inertial addition; equation (3) also includes observer attitude.

3 Local rotating-frame pendulum

Let the body angular velocity be Ω^B and set

$$\Omega_n = \Omega^B \cdot n^B.$$

On the oriented tangent plane define

$$Jv = n^B \times v, \quad J^2 = -I.$$

The following is a declared reduced model, not the exact spherical pendulum.

Model 3.1 (Linear Foucault layer). For pendulum length $\ell_p > 0$, $\omega_0^2 = g_{\text{eff}}/\ell_p$, and damping rate $\beta \geq 0$, the body-fixed horizontal displacement obeys

$$\ddot{u} + 2\beta\dot{u} + 2\Omega_n J\dot{u} + \omega_0^2 u = 0. \quad (4)$$

The model assumes a fixed local vertical, small displacement, constant Ω_n , and neglects terms beyond the horizontal Coriolis projection.

Theorem 3.2 (Signed precession in the undamped reduced model). *Under model 3.1 with $\beta = 0$, identify the oriented tangent plane with $\mathbb{C}$, so J acts as multiplication by i . Every solution has the form*

$$z(t) = e^{-i\Omega_n t} (C_+ \cos(\omega_c t) + C_- \sin(\omega_c t)), \quad \omega_c = \sqrt{\omega_0^2 + \Omega_n^2}.$$

Consequently a linearly polarized oscillation plane precesses in the body frame at signed rate

$$\dot{\alpha} = -\Omega_n$$

for the stated orientation convention.

Proof. With $z = e^{-i\Omega_n t} w$, the complex form

$$\ddot{z} + 2i\Omega_n \dot{z} + \omega_0^2 z = 0$$

reduces to $\ddot{w} + (\omega_0^2 + \Omega_n^2)w = 0$. □

For a spherical body whose spin axis defines latitude λ ,

$$\Omega_n = \Omega \sin \lambda.$$

Hence the magnitude vanishes at the equator and equals $|\Omega|$ at the poles. The sign depends on the orientation of n, e_1, e_2 and on the sign of the body's rotation; only after these are declared is “clockwise” meaningful.

4 Dissipation and energy

Define the reduced mechanical energy

$$E_{\text{loc}} = \frac{m}{2} \|\dot{u}\|^2 + \frac{m\omega_0^2}{2} \|u\|^2.$$

Proposition 4.1 (Coriolis work and damping sign). *Every solution of equation (4) satisfies*

$$\dot{E}_{\text{loc}} = -2m\beta \|\dot{u}\|^2. \quad (5)$$

Thus the Coriolis term performs no work and

$$P_{\text{diss}} = 2m\beta \|\dot{u}\|^2 \geq 0.$$

Proof. Dot equation (4) with mi . Since $\dot{u} \cdot J\dot{u} = 0$, the Coriolis contribution vanishes. $\square$

Equation (5) does not imply that the full body-pendulum-support energy is autonomous. Body rotation and suspension control are externalized by the reduced model.

5 Nested torus layer

Let the active phases be

$$\Theta(t) = \Theta_0 + \omega t \pmod{2\pi}, \quad \Theta \in \mathbb{T}^m.$$

A finite circular hierarchy and local oscillation can be represented by a continuous map $F : \mathbb{T}^m \rightarrow \mathbb{R}^3$, with ideal trace

$$Y(t) = F(\Theta(t)).$$

This includes finite Fourier readouts

$$Y(t) = \sum_{k \in \mathcal{K}} c_k e^{i(k \cdot \omega)t}, \quad \mathcal{K} \subset \mathbb{Z}^m \text{ finite.}$$

Theorem 5.1 (Orbit closure of the phase flow). *Let*

$$\Lambda_\omega = \{k \in \mathbb{Z}^m : k \cdot \omega = 0\}.$$

The closure of $\Theta(t)$ is the translate by Θ_0 of the closed subtorus whose annihilator is Λ_ω . In particular:

- (i) *the flow is periodic iff some $T > 0$ satisfies $T\omega \in 2\pi\mathbb{Z}^m$;*
- (ii) *if the components of ω are rationally independent, the phase orbit is dense in $\mathbb{T}^m$;*
- (iii) $\text{cl}\{Y(t)\} = F(\text{cl}\{\Theta(t)\})$.

Proof. The first two statements are the standard character description of a one-parameter subgroup of a torus. The last follows because the phase closure is compact and F is continuous. $\square$

The map F can collapse dimensions or frequency amplitudes. Therefore a full phase torus need not be identifiable from one projected trace.

6 Transformation audit

Quantity or structure	Declared transformation	Status
$\Omega^B \cdot n^B$	common $A \in \text{SO}(3)$ applied to both vectors	invariant scalar
Endpoint p^I	affine Euclidean frame change	covariant point, not invariant components

Quantity or structure	Declared transformation	Status
Local energy E_{loc}	static tangent-plane orthogonal basis change	invariant
Orbit-closure topology	static observer-space homeomorphism	preserved
Frequency set	static Euclidean isometry and time-origin shift	preserved up to zero/trend conventions
Frequency amplitudes	camera projection	may be suppressed
Frequency set	arbitrary time-dependent rotating observer	not invariant; modulation and shifts may occur
Signed precession	reversal of tangent-plane orientation	changes sign

Boundary 6.1 (No universal spectral invariant). “Frequency support” is not an invariant under every observer choice. The admissible statement requires a fixed clock and a static Euclidean change of observer, or a specified correction for the observer motion.

7 Non-identifiability

Proposition 7.1 (One trace does not identify one hierarchy). *For a single linear camera projection and finite observation horizon, the map from hierarchy parameters, local site state, and phase amplitudes to the projected trace is not injective in general.*

Proof. Any component in the kernel of the camera projection is invisible. Furthermore, two latent Fourier channels whose projected coefficient sum is equal produce the same projected signal when their frequencies coincide. Finite sampled data additionally admit aliases separated by integer multiples of the sampling angular frequency. Hence distinct hidden records can have the same observation. $\square$

Recoverability requires extra hypotheses: multiple noncoplanar camera channels, multiple sites, known body attitudes, anti-alias sampling, or external ephemeris constraints. The present document proves none of these sufficient in a realistic network.

8 Analytic sanity checks

Using the conventional nominal terrestrial rotation magnitude $\Omega_{\oplus} = 7.2921150 \times 10^{-5} \text{ s}^{-1}$, the reduced-model precession period is

$$T_F(\lambda) = \frac{2\pi}{\Omega_{\oplus} |\sin \lambda|}$$

when $\sin \lambda \neq 0$. This gives:

Latitude	$\Omega_n/\Omega_\oplus$	T_F
0°	0	no reduced-model precession
45°	$1/\sqrt{2}$	33.85 h
90°	1	23.93 h

These are formula checks, not predictions for a real installation. The nominal rotation value does not encode polar motion or length-of-day variations.

9 Claim boundary

The strict contributions are frame correction, typed observation maps, a proved precession statement inside a declared reduced model, and qualified transformation laws. This document does not claim:

- a new derivation of the Foucault effect;
- a KAM theorem for the Solar System;
- a realistic pendulum on another celestial body;
- recovery of a celestial hierarchy from one trace;
- an empirical test or a new physical parameter.

References

- [1] V. I. Arnold, *Mathematical Methods of Classical Mechanics*, 2nd ed., Springer, 1989. doi:10.1007/978-1-4757-2063-1.
- [2] G. Rizzi and M. L. Ruggiero, “The Foucault pendulum and the Sagnac effect,” *General Relativity and Gravitation* 35 (2003), 1743–1764. doi:10.1023/A:1021932917936.
- [3] A. Katok and B. Hasselblatt, *Introduction to the Modern Theory of Dynamical Systems*, Cambridge University Press, 1995.

Billiards as a Typed Geometry of Observation Laboratory

Collision reduction, symbolic channels, and spectral quantities

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Strict GO Core reference revision v1.1 — 28 July 2026

Status. This document supersedes the bilingual working note v1.0. It is an application-layer reference migrated to GO Core v0.2. It does not claim a new theorem in billiard dynamics, quantum chaos, or inverse spectral geometry. The new results internal to this note are typing, factorization, identifiability, and benchmark statements for explicitly declared observation channels.

Abstract

Let $\Omega \subset \mathbb{R}^2$ be a bounded billiard table. We separate four objects that are often conflated: the continuous billiard flow, its collision-section reduction, a deterministic observation of collision data, and a finite or stochastic recorded channel. On the regular collision space

$$\mathcal{M}_\partial = \{(s, \varphi) : s \in \partial\Omega_{\text{reg}}, |\varphi| < \pi/2\},$$

the billiard map is invertible almost everywhere and preserves

$$d\mu_\partial = \frac{\cos \varphi}{2|\partial\Omega|} ds d\varphi = \frac{1}{2|\partial\Omega|} ds dp, \quad p = \sin \varphi.$$

Therefore the impact map itself is not an information-loss map. Loss occurs only after a declared projection, partition, finite horizon, missing-event policy, or stochastic channel. For a known strictly convex table, two consecutive regular impact positions recover the outgoing direction, whereas one impact position does not. Finite symbolic words are necessarily non-injective on a continuum of initial states, but an infinite coding may be generating; marginal symbol entropy is neither entropy rate nor Kolmogorov-Sinai entropy. For the quantum layer, Laplacian eigenvalues λ_n have dimension L^{-2} , while physical energies are

$$E_n = \frac{\hbar^2}{2m} \lambda_n.$$

Rigid-motion invariance, similarity covariance, Weyl recovery of area, and isospectral non-identifiability are stated with their hypotheses. The result is a strict observation interface and reproducible benchmark, not a new billiard theory.

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1 Scope, conventions, and claim firewall

1.1 Regular table class

Unless a stronger hypothesis is stated, $\Omega \subset \mathbb{R}^2$ is bounded, connected, and open, with boundary equal to a finite union of piecewise C^3 arcs. The set of corners is $\partial\Omega_{\text{sing}}$; the regular part is $\partial\Omega_{\text{reg}}$. A trajectory hitting a corner, grazing the boundary, or accumulating infinitely many collisions in finite time is outside the default regular orbit class. Convexity, strict convexity, dispersing curvature, focusing curvature, and finite horizon are additional hypotheses, never defaults.

The dimensional model has a declared speed $c_0 > 0$. A regular hidden state is

$$z = (q, u) \in X := \Omega \times \mathbb{S}^1, \quad v = c_0 u, \quad \|u\| = 1.$$

Between collisions,

$$\dot{q} = c_0 u, \quad \dot{u} = 0. \tag{1}$$

At $q \in \partial\Omega_{\text{reg}}$, let $n(q)$ be the inward unit normal. If $u^- \cdot n < 0$, elastic specular reflection is

$$u^+ = u^- - 2(u^- \cdot n)n. \tag{2}$$

The convention is part of the model: changing to an outward normal changes the inequalities but not the physical reflected vector.

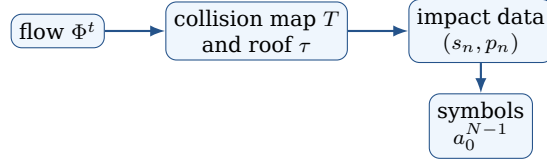
Boundary 1.1 (No universal billiard claim). Properties of circular, elliptic, dispersing, polygonal, and stadium billiards are not interchangeable. In particular, ergodicity, mixing, positive Lyapunov exponents, decay of correlations, and integrability require separate table classes and hypotheses.

1.2 Typed observation chain

For a protocol record λ , the complete chain is

$$X \xrightarrow{S} \mathcal{M}_\partial \xrightarrow{\mathcal{O}_\lambda} Z_\lambda \xrightarrow{\mathcal{K}_\lambda} \text{Law}(Y_\lambda \mid Z_\lambda) \xrightarrow{\hat{\theta}_\lambda} \hat{\theta}. \tag{3}$$

Here $\mathcal{S}$ is collision-section reduction, not a sensor; $\mathcal{O}_\lambda$ is deterministic; $\mathcal{K}_\lambda$ is either a declared stochastic channel or the identity kernel; and $\hat{\theta}_\lambda$ is an estimator or direct readout. The continuous flow is related to the collision map by a roof function and is not identified with the discrete map.



2 Collision space, singularities, and invariant measure

Parameterize each regular boundary component by arclength s , with positively oriented unit tangent $t(s)$. For an outgoing unit vector write

$$u = \cos \varphi n(s) + \sin \varphi t(s), \quad \varphi \in (-\pi/2, \pi/2),$$

and set $p = \sin \varphi \in (-1, 1)$.

Definition 2.1 (Regular collision space and map).

$$\mathcal{M}_\partial = \{(s, \varphi) : s \in \partial\Omega_{\text{reg}}, |\varphi| < \pi/2\}.$$

Let $\mathcal{S}_1$ contain states whose next collision is a corner, grazing, non-unique, or undefined. The collision map is

$$T : \mathcal{M}_\partial \setminus \mathcal{S}_1 \longrightarrow \mathcal{M}_\partial, \quad T(s, \varphi) = (s', \varphi').$$

The roof function

$$\tau(s, \varphi) = \frac{\|q(s') - q(s)\|}{c_0} > 0$$

is the physical flight time to the next collision.

Proposition 2.2 (Regular invertibility and invariant measure). *For the reversible elastic billiard, T is invertible away from forward and backward singularity sets. It preserves the two-form*

$$\omega_\partial = \cos \varphi \, ds \wedge d\varphi = ds \wedge dp$$

and the normalized probability measure

$$d\mu_\partial = \frac{\cos \varphi}{2L_\partial} \, ds \, d\varphi = \frac{1}{2L_\partial} \, ds \, dp, \quad L_\partial := |\partial\Omega|. \quad (4)$$

Proof. Specular reflection is time reversible, so reversing the outgoing velocity at a regular collision supplies the inverse branch. The preservation of ω_∂ is the standard Poincare-section restriction of Liouville measure for billiards. Normalization is direct:

$$\frac{1}{2L_\partial} \int_{\partial\Omega} ds \int_{-\pi/2}^{\pi/2} \cos \varphi \, d\varphi = 1.$$

□

Boundary 2.3 (Reduction is not observation loss). The full pair (s_n, p_n) evolves under an almost-everywhere invertible, measure-preserving map. Calling T itself a compression is incorrect. The flow requires the roof τ , and information loss must be assigned to a subsequent observation channel.

3 Impact-position observation and identifiability

Define the instantaneous position readout

$$\pi_s : \mathcal{M}_\partial \rightarrow \partial\Omega_{\text{reg}}, \quad \pi_s(s, p) = s.$$

This map is non-injective because all $p \in (-1, 1)$ share the same output at fixed s . Sequential observations have different identifiability.

Proposition 3.1 (Two-impact reconstruction). *Let Ω be a known strictly convex table with C^1 boundary. For two distinct consecutive regular impact points*

$$q_n = q(s_n), \quad q_{n+1} = q(s_{n+1}),$$

the outgoing direction and collision coordinate at q_n are uniquely

$$u_n^+ = \frac{q_{n+1} - q_n}{\|q_{n+1} - q_n\|}, \quad p_n = u_n^+ \cdot t(s_n). \quad (5)$$

Thus the two-step position observation (s_n, s_{n+1}) reconstructs (s_n, p_n) on the stated regular class.

Proof. Free flight is a straight segment and strict convexity makes the open chord lie in Ω . Its oriented endpoints therefore determine the unit direction. Projection on the declared tangent gives p_n . $\square$

Remark 3.2. Unknown geometry, unordered impacts, missed impacts, coincident points, finite spatial resolution, non-convex visibility, and stochastic misclassification enlarge the observation fiber. The proposition does not survive those changes without additional hypotheses.

For a horizon N , useful deterministic channels are

$$\mathcal{O}_{\text{full}, N}(z) = (s_k, p_k)_{k=0}^{N-1}, \quad (6)$$

$$\mathcal{O}_{\text{pos}, N}(z) = (s_k)_{k=0}^{N-1}, \quad (7)$$

$$\mathcal{O}_{\alpha, N}(z) = (a_k)_{k=0}^{N-1}, \quad a_k = \alpha(s_k), \quad (8)$$

where $\alpha : \partial\Omega \rightarrow \{1, \dots, m\}$ is a declared measurable partition with a boundary-assignment rule.

4 Finite symbolic channels and entropy semantics

Proposition 4.1 (Finite-horizon non-injectivity). *Let $U \subset \mathcal{M}_\partial$ be uncountable and suppose $\mathcal{O}_{\alpha, N}$ is defined on U , with $m < \infty$ and $N < \infty$. Then*

$$\mathcal{O}_{\alpha, N} : U \rightarrow \{1, \dots, m\}^N$$

is not injective.

Proof. The codomain contains at most m^N words, while U is uncountable. $\square$

This proposition does not imply that every infinite symbolic coding is lossy. A generating partition may identify almost every orbit from its bi-infinite itinerary, modulo partition boundaries.

Let $Z_0 \sim \mu_\partial$, $Z_k = T^k Z_0$, and $A_k = \alpha(\pi_s Z_k)$. The process is stationary. For a logarithm base $b > 1$, define

$$H_b(A_0) = - \sum_{j: p_j > 0} p_j \log_b p_j, \quad (9)$$

$$H_b(A_0^{n-1}) = - \sum_{w: P_n(w) > 0} P_n(w) \log_b P_n(w), \quad (10)$$

$$h_b(T, \alpha) = \lim_{n \rightarrow \infty} \frac{1}{n} H_b(A_0^{n-1}), \quad (11)$$

where the last limit exists for a stationary finite-alphabet process. The Kolmogorov-Sinai entropy is

$$h_{\mu_\partial, b}(T) = \sup_{\alpha} h_b(T, \alpha).$$

Equality for one partition requires a generating hypothesis.

Boundary 4.2 (Marginal entropy is not dynamical entropy). The empirical quantity

$$\hat{H}_b = - \sum_j \hat{p}_j \log_b \hat{p}_j$$

uses only one-symbol frequencies. It ignores order, is partition-dependent, and is neither an entropy-rate estimate nor a KS-entropy estimate without an explicit stochastic and asymptotic protocol.

5 Exact integrable reference: the disk

Let $\Omega = B_R(0)$, $L_\partial = 2\pi R$, and retain $p = \sin \varphi$.

Theorem 5.1 (Disk collision map). *On the regular disk collision space,*

$$T_R(s, p) = (s + 2R \arccos p \pmod{2\pi R}, p). \quad (12)$$

The reduced angular-momentum coordinate

$$j_z := q \times u = Rp$$

is conserved and has the dimension of length. The physical specific angular momentum is $q \times v = c_0 j_z$. The rotation number and roof time are

$$\rho(p) = \frac{\arccos p}{\pi} \pmod{1}, \quad \tau_R(p) = \frac{2R\sqrt{1-p^2}}{c_0}. \quad (13)$$

Proof. At a collision, reflection reverses the normal component of u and preserves its tangential component p . Since $q = Re_r$, $q \times u = R(u \cdot e_\theta) = Rp$, while multiplication by c_0 restores the physical specific angular momentum. The chord subtends the central angle $\pi - 2\varphi = 2 \arccos p$, giving equation (12). Its length is $2R \cos \varphi = 2R\sqrt{1-p^2}$, which gives the roof. $\square$

The disk is an exact benchmark, not evidence that a generic billiard is integrable. For an ellipse, tangency to a member of a confocal caustic family is a classical invariant on regular trajectories; this statement is retained only as a standard example and is not extended to arbitrary convex tables.

6 Chaotic tables and finite-time diagnostics

The Bunimovich stadium is a standard focusing table for which ergodicity and hyperbolic behavior are classical results under the stadium hypotheses. Those results do not justify reading a Lyapunov exponent from an arbitrary two-trajectory plot.

For z outside singularities and a tangent vector $\xi \neq 0$, the mathematical exponent is

$$\lambda_+(z, \xi) = \lim_{N \rightarrow \infty} \frac{1}{N} \log \frac{\|DT_z^N \xi\|}{\|\xi\|}, \quad (14)$$

when the limit exists. A finite-window diagnostic must instead be declared as

$$\hat{\lambda}_{N_0, N_1} = \frac{\sum_{k=N_0}^{N_1} w_k(k - \bar{k}) \log(d_k/d_*)}{\sum_{k=N_0}^{N_1} w_k(k - \bar{k})^2}, \quad (15)$$

where d_k is computed in a declared collision-space metric, $d_* > 0$ is a reference distance, $w_k \geq 0$, and $\bar{k} = \sum w_k k / \sum w_k$. The fit must state the window, renormalization rule, saturation cutoff, singularity handling, and uncertainty model. Without those fields, $\hat{\lambda}$ is only a plot slope.

Boundary 6.1 (No chaos from one positive fit). A positive finite-window slope does not prove a positive asymptotic Lyapunov exponent, ergodicity, mixing, Bernoulli behavior, or positive KS entropy. These are distinct claims.

7 Quantum billiards: operator, dimensions, and spectrum

For a bounded open Ω , define the Dirichlet quadratic form

$$q_\Omega[\psi] = \int_\Omega \|\nabla \psi\|^2 dx, \quad \psi \in H_0^1(\Omega).$$

Its Friedrichs extension is the nonnegative self-adjoint Dirichlet Laplacian $-\Delta_\Omega^D$. It has compact resolvent and spectrum

$$0 < \lambda_1(\Omega) \leq \lambda_2(\Omega) \leq \dots, \quad \lambda_n \rightarrow \infty, \quad [\lambda_n] = L^{-2}, \quad (16)$$

with multiplicity.

Proposition 7.1 (Laplacian-to-energy bridge). *For a nonrelativistic particle of declared mass $m > 0$,*

$$\hat{H}_\Omega = -\frac{\hbar^2}{2m} \Delta_\Omega^D, \quad E_n = \frac{\hbar^2}{2m} \lambda_n.$$

Thus λ_n is not an energy. The bridge has SI dimension

$$\frac{[\hbar]^2 [\lambda_n]}{[m]} = \frac{(ML^2T^{-1})^2 L^{-2}}{M} = ML^2T^{-2}.$$

Proposition 7.2 (Rigid invariance and scale covariance). *For a Euclidean isometry g and scale $a > 0$,*

$$\lambda_n(g\Omega) = \lambda_n(\Omega), \quad \lambda_n(a\Omega) = a^{-2} \lambda_n(\Omega). \quad (17)$$

Consequently, in two dimensions,

$$\tilde{\lambda}_n(\Omega) := |\Omega| \lambda_n(\Omega) \quad (18)$$

is invariant under similarities.

Proof. Rigid motions induce a unitary pullback on L^2 . Under $x = ay$, the Laplacian becomes $a^{-2}\Delta_y$, while $|a\Omega| = a^2|\Omega|$. $\square$

Theorem 7.3 (Weyl area readout). *Let $N_\Omega(\Lambda) = \#\{n : \lambda_n < \Lambda\}$. For a planar open set of finite area,*

$$N_\Omega(\Lambda) \sim \frac{|\Omega|}{4\pi} \Lambda \quad (\Lambda \rightarrow \infty). \quad (19)$$

Hence the full Dirichlet spectrum determines area. Boundary-length terms require additional boundary regularity and a stated asymptotic theorem; they are not inferred from equation (19) alone.

Theorem 7.4 (Spectral non-identifiability). *There exist noncongruent planar domains with the same Dirichlet spectrum, including multiplicities. Therefore*

$$\mathcal{O}_{\text{spec}} : \Omega \mapsto (\lambda_n(\Omega))_{n \geq 1}$$

is not injective on the class of all planar domains.

Boundary 7.5 (Classical-quantum firewall). Sharing the domain Ω does not identify the classical collision map with the quantum spectral map. Any semiclassical claim requires an explicit limit, observable class, and theorem. No solution of inverse spectral geometry or quantum chaos is claimed here.

8 Dissipative and non-specular variants

The reference dynamics is elastic. A normal restitution model with $e \in [0, 1]$, unchanged tangential velocity, and particle mass $m > 0$ gives

$$v_n^+ = -e v_n^-, \quad \Delta E_{\text{diss}} = \frac{m}{2}(1 - e^2)(v_n^-)^2 \geq 0.$$

If $e < 1$, speed is no longer constant, so (s, φ) is not a closed state space: speed or energy must be added. Friction, stochastic reflection, rough walls, moving boundaries, and compliant impact each require a new channel and constitutive law. The invariant measure equation (4) is not inherited automatically.

9 Observation invariants and protocol requirements

Definition 9.1 (Recoverable quantity). Let $\mathcal{O} : X \rightarrow Y$. A quantity $I : X \rightarrow V$ is exactly recoverable from $\mathcal{O}$ if there exists $\tilde{I} : Y \rightarrow V$ such that

$$I = \tilde{I} \circ \mathcal{O}.$$

Equivalently, I is constant on every observation fiber $\mathcal{O}^{-1}(y)$.

Definition 9.2 (Group invariant readout). For a group $G \curvearrowright X$, a readout $I : X \rightarrow V$ is G -invariant when $I(gx) = I(x)$ for all admissible g, x . Equivariance, scale covariance, and estimator stability are not called invariance.

A complete billiard protocol records at least:

1. table class, boundary regularity, corner and grazing policy;
2. boundary parameterization, tangent and normal orientation;
3. speed or energy convention, time unit, and reference frame;

Object	Exact status	Required qualification
Collision measure μ_{∂}	Invariant under regular elastic T	Exclude singular orbit set; declare angle convention
Disk coordinate p	First integral of T_R	Disk geometry and elastic reflection
Finite symbolic word	Deterministic coarse readout	Partition, horizon, boundary rule, missing events
Marginal entropy	Channel statistic	Log base, positive counts, not KS entropy
Laplacian spectrum	Rigid-motion invariant; scale covariant	Boundary condition, multiplicity, operator convention
$ \Omega \lambda_n$	Similarity invariant in two dimensions	Area and eigenvalue belong to the same domain

Table 1: No quantity is called an observation invariant without its channel and transformation class.

4. collision section, roof-time treatment, and observation horizon;
5. deterministic readout, partition, spatial and temporal resolution;
6. missing-impact, duplicate-impact, and misclassification policy;
7. stochastic kernel or explicit noiseless assumption;
8. log base, metric, regression window, weights, and zero guards;
9. spectral boundary condition, multiplicity convention, mass, and $\hbar$ convention whenever energy is reported;
10. estimator, uncertainty model, and declared exactness class.

10 Reproducible benchmarks

The release includes executable tests and a machine-readable benchmark table. The exact reference cases are:

1. normalization of μ_{∂} and the change $p = \sin \varphi$;
2. invertibility, conserved p , roof time, and rotation number for the disk map;
3. reconstruction of p_n from two consecutive impacts in a strictly convex disk;
4. two cyclic binary words with identical marginal entropy but different conditional transition entropy;
5. exact rectangular Dirichlet eigenvalues, a^{-2} spectral scaling, and invariance of $|\Omega|\lambda_n$;
6. the dimensional bridge $E_n = \hbar^2 \lambda_n / (2m)$;
7. finite-word non-injectivity and zero-denominator guards.

For the declared disk benchmark $R = 2.3 \text{ m}$, $p = 0.4$, and $c_0 = 3 \text{ m s}^{-1}$,

$$\rho = \frac{\arccos(0.4)}{\pi} \approx 0.36901012, \quad \tau \approx 1.40532321 \text{ s}.$$

For the cyclic words 01010101 and 00110011, both marginal entropies equal one bit, while their empirical first-order conditional entropies are respectively zero and one bit. This is an exact order-sensitivity benchmark, not a chaos diagnostic.

11 Claim register and boundary

Claim	Status	Scope
Collision-map measure preservation	Standard proposition	Regular elastic billiard, singular sets excluded
Two-impact reconstruction	Proposition	Known strictly convex C^1 table, consecutive regular impacts
Finite symbolic non-injectivity	Proposition	Finite alphabet and horizon on uncountable state set
Disk collision formulas	Theorem	Circular table with declared coordinates
Weyl area recovery	Standard theorem	Planar finite-area Dirichlet Laplacian
Isospectral non-identifiability	Standard theorem	General planar-domain class
Positive finite-time slope implies chaos	Rejected	No such implication
Spectrum determines arbitrary shape	Rejected	False on general planar domains
New billiard or quantum-chaos theorem	Rejected	Not claimed

The admissible program-level conclusion is limited:

Mathematical billiards form a controlled laboratory in which continuous flow, collision reduction, deterministic projection, finite symbolic recording, and spectral observation can be compared without identifying them. Recoverability is a property of the declared channel and hypothesis class, not of the word “observation” alone.

References

- [1] G. D. Birkhoff, *Dynamical Systems*, American Mathematical Society, 1927.
- [2] Ya. G. Sinai, “Dynamical systems with elastic reflections: ergodic properties of dispersing billiards,” *Russian Mathematical Surveys* 25(2), 137–189 (1970).
- [3] L. A. Bunimovich, “On the ergodic properties of nowhere dispersing billiards,” *Communications in Mathematical Physics* 65, 295–312 (1979).
- [4] C. P. Dettmann, “Recent advances in open billiards with some open problems,” in *Frontiers in the Study of Chaotic Dynamical Systems with Open Problems*, arXiv:1007.4166 (2010).
- [5] S. Tabachnikov, *Geometry and Billiards*, American Mathematical Society, 2005.
- [6] N. Chernov and R. Markarian, *Chaotic Billiards*, American Mathematical Society, 2006.
- [7] M. Kac, “Can one hear the shape of a drum?” *American Mathematical Monthly* 73(4), 1–23 (1966).
- [8] C. Gordon, D. L. Webb, and S. Wolpert, “One cannot hear the shape of a drum,” *Bulletin of the American Mathematical Society* 27, 134–138 (1992).
- [9] L. Friedlander, “On the Weyl asymptotic formula for Euclidean domains of finite volume,” arXiv:2106.07690 (2021).

Bobsleigh Contact Geometry

Mass distribution, unilateral contact, and typed observation loss

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Status. This document supersedes the bilingual working draft v1.0. It is migrated to GO Core v0.2 and the Frames–Forces–Dissipation Interface v0.1. Contact and friction are diagnostic smooth-regime models. No sled design, ice constitutive law, safety calculation, or engineering validation is claimed.

Abstract

A center-of-mass trace does not determine the contact state of a sled moving on a shaped surface. We formalize this observation using a regular track parametrization, a finite mass realization, a configuration-space mass form, unilateral gaps, normal reactions, and tangential contact forces. The mass form

$$G_{ij}(q) = \sum_a m_a \partial_i x_a(q) \cdot \partial_j x_a(q)$$

is positive semidefinite and is a metric precisely where the realization map is an immersion. Contact uses the fixed convention

$$g_j \geq 0, \quad N_j \geq 0, \quad N_j g_j = 0, \quad \|T_j\| \leq \mu_j N_j.$$

With T_j defined as the tangential force on the sled and $v_{T,j}$ as its velocity relative to the ice, the nonnegative frictional loss is

$$P_{\text{fric}} = - \sum_j T_j \cdot v_{T,j} \geq 0.$$

Consequently the autonomous stationary-track energy balance is $\dot{E} = P_{\text{act}} - P_{\text{fric}}$ in the smooth active-contact regime. This corrects the sign in v1.0. Cone utilization is defined piecewise at zero normal load, and load entropy has an explicit logarithm base and normalization. Center-of-mass, pose, inertial, contact, load, and thermal channels remain distinct observations of a larger hidden state.

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1 Scope and protocol

Let λ specify a track chart, world and sled frames, clock, contact convention, sampling rates, sensor calibration, estimator, and uncertainty model. The observation chain is

$$X_t \xrightarrow{\mathcal{R}} \tilde{X}_t \xrightarrow{\mathcal{O}_\lambda} Z_t \xrightarrow{\mathcal{K}_\lambda} \text{Law}(Y_t | Z_t) \xrightarrow{\hat{\theta}_\lambda} \hat{\theta}_t.$$

A kinematic centerline supplied as input is not a dynamically predicted trajectory. A proxy load rule is not a solved contact reaction. These statuses are recorded separately.

Boundary 1.1 (Smooth-regime boundary). The equations below apply between impacts while the active contact set is fixed and the state is differentiable. Impact impulses, restitution, Painlevé-type inconsistency, stick-slip switching, and ice deformation require a nonsmooth differential inclusion or complementarity time-stepper and are not proved here.

2 Track as a shaped surface

Let $(e_s, e_u, n_{\text{tr}})$ be an orthonormal track frame and let $\gamma(s)$ be a reference curve. A local trough chart is

$$X_{\text{tr}}(s, u) = \gamma(s) + R_{\text{tr}}(s) \begin{pmatrix} 0 \\ u \\ h(s, u) \end{pmatrix}, \quad R_{\text{tr}} = (e_s, e_u, n_{\text{tr}}) \in \text{SO}(3). \quad (1)$$

Its admissible domain $D \subset \mathbb{R}^2$ must satisfy

$$\partial_s X_{\text{tr}} \times \partial_u X_{\text{tr}} \neq 0 \quad \text{on } D. \quad (2)$$

The surface normal is therefore

$$n_\Sigma = \frac{\partial_s X_{\text{tr}} \times \partial_u X_{\text{tr}}}{\|\partial_s X_{\text{tr}} \times \partial_u X_{\text{tr}}\|}.$$

Centerline curvature, banking, transverse trough curvature, and chart regularity are distinct data. A centerline alone does not determine contact geometry.

3 Finite mass realization

Let a rigid sled be represented by masses $m_a > 0$ at body points r_a^B . Its total mass, center of mass, and inertia about the center of mass are

$$M = \sum_{a=1}^n m_a, \quad r_{\text{cm}}^B = \frac{1}{M} \sum_{a=1}^n m_a r_a^B, \quad (3)$$

$$I_B = \sum_{a=1}^n m_a \left(\|\rho_a\|^2 I_3 - \rho_a \rho_a^\top \right), \quad \rho_a = r_a^B - r_{\text{cm}}^B. \quad (4)$$

For world origin $x^I(t)$ and attitude $A_{I \leftarrow B}(t) \in \text{SO}(3)$,

$$x_a^I = x^I + A_{I \leftarrow B} \rho_a.$$

Let $q \in Q$ be local generalized coordinates after a declared reduction, and let $\Phi(q) = (x_1(q), \dots, x_n(q))$. Define

$$G_{ij}(q) = \sum_{a=1}^n m_a \partial_i x_a(q) \cdot \partial_j x_a(q). \quad (5)$$

Proposition 3.1 (When the mass form is a metric). *$G(q)$ is positive semidefinite. It is positive definite if and only if $d\Phi_q$ is injective.*

Proof. For $\xi \in T_q Q$,

$$\xi^\top G(q) \xi = \sum_a m_a \|d\Phi_q(\xi)_a\|^2.$$

□

Thus redundant coordinates produce a degenerate mass form. Calling it a Riemannian metric requires the immersion hypothesis.

4 Unilateral contact and force pullback

For candidate contact j , let $g_j(t, q)$ be the signed gap, with $g_j > 0$ meaning separation. In the smooth force regime:

$$g_j \geq 0, \quad N_j \geq 0, \quad N_j g_j = 0. \quad (6)$$

Let n_j^I be the ice-to-sled unit normal and $P_{T,j} = I - n_j^I (n_j^I)^\top$. The contact force on the sled is

$$F_j^I = N_j n_j^I + T_j^I, \quad T_j^I \cdot n_j^I = 0. \quad (7)$$

If $J_j^I = \partial_q c_j^I$ is the contact-point Jacobian, its generalized force is

$$Q_j = J_j^\top F_j^I \in T_q^* Q.$$

This pullback is required before adding Q_j to other generalized forces.

Let

$$v_{T,j}^I = P_{T,j} (v_{\text{sled},j}^I - v_{\text{ice},j}^I).$$

A Coulomb-type admissibility condition is

$$\|T_j^I\| \leq \mu_j N_j, \quad T_j^I \cdot v_{T,j}^I \leq 0. \quad (8)$$

For sliding, a maximum-dissipation selection is

$$T_j^I = -\mu_j N_j \frac{v_{T,j}^I}{\|v_{T,j}^I\|}.$$

For sticking, $v_{T,j} = 0$ and T_j is set-valued inside the cone.

5 Dynamics and corrected energy balance

In a smooth reduced chart, let

$$L(t, q, \dot{q}) = \frac{1}{2} \dot{q}^\top G(t, q) \dot{q} - V(t, q).$$

The Lagrange-d'Alembert equation is

$$\frac{d}{dt} \partial_{\dot{q}} L - \partial_q L = Q_{\text{act}} + Q_{\text{fric}} + Q_{\text{con}}. \quad (9)$$

With

$$E_L = \partial_{\dot{q}} L \cdot \dot{q} - L,$$

the frame-force interface gives

$$\dot{E}_L = P_{\text{act}} - P_{\text{fric}} + P_{\text{con}} - \partial_t L, \quad (10)$$

where the nonnegative frictional dissipation is

$$\boxed{P_{\text{fric}} = - \sum_j T_j^I \cdot v_{T,j}^I \geq 0.} \quad (11)$$

Corollary 5.1 (Stationary ideal support). *If the track is stationary, normal constraints are ideal, L is autonomous, and no other nonconservative forces act, then between impacts*

$$\dot{E}_L = -P_{\text{fric}} \leq 0.$$

Remark 5.2 (Correction to v1.0). Version v1.0 defined T_j as the tangential force on the sled but wrote $\dot{E} = -P_{\text{fric}}$ with $P_{\text{fric}} = \sum_j T_j \cdot v_{j,\text{rel}}$. Under dissipative friction that sum is nonpositive, so the displayed energy law increased energy. Equation equation (11) defines the loss with the required leading minus sign.

For a moving support, normal reactions can exchange power and P_{con} need not vanish. The same support work must not also be inserted through a reduced time-dependent L .

6 Guarded diagnostics

6.1 Cone utilization

Define

$$u_j = \begin{cases} \|T_j\| / (\mu_j N_j), & \mu_j N_j > 0, \\ 0, & \mu_j N_j = 0 \text{ and } T_j = 0. \end{cases}$$

The cone law excludes $\mu_j N_j = 0, T_j \neq 0$. For a finite candidate set J ,

$$S_{\text{slip}} = 1 - \max_{j \in J} u_j, \quad \max \emptyset := 0.$$

Then $0 \leq S_{\text{slip}} \leq 1$ on admissible states. This is a cone margin, not a probability of slipping.

6.2 Load distribution and entropy

If $N_{\text{tot}} = \sum_{j=1}^K N_j > 0$, set

$$p_j = \frac{N_j}{N_{\text{tot}}}, \quad H_b(N) = - \sum_{j:p_j>0} p_j \log_b p_j, \quad b > 1.$$

For a declared $K \geq 2$, the normalized diagnostic is

$$\overline{H}_b = \frac{H_b(N)}{\log_b K} \in [0, 1].$$

It is independent of the logarithm base. If $N_{\text{tot}} = 0$, the load distribution and entropy are undefined rather than silently set to zero.

6.3 Two-runner imbalance

For $N_L + N_R > 0$,

$$\Delta_N = \frac{|N_R - N_L|}{N_R + N_L} \in [0, 1].$$

The diagnostic is undefined in free flight.

6.4 Geometric demand and banking mismatch

For a centerline with curvature κ and speed v ,

$$a_\kappa = v^2 \kappa$$

has acceleration dimension. If $f_{\text{req}} = m(a - g) \neq 0$, define the unsigned mismatch

$$\Delta_b = \text{atan2}(\|n_{\text{tr}} \times f_{\text{req}}\|, n_{\text{tr}} \cdot f_{\text{req}}) \in [0, \pi].$$

This is a directional diagnostic; it does not solve the contact forces.

7 Observation channels

The hidden record may contain

$$X_t = (\Sigma, \{m_a, r_a^B\}, q, \dot{q}, \mathcal{A}_t, \{c_j, N_j, T_j\}_{j \in \mathcal{A}_t}, \vartheta),$$

where $\mathcal{A}_t$ is the active contact set and ϑ collects constitutive parameters. Useful deterministic channels include

$$\begin{aligned} \mathcal{O}_{\text{cm}}(X_t) &= x_{\text{cm}}^I(t), & \mathcal{O}_{\text{pose}}(X_t) &= A_{I \leftarrow B}(t), \\ \mathcal{O}_{\text{imu}}(X_t) &= (f^B, \Omega^B), & \mathcal{O}_{\text{load}}(X_t) &= (N_j)_{j \in \mathcal{A}_t}, \\ \mathcal{O}_{\text{contact}}(X_t) &= (c_j)_{j \in \mathcal{A}_t}, & \mathcal{O}_{\text{heat}}(X_t) &= P_{\text{fric}}(t). \end{aligned}$$

No equality between these channels is implied.

Proposition 7.1 (Conditional center-of-mass non-identifiability). *Assume the admissible hidden-state class contains two records $X_t^{(1)} \neq X_t^{(2)}$ with the same center-of-mass*

curve but different mass realization or contact record. Then $\mathcal{O}_{\text{cm}}$ is not injective on that class.

Proof. By hypothesis, $\mathcal{O}_{\text{cm}}(X_t^{(1)}) = \mathcal{O}_{\text{cm}}(X_t^{(2)})$ while the arguments differ. $\square$

This proposition is deliberately conditional. Showing that two specific sleds admit the same trajectory under a fixed constitutive model is a separate dynamical existence problem.

8 Reproducibility record

A numerical benchmark is admissible only if it reports:

- (i) the surface chart and its regularity domain;
- (ii) mass points, center of mass, inertia, and coordinate chart;
- (iii) active-set and contact-law solver;
- (iv) ice-motion convention and friction parameters;
- (v) integration method, tolerances, event handling, and convergence study;
- (vi) sensor map, sampling, noise, and estimator;
- (vii) which outputs are solved reactions and which are proxies.

The proxy values in v1.0 are not retained as validated mechanics because no complementarity/contact solver or convergence study was attached.

9 Claim boundary

The strict result is a corrected and typed interface to standard constrained mechanics. It does not claim:

- a predictive ice-runner friction law;
- uniqueness of the contact solution;
- engineering accuracy or safety certification;
- physiological rider limits;
- a new theorem beyond the stated interface propositions.

References

- [1] B. Brogliato, *Nonsmooth Mechanics: Models, Dynamics and Control*, 3rd ed., Springer, 2016. doi:10.1007/978-3-319-28664-8.
- [2] A. M. Bloch, *Nonholonomic Mechanics and Control*, 2nd ed., Springer, 2015. doi:10.1007/978-1-4939-3017-3.
- [3] V. Acary and B. Brogliato, *Numerical Methods for Nonsmooth Dynamical Systems*, Springer, 2008. doi:10.1007/978-3-540-75392-6.

Roller-Coaster Geometry

Framed curves, rider observations, and regular rail offsets

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Abstract

A roller-coaster centerline does not encode vehicle orientation, rail-contact paths, or rider measurements. We define a track as a unit-speed curve $\gamma(s)$, a nonsingular adapted frame $R(s) \in \text{SO}(3)$, and either a prescribed speed profile or a declared dynamical law. With $R^\top R' = [\kappa_R]_\times$, the vehicle angular velocity is $\Omega^B = v\kappa_R$. The inertial acceleration is

$$a^I = v \frac{dv}{ds} T + v^2 T',$$

while an ideal vehicle accelerometer reads the body-frame specific force

$$f^B = R^\top(a^I - g^I).$$

The dimensionless load vector $G = f^B/g_0$ and its scalar norm $G = \|G\|$ are separate quantities. Body-frame jerk includes the frame-rotation correction and is not merely $R^\top \dot{a}$. Ideal rail offsets $\gamma_\pm = \gamma \pm (d/2)e_2$ are regular under the sufficient bound $(d/2)\sup\|e'_2\| < 1$; global nonintersection additionally requires a tubular-neighborhood hypothesis. The labeled length split $\Delta L = L_+ - L_-$ changes sign under rail exchange, whereas $|\Delta L|$ is exchange invariant. Dissipation and moving support work follow the common energy contract. All claims are qualified by their transformation group and model status.

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1 Scope and protocol

An observation protocol λ specifies the world frame, adapted-frame convention, clock, arc-length chart, sensor axes, sampling, calibration, stochastic/noiseless channel, estimator, and uncertainty. The chain

$$X_t \xrightarrow{\mathcal{R}} \tilde{X}_t \xrightarrow{\mathcal{O}_\lambda} Z_t \xrightarrow{\mathcal{K}_\lambda} \text{Law}(Y_t | Z_t) \xrightarrow{\hat{\theta}_\lambda} \hat{\theta}_t$$

keeps the physical state, ideal sensor signal, recorded data, and estimate distinct.

Boundary 1.1 (Two admissible model modes). A prescribed pair (γ, v) defines a kinematic benchmark. A dynamical prediction requires mass, forces, constraints, drive, dissipation, and initial data. The speed may not be simultaneously prescribed and claimed as the output of an unspecified force law.

2 Track as an adapted framed curve

Let

$$\gamma : [0, L] \rightarrow \mathbb{R}^3, \quad \|\gamma'(s)\| = 1,$$

be C^3 , and set $T = \gamma'$. An adapted frame is

$$R(s) = (e_1, e_2, e_3) \in \text{SO}(3), \quad e_1 = T.$$

It may be a Bishop/rotation-minimizing frame with an added roll profile, or any other declared C^2 adapted frame. Unlike the Frenet frame, this definition remains meaningful at points where $\gamma'' = 0$, provided R is supplied smoothly.

Definition 2.1 (Frame strain). The spatial frame strain $\kappa_R^B(s)$ is defined by

$$R^\top R' = [\kappa_R^B]_\times, \quad [\kappa]_\times u = \kappa \times u. \quad (1)$$

It has dimension L^{-1} .

Curvature magnitude is

$$\kappa(s) = \|T'(s)\|.$$

Torsion is defined through the Frenet frame only where $\kappa > 0$ with sufficient regularity. The supplied roll/frame profile remains the primary apparatus orientation.

3 Time kinematics

Let $s = s(t)$ with $v = \dot{s} > 0$. Then

$$\dot{\gamma} = vT, \quad (2)$$

$$a^I = \ddot{\gamma} = \dot{v}T + v^2T' = v \frac{dv}{ds}T + v^2T'. \quad (3)$$

Since

$$R^\top \dot{R} = vR^\top R' = [v\kappa_R^B]_\times,$$

the vehicle-frame angular velocity is

$$\Omega^B = v\kappa_R^B. \quad (4)$$

Let g^I be gravitational acceleration and let $g_0 > 0$ be a declared reference acceleration. The ideal accelerometer specific force is

$$f^B = R^\top(a^I - g^I). \quad (5)$$

Definition 3.1 (Vector and scalar load).

$$G(s) = \frac{f^B(s)}{g_0}, \quad G(s) = \|G(s)\|. \quad (6)$$

G is a dimensionless body-frame vector; G is a dimensionless scalar. They cannot share one quantity identifier.

Definition 3.2 (Body-frame jerk signal). The derivative of the measured specific-force components is

$$j_f^B = \frac{df^B}{dt} = R^\top(\dot{a}^I - \dot{g}^I) - \Omega^B \times f^B. \quad (7)$$

The last term is required because the sensor axes rotate. The inertial jerk $\dot{a}^I$, the derivative of specific force, and the derivative of body components are different observables.

4 Rail offsets and their domain

Let $d > 0$ be the ideal rail gauge and let $e_2(s)$ be the labeled lateral frame direction. Define

$$\gamma_\pm(s) = \gamma(s) \pm \frac{d}{2}e_2(s). \quad (8)$$

Proposition 4.1 (Sufficient local regularity bound). *If*

$$\frac{d}{2} \sup_{s \in [0, L]} \|e'_2(s)\| < 1, \quad (9)$$

then both offset curves in equation (8) are regular.

Proof.

$$\|\gamma'_\pm\| = \|T \pm (d/2)e'_2\| \geq 1 - (d/2)\|e'_2\| > 0.$$

□

The bound is sufficient, not necessary. It prevents a local offset cusp but not a global self-intersection. To interpret the offsets as two distinct rails, the strip map

$$\Psi(s, u) = \gamma(s) + ue_2(s), \quad |u| \leq d/2,$$

must additionally be an embedding (or be restricted to a declared tubular neighborhood).

The rail lengths are

$$L_\pm = \int_0^L \|\gamma'_\pm(s)\| ds.$$

For a labeled frame define

$$\Delta L_{\text{or}} = L_+ - L_-, \quad \Delta L_{\text{abs}} = |L_+ - L_-|.$$

Proposition 4.2 (Transformation law of the split). *Under a static Euclidean isometry applied to the entire framed track, $L_{\pm}$, ΔL_{or} , and ΔL_{abs} are unchanged. Under the label exchange $+ \leftrightarrow -$,*

$$\Delta L_{\text{or}} \mapsto -\Delta L_{\text{or}}, \quad \Delta L_{\text{abs}} \mapsto \Delta L_{\text{abs}}.$$

Thus the signed split is a covariant of an oriented/labeled rail system, not an unqualified invariant.

5 Rolling count and contact qualification

For constant effective wheel radius $r_w > 0$, pure rolling without slip along rail $\gamma_{\pm}$ gives the ideal revolution count

$$N_{\pm}^{\text{ideal}} = \frac{L_{\pm}}{2\pi r_w}.$$

This formula is dimensionless and parameterization independent. It is not an actual wheel count if slip, flange contact, compliance, wear, variable rolling radius, or loss of contact occurs. Those effects require a contact model and a wheel-angle observation channel.

6 Force and dissipation layer

Let F_d^I be the total dissipative spatial force on a car at a declared application point and v_{rel}^I its velocity relative to the corresponding support or medium. The common sign convention is

$$P_{\text{diss}} = -F_d^I \cdot v_{\text{rel}}^I \geq 0.$$

For a stationary ideal rail constraint, the normal reaction performs no work along an admissible motion. With generalized Lagrangian energy E_L ,

$$\dot{E}_L = P_{\text{drive}} - P_{\text{diss}} + P_{\text{con}} - \partial_t L.$$

For a static autonomous track with ideal normal constraints, $P_{\text{con}} = 0$. A launch system, moving track element, or active brake is not covered by that simplification.

Boundary 6.1 (No force from geometry alone). Curvature determines the kinematic acceleration required by a chosen speed. It does not by itself determine wheel loads, structural stress, or loss of contact. Those require a mass/inertia model and solved constraint reactions.

7 Observation loss

Consider the scalar load observation

$$\mathcal{O}_G : (\gamma, R, v) \mapsto G(s) = \frac{\|a^I(s) - g^I\|}{g_0}.$$

Proposition 7.1 (Scalar load is not a complete framed-track observation). *Fix a centerline γ and speed v . Any two smooth adapted frames R_1, R_2 with the same tangent but distinct roll profiles produce the same scalar load G . Hence $\mathcal{O}_G$ is not injective on framed tracks.*

Proof. Equation (3) depends on γ, v , not on the choice of lateral/up axes. Orthogonality gives

$$\|R_i^T(a^I - g^I)\| = \|a^I - g^I\|$$

for both frames, although their body-component vectors generally differ. $\square$

The full vector G , angular velocity, visual flow, contact signals, and known speed add information but are still protocol outputs, not automatically a complete reconstruction of the track.

8 Transformation audit

Object	Transformation group	Status
Arc length and curvature magnitude	static Euclidean isometries	invariant
Frame components G, Ω^B	declared apparatus-axis change	covariant vectors
Scalar load G	apparatus-axis orthogonal change	invariant
Signed rail split ΔL_{or}	rail-label exchange	sign-covariant
Absolute split ΔL_{abs}	rail-label exchange	invariant
Ideal wheel count	curve reparameterization	invariant under orientation-preserving regular reparameterization
Body jerk components	time-dependent apparatus frame	require the correction in equation (7)

9 Reproducibility and claim boundary

A numerical benchmark must report:

- (i) γ, R, v and their differentiability;
- (ii) sampling and the derivative/smoothing method;
- (iii) the bound equation (9) and global strip validation;
- (iv) gravity, sensor axes, and force/contact status;
- (v) numerical convergence.

The v1.0 synthetic numbers are not retained as engineering results: the layout supplied no clearance, structural, contact, or convergence certificate.

This document does not claim:

- a realizable or safe track;
- a multibody wheel-rail solution;

- structural, fatigue, restraint, or human-tolerance validation;
- recovery of a track from one rider signal;
- novelty of the standard curve and accelerometer formulas.

References

- [1] R. L. Bishop, "There is more than one way to frame a curve," *American Mathematical Monthly* 82 (1975), 246–251. doi:10.2307/2319846.
- [2] M. P. do Carmo, *Differential Geometry of Curves and Surfaces*, Prentice-Hall, 1976.
- [3] B. Brogliato, *Nonsmooth Mechanics: Models, Dynamics and Control*, 3rd ed., Springer, 2016. doi:10.1007/978-3-319-28664-8.

Gear Contact Geometry

A typed finite observation system

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Abstract

Two ideal gears with positive tooth counts N_1, N_2 admit an exact finite event model on

$$X_{N_1, N_2} = \mathbb{Z}_{N_1} \times \mathbb{Z}_{N_2}, \quad T_\sigma(i, j) = (i + 1, j + \sigma),$$

where $\sigma = -1$ denotes an external mesh and $\sigma = +1$ an internal mesh under the declared common angular orientation. Every orbit has length $\text{lcm}(N_1, N_2)$, the state space has $\text{gcd}(N_1, N_2)$ orbits, and one orbit covers every ordered tooth pair exactly when the counts are coprime. We separate this exact combinatorial layer from the active contact set, contact ratio, normal load, sliding, dissipation, and material wear. The signed speed ratio

$$\frac{\omega_2}{\omega_1} = \sigma \frac{N_1}{N_2}$$

identifies only the coprime count ratio, not the common scale that controls the cycle decomposition. For a planetary set, uniform subtraction of frame angular speed is formalized as an $(\mathbb{R}, +)$ -action; only a relative-observation protocol passes to its quotient. For a periodic non-circular kinematic model, we define the driver-period return map, its physical lift, invariant measures, and the Poincare translation number. No KAM conclusion is drawn. The result is a strict observation interface and a reproducible finite benchmark, not a new gear-design or tribology theory.

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1 Scope and observation protocol

The system descriptor is

$$d = (N_1, N_2, \sigma, \mathbf{g}, \mathbf{m}, \vartheta),$$

where $N_a \in \mathbb{N}_{>0}$, $\sigma \in \{-1, +1\}$, $\mathbf{g}$ records the declared geometry and frame conventions, $\mathbf{m}$ records the mesh class, and ϑ contains any constitutive parameters. The complete protocol is the typed chain

$$X_t \xrightarrow{\mathcal{R}_{\text{red}}} \tilde{X}_t \xrightarrow{\mathcal{O}_\lambda} Z_t \xrightarrow{\mathcal{K}_\lambda} \text{Law}(Y_t | Z_t) \xrightarrow{\hat{\theta}_\lambda} \hat{\theta}_t.$$

The record λ must state the tooth labels, event clock, reference frames, angular orientation, observation horizon, missed- event and misclassification policy, load calibration, sampling rates, noise model, and estimator.

Boundary 1.1 (Three layers that must not be identified). The pitch-event state x_k , the simultaneous active contact set $\mathcal{A}(t)$, and a material wear field are different objects. The first is finite combinatorics. The second requires tooth geometry and contact ratio. The third additionally requires loads, sliding, material data, lubrication, and a constitutive evolution law.

2 Fixed-axis circular kinematics

Let both angular coordinates be positive counterclockwise in one declared inertial frame. Put

$$\sigma = \begin{cases} -1, & \text{external mesh,} \\ +1, & \text{internal mesh.} \end{cases}$$

For an ideal constant-ratio pair, the phase constraint is

$$h(\theta_1, \theta_2) = N_1\theta_1 - \sigma N_2\theta_2 - \phi_0 = 0 \pmod{2\pi}. \quad (1)$$

On every smooth branch,

$$N_1\omega_1 - \sigma N_2\omega_2 = 0, \quad \rho_{2 \leftarrow 1} := \frac{\omega_2}{\omega_1} = \sigma \frac{N_1}{N_2}, \quad (2)$$

provided $\omega_1 \neq 0$. The direction in $\rho_{2 \leftarrow 1}$ is part of the symbol; writing only “gear ratio” is rejected.

Proposition 2.1 (Ideal mesh-reaction power). *Let $Q_c = \lambda dh$ be the reaction covector of the ideal constraint equation (1). Then its total generalized power vanishes on admissible velocities:*

$$\langle Q_c, (\omega_1, \omega_2) \rangle = \lambda(N_1\omega_1 - \sigma N_2\omega_2) = 0.$$

This statement concerns the ideal constraint reaction. Sliding friction, lubricant shear, impacts, backlash, bearing losses, and profile errors belong to separate force and dissipation terms.

3 Finite pitch-event dynamics

Definition 3.1 (Pitch-event state). Choose an event section that records one reference tooth-pitch crossing per advance of gear 1. The label state is

$$X_{N_1, N_2} := \mathbb{Z}_{N_1} \times \mathbb{Z}_{N_2}.$$

Under equation (2), its ideal event map is

$$T_\sigma(i, j) = (i + 1, j + \sigma) \pmod{(N_1, N_2)}. \quad (3)$$

Theorem 3.2 (Orbit length and cycle decomposition). *Every orbit of T_σ has length*

$$P = \text{lcm}(N_1, N_2).$$

The state space is the disjoint union of

$$g = \text{gcd}(N_1, N_2)$$

directed cycles of length P . A single orbit contains all $N_1 N_2$ ordered tooth pairs if and only if $g = 1$.

Proof. The period is the order of $(1, \sigma)$ in $\mathbb{Z}_{N_1} \times \mathbb{Z}_{N_2}$. It is the least positive k satisfying

$$k = 0 \pmod{N_1}, \quad \sigma k = 0 \pmod{N_2}.$$

Because $\sigma = \pm 1$, this is $\text{lcm}(N_1, N_2)$. Translation acts freely on each orbit, so every orbit has this size. The orbit count is therefore

$$\frac{N_1 N_2}{\text{lcm}(N_1, N_2)} = \text{gcd}(N_1, N_2).$$

□

Corollary 3.3 (Hunting-tooth criterion). *Every tooth of gear 1 is paired with every tooth of gear 2 on one event orbit exactly when $\text{gcd}(N_1, N_2) = 1$.*

The directed graph

$$G_{N_1, N_2, \sigma} = (X_{N_1, N_2}, \{(x, T_\sigma x) : x \in X_{N_1, N_2}\})$$

is consequently a union of equal directed cycles. Label translations $\ell_{a,b}(i, j) = (i + a, j + b)$ commute with T_σ , while $R(i, j) = (i, -j)$ conjugates T_σ to $T_{-\sigma}$. Thus P , g , and the cycle multiset are invariant under tooth relabeling and orientation reversal. The ordered label sequence is not invariant under those transformations.

Boundary 3.4 (Event-model boundary). When transverse or overlap contact ratio exceeds one, several tooth pairs may carry load simultaneously. The map T_σ then records a chosen pitch event, not the full active contact set. Undercutting, interference, flank modification, elastic deflection, and loss of contact are also absent from equation (3).

4 Ratio observation and identifiability

Let

$$\mathcal{O}_{\text{ratio}}(N_1, N_2, \sigma) = \sigma \frac{N_1}{N_2}.$$

This is a deterministic observation of the descriptor, not of an individual tooth-pair state.

Proposition 4.1 (Fiber of the magnitude-ratio observation). *Suppose*

$$|\mathcal{O}_{\text{ratio}}| = \frac{u}{v}, \quad \gcd(u, v) = 1.$$

Then every compatible positive tooth-count pair is uniquely

$$(N_1, N_2) = (cu, cv), \quad c \in \mathbb{N}_{>0}.$$

For this pair,

$$\gcd(N_1, N_2) = c, \quad \text{lcm}(N_1, N_2) = cuv.$$

Hence the speed-ratio magnitude does not identify contact-cycle count or return period.

Proof. The equality $N_1/N_2 = u/v$ gives $vN_1 = uN_2$. Coprimality implies $u \mid N_1$ and $v \mid N_2$, with one common positive multiplier c . The gcd and lcm formulas follow. $\square$

On a bounded design class

$$\mathcal{D}_M = \{(N_1, N_2, \sigma) : 1 \leq N_1, N_2 \leq M\},$$

the observation fiber is finite. On the unbounded class it is countably infinite. Therefore an information-loss statement must declare the descriptor domain; it cannot be inferred from a ratio alone.

Useful deterministic readouts include

$\mathcal{O}_{\text{pair}}(x_k) = (i_k, j_k)$	only with calibrated tooth labels,
$\mathcal{O}_{\text{phase}}(q) = (\theta_1, \theta_2) \pmod{2\pi}$	with declared encoders and frames,
$\mathcal{O}_{\text{ratio}}(d) = \rho_{2 \leftarrow 1}$	with signed orientation,
$\mathcal{O}_{\text{active}}(X_t) = \mathcal{A}(t)$	with contact-resolution policy,
$\mathcal{O}_{\text{load}}(X_t) = (N_\alpha, T_\alpha)_{\alpha \in \mathcal{A}(t)}$	with force calibration.

None of these channels is silently substituted for another.

5 Contact force, sliding, and dissipation

For an active candidate pair $\alpha = (i, j)$, let $g_\alpha(q) \geq 0$ mean separation and let $N_\alpha \geq 0$ be the compressive normal-force magnitude. In the smooth force regime,

$$g_\alpha \geq 0, \quad N_\alpha \geq 0, \quad N_\alpha g_\alpha = 0. \quad (4)$$

Let n_α point from gear 1 toward gear 2, and let $F_\alpha = N_\alpha n_\alpha + T_\alpha$ be the force on gear 2. The force on gear 1 is $-F_\alpha$. With

$$v_{\text{rel},\alpha} = v_{2,\alpha} - v_{1,\alpha}, \quad v_{T,\alpha} = (I - n_\alpha n_\alpha^\top) v_{\text{rel},\alpha},$$

a dissipative tangential law satisfies

$$T_\alpha \cdot v_{T,\alpha} \leq 0, \quad P_{\text{fric},\alpha} := -T_\alpha \cdot v_{T,\alpha} \geq 0. \quad (5)$$

If $J_{\text{rel},\alpha} \dot{q} = v_{\text{rel},\alpha}$, the generalized contact covector is

$$Q_\alpha = J_{\text{rel},\alpha}^\top F_\alpha \in T_q^* Q. \quad (6)$$

This pullback is required before contact forces are combined with other generalized forces.

In a smooth active-set interval, the nonnegative total frictional loss is

$$P_{\text{fric}} = - \sum_{\alpha \in \mathcal{A}(t)} T_{\alpha} \cdot v_{T,\alpha} \geq 0. \quad (7)$$

For a stationary autonomous model with ideal normal constraints, its energy balance has the interface form

$$\dot{E}_L = P_{\text{act}} - P_{\text{fric}} - P_{\text{other}}, \quad P_{\text{other}} \geq 0.$$

Moving supports, time-dependent tooth geometry, or imposed carrier motion require the explicit support-power or $-\partial_t L$ terms of the common mechanics interface.

Boundary 5.1 (Nonsmooth boundary). Contact onset, separation, impact, backlash switching, stick-slip transitions, and inconsistent rigid Coulomb contact require a nonsmooth inclusion or complementarity time-stepper. Existence, uniqueness, and numerical convergence are not proved here.

6 Counts, exposure, dissipation, and wear

For the pitch-event orbit $x_k = T_{\sigma}^k x_0$, define the count

$$C_{ij}(K) = \sum_{k=0}^{K-1} \mathbf{1}_{\{x_k=(i,j)\}}.$$

For a resolved continuous active set, define instead

$$I_{ij}(T) = \int_0^T \mathbf{1}_{\{(i,j) \in \mathcal{A}(t)\}} N_{ij}(t) dt, \quad (8)$$

$$D_{ij}(T) = \int_0^T \mathbf{1}_{\{(i,j) \in \mathcal{A}(t)\}} [-T_{ij}(t) \cdot v_{T,ij}(t)] dt. \quad (9)$$

Here C_{ij} is a count, I_{ij} has impulse dimension, and D_{ij} has energy dimension. They cannot be added or called equivalent.

For any declared nonnegative channel w_{α} with $W = \sum_{\alpha} w_{\alpha} > 0$, put

$$p_{\alpha} = \frac{w_{\alpha}}{W}, \quad H_b(p) = - \sum_{\alpha: p_{\alpha} > 0} p_{\alpha} \log_b p_{\alpha}, \quad b > 1.$$

If $W = 0$, the normalized distribution and its entropy are undefined.

Proposition 6.1 (Full-orbit event entropy). *For one full pitch-event orbit, the empirical distribution is uniform on $P = \text{lcm}(N_1, N_2)$ states. Therefore*

$$H_b = \log_b P.$$

Relative to the uniform distribution on all $N_1 N_2$ ordered pairs, the reachability support defect is

$$\Delta_b = \log_b(N_1 N_2) - \log_b P = \log_b \gcd(N_1, N_2). \quad (10)$$

For $N_1 N_2 > 1$, the normalized defect

$$\overline{\Delta} = \frac{\log_b \gcd(N_1, N_2)}{\log_b(N_1 N_2)}$$

is independent of b .

Neither C_{ij} , I_{ij} , D_{ij} , nor their entropies are a wear law. A wear prediction additionally needs a declared constitutive map with material hardness, local pressure, sliding distance, lubrication, temperature, surface evolution, and uncertainty.

7 Fixed-axis gear trains

Let a gear train have angular-velocity vector $\omega \in \mathbb{R}^m$. For every fixed-axis mesh edge $e = (a, b)$, use

$$N_a \omega_a - \sigma_e N_b \omega_b = 0.$$

For every rigid compound-shaft relation use $\omega_a - \omega_b = 0$. Stacking these rows gives

$$C\omega = 0. \quad (11)$$

The admissible ideal velocity space is $\ker C$; this formulation does not assume in advance that it is one-dimensional.

Along an oriented simple mesh path $v_0 \rightarrow v_1 \rightarrow \cdots \rightarrow v_m$,

$$\frac{\omega_{v_m}}{\omega_{v_0}} = \prod_{r=1}^m \sigma_{e_r} \frac{N_{v_{r-1}}}{N_{v_r}}, \quad (12)$$

when all relevant velocities are nonzero. A closed ideal loop admits a nonzero compatible motion only if the corresponding signed product is one. Idler cancellation in a terminal ratio does not erase the idler's tooth-contact history, bearing loads, inertia, or losses.

8 Planetary frames as an explicit group action

Let $\omega = (\omega_s, \omega_r, \omega_c) \in \mathbb{R}^3$ denote sun, ring, and carrier angular velocities in one common axial orientation. A change to a frame rotating at angular speed α acts by

$$G_\alpha(\omega) = \omega - \alpha \mathbf{1}, \quad \mathbf{1} = (1, 1, 1). \quad (13)$$

It is an action of $(\mathbb{R}, +)$, because

$$G_\alpha \circ G_\beta = G_{\alpha+\beta}, \quad G_0 = \text{id}, \quad G_\alpha^{-1} = G_{-\alpha}.$$

Theorem 8.1 (Relative quotient and Willis relation). *The map*

$$\pi : \mathbb{R}^3 \longrightarrow \mathbb{R}^2, \quad \pi(\omega) = (\omega_s - \omega_c, \omega_r - \omega_c)$$

is constant exactly on the G_α -orbits and induces

$$\mathbb{R}^3 / \text{span}\{\mathbf{1}\} \cong \mathbb{R}^2.$$

For a simple ideal planetary set, the Willis relation is the quotient line

$$N_s(\omega_s - \omega_c) + N_r(\omega_r - \omega_c) = 0, \quad (14)$$

or, when the denominator is nonzero,

$$\frac{\omega_s - \omega_c}{\omega_r - \omega_c} = -\frac{N_r}{N_s}.$$

Proof. $\pi(\omega) = \pi(\omega')$ exactly when $\omega' - \omega = \alpha \mathbf{1}$ for some α . Thus its fibers are the action orbits and its kernel is $\text{span}\{\mathbf{1}\}$. Equation (14) depends only on $\pi(\omega)$. $\square$

Uniform frame subtraction is therefore a physical change of representation. It becomes an observational gauge redundancy only for a protocol of the form $\mathcal{O}_{\text{rel}} = f \circ \pi$, which deliberately discards common absolute spin. A laboratory encoder that measures ω_c separately does not pass to this quotient.

If the ring is fixed in the laboratory, $\omega_r = 0$, then

$$\omega_c = \frac{N_s}{N_s + N_r} \omega_s.$$

For $N_s = 30$, $N_r = 78$, and $\omega_s = 1 \text{ rad s}^{-1}$, this gives $\omega_c = 5/18 \text{ rad s}^{-1}$. For a standard equal-module sun-planet-ring geometry, the separate compatibility condition is $N_r = N_s + 2N_p$, giving $N_p = 24$ in this example.

9 Periodic non-circular kinematics

Let $r_1, r_2 : S^1 \rightarrow (0, \infty)$ be positive C^1 effective pitch radii, and assume the driver angle θ_1 is monotone. A kinematic rolling model is

$$\frac{d\theta_2}{d\theta_1} = \sigma \frac{r_1(\theta_1)}{r_2(\theta_2)}. \quad (15)$$

This equation is a model input. It does not prove manufacturability, absence of interference, force closure, or elastic stability.

For the lifted solution $\tilde{\theta}_2(\theta_1; x)$ with initial value x , define the one-driver-turn return lift

$$\tilde{F}(x) = \tilde{\theta}_2(2\pi; x), \quad \tilde{F}(x + 2\pi) = \tilde{F}(x) + 2\pi. \quad (16)$$

Uniqueness of the scalar C^1 flow makes the induced $F : S^1 \rightarrow S^1$ orientation-preserving, even when the output angle decreases along the driven orbit.

Theorem 9.1 (Translation number with declared lift). *For the orientation-preserving circle homeomorphism F and the physical lift $\tilde{F}$, the limit*

$$\tau(\tilde{F}) = \lim_{n \rightarrow \infty} \frac{\tilde{F}^n(x) - x}{2\pi n} \quad (17)$$

exists and is independent of x . Replacing the lift by $\tilde{F} + 2\pi k$ changes τ by k ; hence $\rho(F) = \tau(\tilde{F}) \bmod \mathbb{Z}$ is lift-independent. If μ is any F -invariant Borel probability measure, then

$$\tau(\tilde{F}) = \frac{1}{2\pi} \int_{S^1} (\tilde{F}(\tilde{x}) - \tilde{x}) d\mu(x),$$

where the displacement is the well-defined periodic function induced by the selected lift.

For constant radii, $\tilde{F}(x) = x + 2\pi\sigma r_1/r_2$, so $\tau = \sigma r_1/r_2$. Elasticity, backlash, impact, or controller hysteresis may destroy the single-valued homeomorphism hypothesis. No KAM persistence, Diophantine stability, or locking theorem follows from equation (15) alone.

10 Reproducible benchmarks and machine gates

(N_1, N_2)	gcd	lcm	orbit fraction	$ \rho_{2 \leftarrow 1} $
(12, 25)	1	300	1	12/25
(12, 18)	6	36	1/6	2/3
(2, 3)	1	6	1	2/3
(4, 6)	2	12	1/2	2/3

The release tests verify the orbit theorem by exhaustive enumeration over a finite tooth-count range, ratio fibers, graph conjugacy, explicit entropy bases, contact complementarity, dissipation signs, force-power covariance, train path products, the planetary group action and quotient, the Willis example, and translation-number convergence for benchmark circle maps.

Gate	Required declaration	Rejected inference
Event semantics	pitch section and label convention	event pair equals simultaneous contact set
Ratio direction	ω_2/ω_1 , sign, and frame	unnamed “gear ratio”
Frame action	G_α and observed quotient	calling every carrier subtraction a gauge
Contact force	gap sign, active set, force-on-body convention	unsigned or double-counted reactions
Dissipation	$P_{\text{diss}} \geq 0$ with relative velocity	event count used as energy loss
Exposure	count, impulse, and energy kept dimensionally distinct	load count called wear
Entropy	base, support, positive-total guard	log without base or zero normalization
Rotation number	return map, lift, and hypotheses	KAM language from a local ratio equation

11 Claim boundary

Statement	Status	Required hypotheses
Finite orbit decomposition	theorem	positive integer tooth counts; pitch-event map equation (3)
Ratio fiber	proposition	exact constant ratio; positive tooth counts
Ideal reaction power	proposition	smooth ideal holonomic constraint
Willis quotient	theorem	common axis convention; ideal simple planetary kinematics
Translation number	classical theorem	orientation-preserving circle homeomorphism and selected lift
Contact friction	constitutive model	declared active set, force convention, smooth force regime
Wear prediction	absent	requires an external validated material-evolution model

This revision establishes no new gear-design standard, load-capacity formula, lubrication model, lifetime law, or safety result. Its exact mathematics is elementary finite-group dynamics plus classical circle rotation theory. The contribution is a strict separation of descriptor, event, contact, frame, dissipation, and observation layers suitable for GO Core validation.

References

- [1] International Organization for Standardization, *ISO 21771-1:2024, Cylindrical involute gears and gear pairs – Part 1: Concepts and geometry*, 2024. ISO 21771-1:2024.
- [2] F. L. Litvin and A. Fuentes, *Gear Geometry and Applied Theory*, 2nd ed., Cambridge University Press, 2004. doi:10.1017/CBO9780511547126.
- [3] F. L. Litvin, A. Fuentes-Aznar, I. Gonzalez-Perez, and K. Hayasaka, *Noncircular Gears: Design and Generation*, Cambridge University Press, 2009, ISBN 978-0-521-76170-3.
- [4] V. I. Arnold, *Mathematical Methods of Classical Mechanics*, 2nd ed., Springer, 1989. doi:10.1007/978-1-4757-2063-1.
- [5] J. E. Marsden and T. S. Ratiu, *Introduction to Mechanics and Symmetry*, 2nd ed., Springer, 1999. doi:10.1007/978-0-387-21792-5.

Conical Intersections as Typed Spectral Singularities

Eigenprojectors, holonomy, nonadiabatic geometry, and observation maps

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Strict GO Core reference revision v1.1 — 28 July 2026

Status. This document supersedes the bilingual working note v1.0. It is an application-layer reference migrated to GO Core v0.2. It does not propose a new electronic-structure method, nonadiabatic dynamics algorithm, conical-intersection theorem, or optical-caustic equivalence. The internal contribution is a typed separation of nuclear coordinates, energy-valued branching coordinates, gauge-invariant projectors, local eigenvector sections, dynamics, and finite-resolution observation.

Abstract

Let q denote physical nuclear coordinates and let $H_e(q)$ be an electronic Hamiltonian family with an isolated two-state spectral cluster. Near a generic real two-state degeneracy, the linear branching-plane model is

$$H_{\text{eff}}(q) = E_0(q)I + x(q)\sigma_z + y(q)\sigma_x, \quad x = g_\mu \delta q^\mu, \quad y = h_\mu \delta q^\mu,$$

where x, y are energies, not nuclear lengths. The seam is the regular preimage $x = y = 0$, hence has codimension two when $\text{rank } D_q(x, y) = 2$. On the punctured branching plane,

$$E_\pm = E_0 \pm r, \quad P_\pm = \frac{1}{2} \left(I \pm \frac{x\sigma_z + y\sigma_x}{r} \right), \quad r = \sqrt{x^2 + y^2}.$$

The unordered spectrum and the two-state cluster remain meaningful at the degeneracy, but neither rank-one projector has a unique limit there. Eigenvectors are only gauge-dependent local sections. A loop of winding number w has real-line holonomy $(-1)^w$ and Berry phase πw modulo 2π . In the energy coordinates, the gauge-invariant quantum metric is

$$g^{(E)} = \frac{1}{4r^4} \begin{pmatrix} y^2 & -xy \\ -xy & x^2 \end{pmatrix}, \quad \text{Tr } g^{(E)} = \frac{1}{4r^2}.$$

Its pullback by $D_q(x, y)$ supplies the correctly dimensioned physical-coordinate coupling. A closing gap invalidates a uniform adiabatic estimate, but does not determine a transition probability; that requires a nuclear path, speed, initial state, and dynamical model. Finally, finite spectral resolution can certify only an unresolved near-degeneracy unless uncertainty and identifiability conditions are supplied. The strict object is a spectral-discriminant singularity. “Observation caustic” is retained only as a qualified program-level analogy.

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1 Scope and typed setting

1.1 Electronic family and spectral cluster

Let $U \subset \mathbb{R}^f$ be a nuclear-coordinate chart. For the strict finite-dimensional reference, $H_e : U \rightarrow \text{Herm}(n)$ is C^2 . The two electronic levels of interest form a cluster separated from the remaining spectrum by a positive external gap on a neighborhood $V \subset U$. The Riesz projector onto that cluster is

$$P_{\text{cl}}(q) = \frac{1}{2\pi i} \oint_{\Gamma} (z - H_e(q))^{-1} dz, \quad (1)$$

where one fixed contour Γ encloses precisely the two cluster eigenvalues. Under these hypotheses P_{cl} is C^2 , even when the two eigenvalues cross internally.

For an unbounded electronic operator, a common dense domain, self-adjointness, resolvent regularity, and an isolated spectral cluster must be stated separately. The finite matrix reference does not silently prove those operator-theoretic hypotheses.

1.2 Physical and energy-valued coordinates

Choose $q_* \in U$, set $\delta q = q - q_*$, and write the linear two-state reduction after a fixed local basis choice as

$$H_{\text{eff}}(q) = E_0(q)I + x(q)\sigma_z + y(q)\sigma_x + O(\|\delta q\|^2), \quad x = g_\mu \delta q^\mu, \quad y = h_\mu \delta q^\mu. \quad (2)$$

Here q^μ is a physical nuclear coordinate, while g_μ, h_μ are energy-gradient covectors. Thus

$$[q^\mu] = L, \quad [g_\mu] = [h_\mu] = EL^{-1}, \quad [x] = [y] = E$$

in the declared length chart. A mass-weighted or curvilinear chart is allowed only with its own coordinate dimensions and Jacobian.

Boundary 1.1 (Coordinate firewall). The notation $H = Q_1\sigma_z + Q_2\sigma_x$ is dimensionally valid only if Q_1, Q_2 are declared energy-valued branching coordinates. It is not valid when the same symbols denote unscaled nuclear lengths.

2 Spectral discriminant and codimension

The exact benchmark drops $E_0 I$ and higher-order terms:

$$H(x, y) = x\sigma_z + y\sigma_x = \begin{pmatrix} x & y \\ y & -x \end{pmatrix}, \quad r = \sqrt{x^2 + y^2}. \quad (3)$$

Proposition 2.1 (Conical spectrum). *For $(x, y) \neq (0, 0)$,*

$$E_{\pm} = \pm r, \quad \Delta := E_+ - E_- = 2r. \quad (4)$$

The repeated eigenvalue occurs precisely at $x = y = 0$.

Proof. $\det(H - EI) = E^2 - x^2 - y^2$, so $E = \pm r$. Equality of the two eigenvalues is equivalent to $r = 0$. $\square$

Let $F = (x, y) : U \rightarrow \mathbb{R}^2$ and $S = F^{-1}(0)$.

Proposition 2.2 (Generic real seam). *If $\text{rank } D_q F(q_*) = 2$, then S is a C^1 submanifold of codimension two near q_* . Its tangent space is*

$$T_{q_*} S = \ker D_q F(q_*).$$

Proof. This is the regular-value theorem applied to F . $\square$

Boundary 2.3 (Symmetry class matters). A general complex Hermitian two-state matrix contains three independent traceless coefficients; its generic degeneracy has codimension three. Symmetry-enforced crossings, spin-orbit coupling, Kramers degeneracy, and higher multiplicity require different discriminants. Codimension two is asserted only for the declared real two-state class with a transverse parameter map.

3 Projectors, eigenlines, and gauge

On $\mathbb{R}^2 \setminus \{0\}$, define the gauge-invariant rank-one projectors

$$P_{\pm}(x, y) = \frac{1}{2} \left(I \pm \frac{x\sigma_z + y\sigma_x}{r} \right). \quad (5)$$

Proposition 3.1 (No rank-one extension through the seam). *Neither P_+ nor P_- has a continuous extension to $(x, y) = (0, 0)$. The cluster projector $P_{\text{cl}} = P_+ + P_- = I_2$ does extend smoothly.*

Proof. Along $y = 0$, the limits of P_- as $x \rightarrow 0^+$ and $x \rightarrow 0^-$ are respectively $\text{diag}(0, 1)$ and $\text{diag}(1, 0)$. They differ. The sum is identically I_2 . $\square$

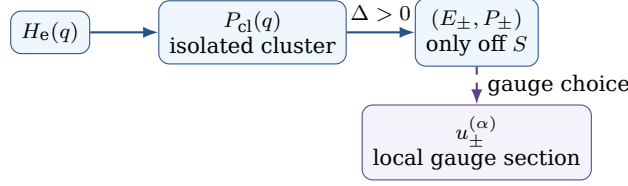
Write $x = r \cos \phi$, $y = r \sin \phi$. One real local lift is

$$u_-(\phi) = \begin{pmatrix} -\sin(\phi/2) \\ \cos(\phi/2) \end{pmatrix}, \quad u_+(\phi) = \begin{pmatrix} \cos(\phi/2) \\ \sin(\phi/2) \end{pmatrix}. \quad (6)$$

It satisfies $u_{\pm}(\phi + 2\pi) = -u_{\pm}(\phi)$. A local eigenvector is not the output of a global gauge-invariant observation map. Under

$$u_a(q) \mapsto e^{i\chi_a(q)} u_a(q),$$

the projector $P_a = |u_a\rangle\langle u_a|$ is unchanged.



4 Berry holonomy without a global eigenvector

Let $C : [0, 1] \rightarrow \mathbb{R}^2 \setminus \{0\}$ be a closed loop and let $w(C, 0) \in \mathbb{Z}$ be its winding number around the origin.

Proposition 4.1 (Real-line holonomy). *For the model equation (3), parallel transport of either real eigenline around C has holonomy*

$$\text{Hol}(C) = (-1)^{w(C,0)}, \quad \gamma(C) = \pi w(C, 0) \pmod{2\pi}. \quad (7)$$

Proof. The polar angle changes by $2\pi w$. Equation (6) therefore changes the real lift by $(-1)^w$. The ray returns to itself, while the lifted line has the stated holonomy. $\square$

A gauge-invariant discrete diagnostic for a sufficiently fine loop $q_0, \dots, q_{N-1}$, with nonzero neighboring overlaps, is

$$W_N = \prod_{j=0}^{N-1} \frac{\langle u_j, u_{j+1} \rangle}{|\langle u_j, u_{j+1} \rangle|}, \quad u_N = u_0, \quad \gamma_N = \arg W_N. \quad (8)$$

Local phase factors telescope in the closed product. In a real gauge, $W_N = -1$ for odd winding and $+1$ for even winding.

Boundary 4.2 (Bundle and curvature boundary). Over a punctured plane, a complex eigenline can be represented by a single-valued complex section, but the Berry connection may have nontrivial holonomy. In the real model, the real line over a loop of odd winding is the nontrivial real line bundle. The smooth Berry curvature vanishes away from the origin in this two-dimensional real model; the phase must not be attributed to an ordinary nonzero local curvature density on the punctured plane.

5 Derivative coupling and quantum metric

For local orthonormal sections, define

$$d_{ab}^{(q)} = \langle u_a, \nabla_q u_b \rangle.$$

For $a \neq b$, a gauge change multiplies d_{ab} by a phase; therefore d_{ab} itself is gauge covariant, while its norm and the quantum metric are gauge invariant.

In the energy coordinates $\xi = (x, y)$, the gauge equation (6) gives

$$d_{+-}^{(E)} = -\frac{1}{2} \nabla_\xi \phi = \frac{1}{2r^2} \begin{pmatrix} y \\ -x \end{pmatrix}, \quad \|d_{+-}^{(E)}\| = \frac{1}{2r}. \quad (9)$$

Its dimension is E^{-1} , not L^{-1} .

The rank-one quantum metric is

$$g_{\mu\nu}^{(E)} = \text{Re} \langle \partial_\mu u_-, (I - P_-) \partial_\nu u_- \rangle = \frac{1}{2} \text{Tr}(\partial_\mu P_- \partial_\nu P_-) = \frac{1}{4r^4} \begin{pmatrix} y^2 & -xy \\ -xy & x^2 \end{pmatrix}_{\mu\nu}. \quad (10)$$

Hence

$$\text{Tr } g^{(E)} = \frac{1}{4r^2} = \|d_{+-}^{(E)}\|^2. \quad (11)$$

Let $J = D_q(x, y)$, with entries of dimension EL^{-1} in the declared length chart. The physical-coordinate quantities are the pullbacks

$$d_{+-}^{(q)} = J^\top d_{+-}^{(E)}, \quad g^{(q)} = J^\top g^{(E)} J, \quad (12)$$

so $[d_{+-}^{(q)}] = L^{-1}$ and $[g^{(q)}] = L^{-2}$. The scalar $1/(2r)$ is therefore not a coordinate-independent physical coupling magnitude; it belongs to the Euclidean metric of the chosen energy-coordinate plane.

6 Born-Oppenheimer reduction and dynamics firewall

For a local adiabatic expansion

$$\Psi(q, r_e) = \sum_a \chi_a(q) u_a(q; r_e),$$

the nuclear kinetic operator contains

$$(T_N)_{ab} = - \sum_A \frac{\hbar^2}{2M_A} \left[\delta_{ab} \nabla_A^2 + 2d_{ab}^{(A)} \cdot \nabla_A + \tau_{ab}^{(A)} \right], \quad \tau_{ab}^{(A)} = \langle u_a, \nabla_A^2 u_b \rangle. \quad (13)$$

Under a coordinate-dependent unitary change within the retained cluster, covariance belongs to the complete matrix operator. Discarding the off-diagonal terms is an approximation whose uniform validity fails near an internal gap closure.

For a prescribed path $q(t)$, a local adiabatic diagnostic is

$$\eta_{ab}(t) = \frac{\hbar \left| \langle u_a, \dot{H}_e u_b \rangle \right|}{|E_b - E_a|^2}. \quad (14)$$

The condition $\eta_{ab} \ll 1$ is a model-dependent sufficiency diagnostic, not a universal error theorem. At a gap closure, the uniform-gap hypothesis of standard adiabatic estimates fails.

Model 6.1 (Landau-Zener control). For

$$H_{\text{LZ}}(t) = vt \sigma_z + b \sigma_x, \quad v > 0, \quad b > 0,$$

with evolution from $t = -\infty$ to $+\infty$, the exact diabatic survival probability is

$$P_D = \exp \left(-\frac{\pi b^2}{\hbar v} \right). \quad (15)$$

This control has minimum gap $2b$, but changing the sweep rate v changes P_D . Thus the static gap, Berry phase, or derivative coupling alone does not determine a molecular branching ratio. Wavepacket shape, masses, path, velocity, decoherence, additional states, and the measurement channel are separate inputs.

7 Observation chain and finite-resolution identifiability

The strict factorization is

$$(H_{\text{mol}}, \Psi) \xrightarrow{\mathcal{S}_{\text{cl}}} (H_{\text{cl}}(q), \chi(q)) \xrightarrow{\mathcal{P}_{\text{ad}}} (E_{\pm}(q), P_{\pm}(q), \chi_{\pm}(q)) \xrightarrow{\mathcal{O}_{\lambda}} Z_{\lambda} \xrightarrow{\mathcal{K}_{\lambda}} \text{Law}(Y_{\lambda} \mid Z_{\lambda}) \xrightarrow{\hat{\Theta}_{\lambda}} \hat{\Theta}. \quad (16)$$

The rank-one adiabatic splitting $\mathcal{P}_{\text{ad}}$ is defined only off the seam. A local eigenvector section is an additional gauge choice, not an observation. A spectroscopic or product channel $\mathcal{O}_{\lambda}$ does not generally reveal the full projector field or Berry holonomy.

Definition 7.1 (Spectral-discriminant singularity). Let $\mathcal{D} \subset \text{Herm}(n)$ be the repeated-eigenvalue discriminant. A conical intersection of the declared real two-state class is a transverse point or seam in

$$S = H_{\text{eff}}^{-1}(\mathcal{D})$$

for which the ordered rank-one eigenprojector map fails to extend through S and the leading eigenvalue splitting is conical.

This is not the same as requiring $\text{rank } DH_{\text{eff}}$ to drop. The singular map is the eigen-decomposition at the discriminant, not necessarily the parameter-to-Hamiltonian map. For a classical optical caustic, the reusable criterion is rank loss of a ray projection; a determinant criterion is valid only in a square local chart.

Boundary 7.2 (Qualified use of “observation caustic”). The term may be used as a GO analogy for singularity of a reduced description. It is not a theorem that conical intersections and optical caustics are equivalent singularities, and no Arnold classification of Lagrangian-map caustics is imported here.

Suppose measured energies are

$$Y_{\pm} = E_{\pm} + \varepsilon_{\pm}, \quad \mathbb{E}[\varepsilon_{\pm}] = 0.$$

Then the measured gap $\hat{\Delta} = Y_{+} - Y_{-}$ has variance

$$\sigma_{\hat{\Delta}}^2 = \sigma_{+}^2 + \sigma_{-}^2 - 2 \text{Cov}(\varepsilon_{+}, \varepsilon_{-}). \quad (17)$$

With instrumental resolution $\epsilon_E \geq 0$, a rule such as

$$|\hat{\Delta}| \leq \kappa \sigma_{\Delta} + \epsilon_E \quad (18)$$

classifies an *unresolved near-degeneracy*. It does not prove an exact degeneracy, conical topology, odd Berry holonomy, or a nonadiabatic transition. Those conclusions require additional sampling, model class, path, and identifiability assumptions.

8 Gap-opening control and reproducible benchmarks

The complex Hermitian control

$$H_{\delta}(x, y) = x\sigma_z + y\sigma_x + \delta\sigma_y, \quad \delta \neq 0, \quad (19)$$

has

$$E_{\pm}^{\delta} = \pm \sqrt{x^2 + y^2 + \delta^2}, \quad \Delta_{\delta} = 2\sqrt{r^2 + \delta^2} \geq 2|\delta|. \quad (20)$$

It opens a gap on the x, y plane by leaving the real-symmetric class. It is therefore a symmetry-breaking control, not the same real conical-intersection model with a harmless numerical regularizer.

For the positively oriented circle $x = R \cos \phi$, $y = R \sin \phi$, and the orientation convention used in the release, the lower-state phase modulo 2π has magnitude

$$\gamma_\delta(R) = \pi \left(1 - \frac{\delta}{\sqrt{R^2 + \delta^2}} \right). \quad (21)$$

It tends to π as $\delta \rightarrow 0^+$ and to zero as $\delta/R \rightarrow +\infty$; unlike the real sign holonomy, it is not quantized for $\delta \neq 0$.

Benchmark	Exact value	Purpose
$H(3, 4)$, energy units	$E_\pm = \pm 5$, $\Delta = 10$	Conical spectrum and dimensions
$r = 0.25$, energy units	$\ d_{+-}^{(E)}\ = 2$	Energy-coordinate coupling scaling
Unit circle, winding 1	$W_N = -1$, $\gamma = \pi$	Gauge-invariant discrete holonomy
Unit circle, winding 2	$W_N = +1$, $\gamma = 0 \bmod 2\pi$	Parity of real-line holonomy
$R = 1, \delta = 1$	$\gamma_\delta = \pi(1 - 1/\sqrt{2})$	Nonquantized gapped control
$\hbar v = b^2$	$P_D = e^{-\pi}$	Dimensionless dynamical exponent

Table 1: Reference values implemented by the P8 regression suite.

The executable release additionally checks projector idempotence and orthogonality, absence of a rank-one limit, cluster smoothness, gauge invariance of the overlap product, quantum-metric identities, coordinate pullback dimensions, the Landau-Zener formula, covariance in equation (17), and zero-gap guards.

9 Protocol requirements and claim register

A complete conical-intersection observation record states at least:

1. nuclear-coordinate chart, coordinate dimensions, reference geometry, and frame;
2. electronic Hamiltonian class, domain assumptions, basis, symmetry class, and retained spectral cluster;
3. external cluster gap, active-state multiplicity, and degeneracy/discriminant convention;
4. physical-to-branching Jacobian $D_q(x, y)$ and its units;
5. whether outputs are unordered eigenvalues, ordered energies, projectors, local eigenvectors, or a diabatic matrix;
6. gauge convention, loop orientation, winding rule, and overlap threshold for a discrete Berry phase;
7. nuclear path, speed, masses, initial state, and dynamical approximation for every transition claim;
8. spectroscopic or product observation map, resolution, stochastic kernel, covariance, and detection threshold;
9. estimator, uncertainty statement, sampling region, and identifiability class;
10. explicit boundary between exact normal-form results, asymptotic diagnostics, numerical evidence, and chemical interpretation.

The strict program-level conclusion is that a conical intersection is a transverse encounter with a spectral discriminant for which the individual adiabatic rank-one splitting becomes singular while the retained two-state cluster can remain regular. Gauge, dynamics, and measurement are separate typed layers.

Claim	Status	Scope
Double-cone spectrum	Proposition	Exact real two-state normal form
Codimension-two seam	Proposition	rank $D_q(x, y) = 2$, real two-state class
No rank-one projector extension	Proposition	Punctured branching plane approaching the origin
Berry sign holonomy	Proposition	Closed loop with declared winding, real eigenline
$\text{Tr } g^{(E)} = 1/(4r^2)$	Proposition	Euclidean energy-coordinate metric
Gap closure fixes transition probability	Rejected	Dynamics and initial data are additionally required
Finite unresolved gap proves a CI	Rejected	Resolution alone does not identify topology
CI equals an optical caustic	Rejected	Only a qualified observation-reduction analogy
New quantum-chemical method or theorem	Rejected	Not claimed

References

- [1] M. Born and R. Oppenheimer, “Zur Quantentheorie der Molekeln,” *Annalen der Physik* 389, 457–484 (1927), doi:10.1002/andp.19273892002.
- [2] H. C. Longuet-Higgins, U. Oepik, M. H. L. Pryce, and R. A. Sack, “Studies of the Jahn-Teller effect. II. The dynamical problem,” *Proceedings of the Royal Society A* 244, 1–16 (1958), doi:10.1098/rspa.1958.0022.
- [3] G. Herzberg and H. C. Longuet-Higgins, “Intersection of potential energy surfaces in polyatomic molecules,” *Discussions of the Faraday Society* 35, 77–82 (1963), doi:10.1039/DF9633500077.
- [4] C. A. Mead and D. G. Truhlar, “On the determination of Born-Oppenheimer nuclear motion wave functions including complications due to conical intersections and identical nuclei,” *Journal of Chemical Physics* 70, 2284–2296 (1979), doi:10.1063/1.437734.
- [5] M. V. Berry, “Quantal phase factors accompanying adiabatic changes,” *Proceedings of the Royal Society A* 392, 45–57 (1984), doi:10.1098/rspa.1984.0023.
- [6] C. Zener, “Non-adiabatic crossing of energy levels,” *Proceedings of the Royal Society A* 137, 696–702 (1932), doi:10.1098/rspa.1932.0165.
- [7] T. Kato, *Perturbation Theory for Linear Operators*, Springer, corrected printing of the second edition, 1995.
- [8] D. R. Yarkony, “Diabolical conical intersections,” *Reviews of Modern Physics* 68, 985–1013 (1996), doi:10.1103/RevModPhys.68.985.

Quantum Chemistry as a Typed Inference Stack

Many-electron states, reduced data, orbital gauge, nuclear geometry, and spectra

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Strict GO Core reference revision v1.1 — 28 July 2026

Status. This document supersedes the bilingual working note v1.0 and is an application-layer reference migrated to GO Core v0.2. It does not introduce a new electronic-structure method, density functional, orbital interpretation, bond definition, or spectroscopic inversion theorem. Its internal contribution is a typed separation of the exact electron-nuclear state, clamped-nuclei electronic models, reduced density data, representation gauge, nuclear reduction, ideal observables, and finite-resolution measurement.

Abstract

Quantum chemistry is not described by a single composable chain $\Psi \rightarrow n \rightarrow E(R) \rightarrow R_* \rightarrow$ chemical behavior. Those arrows have different domains, hypotheses, and information content. This reference replaces that chain by a typed graph. The exact nonrelativistic molecular state depends on electronic and nuclear coordinates. A clamped-nuclei Hamiltonian family $H_e(R)$ generates potential-energy surfaces only after a Born-Oppenheimer reduction. For an N -electron density operator Γ_N , the one-body reduced density matrix $\gamma = N \text{Tr}_{2\dots N} \Gamma_N$ satisfies $\text{Tr} \gamma = N$, and the spatial density integrates to N , but this reduction is non-injective. Occupied-orbital rotations leave a Slater determinant ray and its one-body projector invariant, so localized orbitals, hybridizations, and active-space populations require an explicit representation protocol. At a stationary nuclear geometry, the mass-weighted Hessian has eigenvalues $\lambda_k = \omega_k^2$, not frequencies; five rigid zero modes for a linear molecule or six for a nonlinear molecule must be removed in the isolated, unconstrained case. An ideal transition spectrum depends on energies, populations, a transition operator, and a line profile, while measured data additionally depend on the instrument channel and noise. The resulting formalism preserves the useful observation-geometry interpretation but forbids claims of canonical orbital observability, unique bond projection, exact degeneracy from finite resolution, or chemistry determined by energy differences alone.

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1 Scope, state spaces, and unit firewall

Let N_e electrons and N_n nuclei be described in a declared inertial frame. Electronic spin coordinates are denoted by $x_i = (\mathbf{r}_i, \sigma_i)$, and nuclear Cartesian coordinates by $R = (\mathbf{R}_1, \dots, \mathbf{R}_{N_n})$. Ignoring relativity, external fields, nuclear spin, and finite nuclear size, the nonrelativistic Coulomb Hamiltonian in atomic units is

$$\begin{aligned} \hat{H}_{\text{mol}} = & -\frac{1}{2} \sum_{i=1}^{N_e} \nabla_i^2 - \sum_{A=1}^{N_n} \frac{1}{2M_A} \nabla_A^2 - \sum_{i,A} \frac{Z_A}{r_{iA}} \\ & + \sum_{i<j} \frac{1}{r_{ij}} + \sum_{A<B} \frac{Z_A Z_B}{R_{AB}}. \end{aligned} \quad (1)$$

The nuclear masses M_A are expressed in electron-mass units. A full stationary state is

$$\Psi = \Psi(x_1, \dots, x_{N_e}; R_1, \dots, R_{N_n}), \quad \hat{H}_{\text{mol}} \Psi = E \Psi. \quad (2)$$

Writing only electronic arguments after equation (1) silently changes the problem and is therefore prohibited.

Boundary 1.1 (Unit context). Atomic units suppress $m_e, e, \hbar, 4\pi\epsilon_0$, but do not erase physical dimensions. A numerical atomic-unit expression may enter an SI formula only through an explicit conversion using the declared values of $E_h, a_0, m_e, e, \hbar$. No addition or comparison across SI and atomic contexts is implicit in this reference.

For pure isolated electronic states, the state space is the antisymmetric sector

$$\mathcal{H}_e^{(N)} = \bigwedge^N L^2(\mathbb{R}^3; \mathbb{C}^2).$$

Mixed or open-system electronic states are positive trace-one operators Γ_N on this sector. A wavefunction is therefore not the universal state object; it is a pure-state representative.

2 Exact molecule and Born-Oppenheimer reduction

At fixed nuclear geometry R , define the clamped-nuclei electronic Hamiltonian

$$\hat{H}_e(R) = \hat{T}_e + \hat{V}_{ee} + \hat{V}_{en}(R) + V_{nn}(R). \quad (3)$$

This document includes the scalar nuclear repulsion $V_{nn}(R)$ in $\hat{H}_e(R)$; a convention that excludes it must add it back before calling an eigenvalue a potential-energy surface.

The parametric eigenproblem is

$$\hat{H}_e(R) \psi_k(x; R) = E_k(R) \psi_k(x; R). \quad (4)$$

The map $R \mapsto E_k(R)$ is not obtained by taking a marginal of the exact molecular wavefunction. It is generated from the Hamiltonian family after choosing a clamped-nuclei model and an electronic state label. A Born-Huang expansion has the form

$$\Psi(x, R) = \sum_k \chi_k(R) \psi_k(x; R). \quad (5)$$

Substitution into the full equation yields coupled nuclear equations containing derivative couplings. A one-surface Born-Oppenheimer model is a reduction obtained by neglecting specified off-diagonal terms; it is not a lossless frame transform and it is not uniformly valid at a closing internal electronic gap.

Boundary 2.1 (Conical-intersection inheritance). Near a conical intersection, the individual rank-one adiabatic projectors need not extend through the seam even when the retained two-state cluster remains regular. Branching coordinates, gauge, Berry holonomy, derivative coupling, dynamics, and finite-resolution identifiability are governed by the separate P8 conical-intersection contract. The present document does not replace that contract with the phrase “observation caustic.”

3 Reduced density data and information loss

Definition 3.1 (One-body reduced density matrix). For a normalized N -electron state Γ_N , define

$$\gamma^{(1)} = N \operatorname{Tr}_{2,\dots,N} \Gamma_N. \quad (6)$$

Its spatial density is

$$n(\mathbf{r}) = \sum_{\sigma} \langle \mathbf{r}\sigma | \gamma^{(1)} | \mathbf{r}\sigma \rangle. \quad (7)$$

Proposition 3.2 (Normalization and one-body observables). *The reduction in equation (6) obeys*

$$\operatorname{Tr} \gamma^{(1)} = N, \quad \int_{\mathbb{R}^3} n(\mathbf{r}) \, d^3r = N. \quad (8)$$

For every one-body observable $\hat{A} = \sum_{i=1}^N a(i)$,

$$\operatorname{Tr}(\Gamma_N \hat{A}) = \operatorname{Tr}(\gamma^{(1)} a). \quad (9)$$

Proof. The partial trace preserves the trace, so $\operatorname{Tr} \gamma^{(1)} = N \operatorname{Tr} \Gamma_N = N$. Resolving the one-particle trace in the position-spin basis gives the density integral. Permutation symmetry of Γ_N makes all N terms in $\hat{A}$ equal after tracing the other particles, which yields equation (9). $\square$

Proposition 3.3 (The one-body reduction is non-injective). *Let $|ij\rangle = a_i^\dagger a_j^\dagger |0\rangle$ in $\wedge^2 \mathbb{C}^4$, and define*

$$|\Psi_{\pm}\rangle = \frac{|12\rangle \pm |34\rangle}{\sqrt{2}}.$$

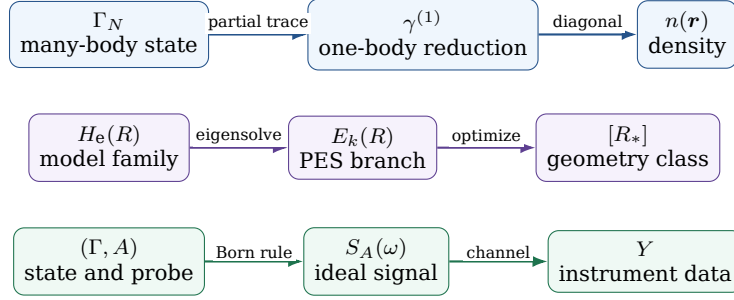
Both states have $\gamma^{(1)} = \frac{1}{2} I_4$, but the two-body coherence

$$\hat{B} = |12\rangle\langle 34| + |34\rangle\langle 12|$$

has expectation +1 in Ψ_+ and -1 in Ψ_- .

Proof. Each spin orbital occurs with probability 1/2, and all one-body cross terms vanish because the determinants differ by two orbitals. The operator $\hat{B}$ exchanges the two determinants, producing the stated signs. $\square$

Thus neither $\Gamma_N \mapsto \gamma^{(1)}$ nor $\Gamma_N \mapsto n$ is generally invertible. Electron correlation is not defined as an information-theoretic defect of this map. It is method- and Hamiltonian-specific unless a precise correlation functional or entanglement quantity is declared.



The three rows are related by a declared scientific workflow, but they are not a single automatic composition.

4 Finite bases, Hartree-Fock, and orbital gauge

Let $\{\chi_\mu\}_{\mu=1}^K$ be a linearly independent, generally nonorthogonal one-particle basis with overlap matrix

$$S_{\mu\nu} = \langle \chi_\mu, \chi_\nu \rangle, \quad S = S^\dagger > 0.$$

Roothaan–Hall equations have the generalized form

$$FC = SC\varepsilon, \quad C^\dagger SC = I. \quad (10)$$

Writing $FC = C\varepsilon$ without either an orthonormal basis or the overlap matrix changes the discrete problem. Small eigenvalues of S also require a declared linear-dependence threshold.

For occupied columns C_o , define the AO density matrix

$$P = C_o C_o^\dagger, \quad \text{Tr}(PS) = N_{\text{occ}} \quad (11)$$

for one electron per occupied spin orbital. Closed-shell spatial orbital conventions add their declared occupation factor.

Proposition 4.1 (Occupied-orbital gauge). *If $U \in U(N_{\text{occ}})$, then*

$$C'_o = C_o U \implies P' = P.$$

The associated normalized Slater determinant changes only by $\det U$, hence represents the same quantum ray.

Proof. $P' = C_o U U^\dagger C_o^\dagger = P$. Multilinearity of the exterior product gives $\phi'_1 \wedge \cdots \wedge \phi'_{N_{\text{occ}}} = (\det U) \phi_1 \wedge \cdots \wedge \phi_{N_{\text{occ}}}$, and $|\det U| = 1$. $\square$

Consequently, localized occupied orbitals, hybrid orbitals, and orbital-by-orbital bond pictures are representation choices unless the localization or decomposition functional is part of the protocol. Canonical orbitals diagonalize a chosen effective operator; that gauge choice does not turn every orbital or orbital energy into a many-body observable.

Hartree–Fock is the variational problem

$$E_{\text{HF}} = \inf_{\Phi \text{ Slater}, \|\Phi\|=1} \langle \Phi | \hat{H}_e | \Phi \rangle. \quad (12)$$

The nonlinear SCF equations characterize stationary solutions. An iteration may converge to a local stationary point, oscillate, or break an exact symmetry; convergence alone does not certify the global Hartree-Fock minimum.

For the same Hamiltonian and one-particle space,

$$E_{\text{corr}} = E_0 - E_{\text{HF}} \leq 0, \quad (13)$$

where E_0 is the exact or full-CI ground energy in the declared space. Cross-method correlation energies are not comparable unless Hamiltonian, basis, frozen-core convention, geometry, and reference state agree.

5 Density-functional theory firewall

The density map is a partial trace followed by a diagonal readout. Ground-state density-functional theory adds a separate variational statement. For fixed particle number and electron-electron interaction, a constrained-search functional can be written

$$F[n] = \inf_{\Gamma \mapsto n} \text{Tr} \left[\Gamma (\hat{T} + \hat{W}) \right]. \quad (14)$$

The external-potential ground energy is then

$$E_v = \inf_n \left\{ F[n] + \int v_{\text{ext}}(\mathbf{r}) n(\mathbf{r}) \, \mathrm{d}^3 r \right\}, \quad (15)$$

over the declared admissible density class.

Boundary 5.1 (Hohenberg-Kohn scope). In its original ground-state setting, the density determines the external scalar potential up to an additive constant under fixed particle number and interaction, subject to the theorem’s representability and degeneracy hypotheses. This does not invert the density map for an arbitrary excited, time-dependent, mixed, or open-system state. Degenerate and ensemble cases require their own formulation.

The Kohn-Sham construction introduces auxiliary orbitals with

$$n(\mathbf{r}) = \sum_i f_i |\varphi_i(\mathbf{r})|^2, \quad \left[-\frac{1}{2} \nabla^2 + v_{\text{eff}}[n] \right] \varphi_i = \varepsilon_i \varphi_i \quad (16)$$

in atomic units. The orbitals reproduce the target density within the model. General Kohn-Sham eigenvalue differences are not equal by definition to interacting excitation energies, and approximate exchange-correlation functionals introduce model error not encoded by the density normalization.

6 Nuclear geometry and the normal-mode theorem

For an isolated molecule, absolute translation and rotation do not define internal geometry. Away from collisions and singular configurations, an internal configuration is an equivalence class

$$[R] \in \mathcal{Q}_{\text{int}} = ((\mathbb{R}^{3N_n} \setminus \Delta) / \text{SE}(3)) / \Pi_{\text{identical}},$$

where the final quotient is included when identical-nucleus permutations are physically identified. A Cartesian minimizer is therefore not unique before this quotient or a gauge-fixing alignment.

Let R_* be a stationary geometry of a smooth PES, $H_R = \nabla_R^2 E(R_*)$ its Cartesian Hessian, and $\mathbf{M} = \text{diag}(M_1, M_1, M_1, \dots)$. The mass-weighted dynamical matrix is

$$\mathbf{K} = \mathbf{M}^{-1/2} H_R \mathbf{M}^{-1/2}. \quad (17)$$

Theorem 6.1 (Harmonic normal modes). *If $\mathbf{K}u_k = \lambda_k u_k$, then the linearized nuclear equation has modal solutions with*

$$\lambda_k = \omega_k^2, \quad \nu_k = \frac{\omega_k}{2\pi}, \quad \tilde{\nu}_k = \frac{\omega_k}{2\pi c}. \quad (18)$$

For an unconstrained isolated nonlinear molecule, six rigid translation/rotation directions are removed; for a linear molecule, five are removed. A strict harmonic minimum has $\lambda_k > 0$ on the remaining vibrational subspace. A first-order transition state has exactly one negative vibrational eigenvalue there.

Proof. With displacement δR , the linearized equation is $\mathbf{M}\delta\ddot{R} + H_R\delta R = 0$. Setting $q = \mathbf{M}^{1/2}\delta R$ gives $\ddot{q} + \mathbf{K}q = 0$. Substitution $q(t) = u_k e^{i\omega t}$ yields $\mathbf{K}u_k = \omega^2 u_k$. Euclidean invariance makes translations and infinitesimal rotations null directions at a stationary geometry. Rotation about the molecular axis displaces no nucleus for a linear geometry, leaving five rather than six independent rigid directions. $\square$

Boundary 6.2 (Qualification). Atoms, constraints, periodic cells, external fields, nonstationary points, exact symmetries beyond rigid motion, and numerical finite-difference Hessians change the null-space analysis. A small computed value is not automatically an exact zero mode; the threshold must be scaled and reported.

7 Active spaces, valence, and bond interpretations

A mathematically typed active-space reduction begins with a declared one-particle projector P_A :

$$\gamma_A = P_A \gamma^{(1)} P_A, \quad N_A = \text{Tr}(\gamma^{(1)} P_A). \quad (19)$$

The projector may be generated by atomic valence functions, energy windows, natural occupations, localized orbitals, or another stated algorithm. Different admissible choices need not give the same N_A , oxidation assignment, or bond decomposition.

Definition 7.1 (Interpretation map). A population, partial charge, bond order, hybridization, or “valence projection” is an interpretation map

$$\mathfrak{I}_\alpha : (\Gamma, H, \mathcal{B}, \mathcal{A}, \text{conventions}) \mapsto z_\alpha$$

with a named scheme α , basis $\mathcal{B}$, active space $\mathcal{A}$, and normalization. It is not promoted to a scheme-independent observable without an invariance theorem.

This distinction does not make such quantities useless. It makes comparisons conditional: trends are meaningful within a controlled protocol, while equality across unrelated partition schemes is not assumed.

8 Ideal spectra and finite-resolution data

Let $\{|i\rangle, E_i\}$ be stationary states, p_i initial populations, and $\hat{A}$ the probe-dependent transition operator. With a normalized line profile $\int L_\Gamma(\omega) d\omega = 1$, define

$$S_A(\omega) = \sum_{i,f} p_i \left| \langle f | \hat{A} | i \rangle \right|^2 L_\Gamma \left(\omega - \frac{E_f - E_i}{\hbar} \right). \quad (20)$$

The observed record is a separate channel, schematically

$$Y = \mathcal{D}_\epsilon \mathcal{C}_{\text{inst}}[S_A] + \eta, \quad (21)$$

where $\mathcal{C}_{\text{inst}}$ includes response and convolution, $\mathcal{D}_\epsilon$ sampling or quantization, and η a declared stochastic error process.

Energy differences determine line locations only after a dynamical model identifies the relevant states. Intensities also require populations and matrix elements. Broadening, unresolved blends, calibration, baseline, and selection bias enter the inference from Y back to a molecular model.

If a symmetry product contains the totally symmetric representation, a transition is symmetry-allowed for the declared operator. This is a necessary symmetry condition, not a guarantee that the matrix element is numerically nonzero: additional quantum numbers, cancellation, population, or experimental geometry may suppress the line.

9 Approximation and identifiability ledger

A reproducible quantum-chemical claim separates at least the following errors:

1. **physical-model error:** nonrelativistic Hamiltonian, nuclear model, environmental and field terms;
2. **electron-nuclear reduction error:** Born-Oppenheimer, retained surfaces, derivative couplings, nuclear quantum effects;
3. **discretization error:** one-particle basis, active space, grid, pseudopotential, boundary conditions;
4. **many-body model error:** determinant ansatz, excitation rank, density functional, embedding approximation;
5. **numerical error:** SCF stationarity, integral and diagonalization tolerances, geometry and Hessian convergence;
6. **property-model error:** response order, transition operator, gauge, relativistic picture-change corrections;
7. **observation error:** instrument response, resolution, calibration, noise, estimator, and assignment ambiguity.

Agreement of one final scalar with experiment does not identify these components separately.

Boundary 9.1 (Relativistic boundary). For heavy elements, a relativistic model must state its nuclear charge distribution, chosen positive-energy or no-pair space, two-electron terms, and finite-basis kinetic balance. An unqualified nonrelativistic Rayleigh-Ritz argument cannot be transferred to the Dirac operator across its negative-energy continuum.

10 Reproducible controls and claim register

The strict program-level result is a typed inference graph: Hamiltonian selection, state preparation, many-body reduction, representation choice, nuclear reduction, ideal property evaluation, and measurement are distinct operations. Structural analogy is retained only where their domains and invariants are explicit.

Control	Exact result	Purpose
$ \Psi_{\pm}\rangle = (12\rangle \pm 34\rangle)/\sqrt{2}$	$\gamma^{(1)} = \frac{1}{2}I_4, \langle B \rangle = \pm 1$	Non-injectivity of one-body reduction
$C_0 \mapsto C_0 U$	$P' = P$, Slater ray unchanged	Occupied-orbital gauge
Diatomic spring k , masses m_1, m_2	$\omega^2 = k(m_1^{-1} + m_2^{-1})$	Frequency-square and rigid-mode test
Two-determinant matrix $\begin{pmatrix} 0 & V \\ V & \Delta \end{pmatrix}$	$E_0 = (\Delta - \sqrt{\Delta^2 + 4V^2})/2 \leq 0$	Variational correlation sign
Gaussian L_{σ}	$\int L_{\sigma}(\omega - \omega_0) d\omega = 1$	Line-profile normalization
Nested Ritz spaces	$E_0^{(K+1)} \leq E_0^{(K)}$	Finite-basis variational control

Table 1: Reference values implemented by the P9 regression suite.

Claim	Status	Scope
$\text{Tr} \gamma^{(1)} = N$ and $\int n = N$	Proposition	Normalized N -electron state
One-body reduction is non-injective	Proposition	Explicit two-electron four-orbital control
Occupied rotations preserve P	Proposition	Unitary rotations within occupied space
$\lambda_k = \omega_k^2$	Theorem	Harmonic mass-weighted Hessian
Five or six rigid zero modes	Qualified theorem	Isolated linear or nonlinear molecule
Density uniquely reconstructs any state	Rejected	Hohenberg-Kohn is a ground-state theorem
Orbitals or bonds are unique observables	Rejected	Representation and scheme must be declared
Energy differences alone determine a spectrum	Rejected	Probe, populations, line shape, and channel are required
New quantum-chemical method	Rejected	Not claimed

Protocol minimum

A complete record states at least: particle numbers and isotope masses; Hamiltonian and unit context; frame, geometry convention, and symmetry; pure or mixed state; electronic and nuclear reduction; basis, overlap threshold, active space, and frozen sectors; orbital gauge or localization rule; correlation or density-functional model; SCF and geometry stationarity criteria; Hessian projection and zero-mode threshold; property operator and gauge; populations, line profile, instrument response, noise law, estimator, uncertainty, and identifiability scope.

References

- [1] M. Born and R. Oppenheimer, "Zur Quantentheorie der Molekeln," *Annalen der Physik* 389, 457–484 (1927), doi:10.1002/andp.19273892002.
- [2] J. C. Slater, "The Theory of Complex Spectra," *Physical Review* 34, 1293–1322 (1929), doi:10.1103/PhysRev.34.1293.
- [3] C. C. J. Roothaan, "New Developments in Molecular Orbital Theory," *Reviews of Modern Physics* 23, 69–89 (1951), doi:10.1103/RevModPhys.23.69.
- [4] P.-O. Lowdin, "Quantum Theory of Many-Particle Systems. I. Physical Interpretations by Means of Density Matrices, Natural Spin-Orbitals, and Convergence Problems in the Method of Configurational Interaction," *Physical Review* 97, 1474–1489 (1955), doi:10.1103/PhysRev.97.1474.
- [5] A. J. Coleman, "Structure of Fermion Density Matrices," *Reviews of Modern Physics* 35, 668–686 (1963), doi:10.1103/RevModPhys.35.668.
- [6] P. Hohenberg and W. Kohn, "Inhomogeneous Electron Gas," *Physical Review* 136, B864–B871 (1964), doi:10.1103/PhysRev.136.B864.
- [7] W. Kohn and L. J. Sham, "Self-Consistent Equations Including Exchange and Correlation Effects," *Physical Review* 140, A1133–A1138 (1965), doi:10.1103/PhysRev.140.A1133.
- [8] M. Levy, "Universal variational functionals of electron densities, first-order density matrices, and natural spin-orbitals and solution of the v -representability problem," *Proceedings of the National Academy of Sciences USA* 76, 6062–6065 (1979), doi:10.1073/pnas.76.12.6062.
- [9] C. A. Mead and D. G. Truhlar, "On the determination of Born–Oppenheimer nuclear motion wave functions including complications due to conical intersections and identical nuclei," *Journal of Chemical Physics* 70, 2284–2296 (1979), doi:10.1063/1.437734.

Relativistic Beam Paths as Observation Geometry

From synchrotron phase space to typed detector invariants

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Strict GO Core reference revision v1.3 — 28 July 2026

Status. This document supersedes the bilingual working draft v1.2. It is migrated to GO Core v0.2 and the SI-HEP Quantity Passport v0.5. The physical reference is standard Run 3 proton-proton operation; on the document date the LHC is in Long Shutdown 3. No new accelerator model or particle-physics prediction is claimed.

Abstract

An accelerator is not observed as one visible trajectory. Its hidden state contains a lattice-constrained reference orbit, time-dependent beam distributions, RF and betatron phases, magnet settings, collision states, and detector response variables. We type the corresponding chain as

$$X_t \xrightarrow{\mathcal{R}} \tilde{X}_t \xrightarrow{\mathcal{O}_\lambda} Z_t \xrightarrow{\mathcal{K}_\lambda} \text{Law}(Y_t | Z_t) \xrightarrow{\hat{\theta}_\lambda} \hat{\theta}_t,$$

where reduction, ideal observation, stochastic measurement, and estimation are distinct maps. The SI-HEP boundary is explicit: $p_{\text{SI}}^\mu = (E/c, \mathbf{p})$, $\Pi^\mu = (E, c\mathbf{p})$, and the HEP numerical four-vector is $\Pi^\mu/\mathcal{E}_0$. This removes the ambiguity in $E^2 = \mathbf{p}^2 + m^2$. For charged beam transport, SI magnetic rigidity and HEP momentum are connected by the exact declared-unit identity

$$p[\text{GeV}/c] = 0.299792458 z B\rho[\text{T m}].$$

At a Run 3 reference total energy of 6.8 TeV, the proton has $\gamma \approx 7.2474 \times 10^3$, $1 - \beta \approx 9.5194 \times 10^{-9}$, and $B\rho \approx 2.26824 \times 10^4 \text{ T m}$. The manuscript separates Lorentz invariants, longitudinal-boost covariants, symplectic invariants, operational observables, and estimator outputs. In particular, geometric emittance is invariant only under a specified ideal symplectic transport, while normalized emittance is the relevant ideal adiabatic invariant during acceleration. Equal-time Lorentz contraction and Penrose–Terrell camera appearance are retained as different observation maps, neither being a direct imaging model for LHC protons.

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1 Scope and claim boundary

The manuscript is a typed interface between standard accelerator physics and Geometry of Observation. It is not:

- a lattice design, collimation, impedance, beam-loss, or machine-protection model;
- a replacement for tracking, beam-beam, or detector simulation;
- a hard-body model of the proton;
- a detector reconstruction algorithm;
- a claim that a hadron's Lorentz-contracted equal-time profile is photographed in the LHC;
- a new Standard Model result.

The strict contribution is infrastructural: it identifies the objects, maps, unit charts, transformation groups, and claim conditions needed to state accelerator observations without type or dimension gaps.

2 Observation protocol

Definition 2.1 (Hidden accelerator state). A minimal hidden state at machine time t is a record

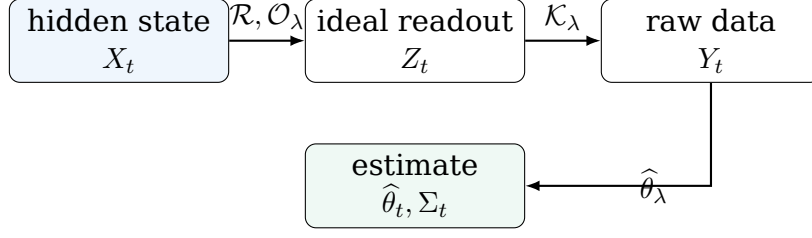
$$X_t = (\Gamma_0, \mathcal{L}_{\text{mag}}(t), \Phi_{\text{RF}}(t), f_1(t), f_2(t), \mathcal{C}_t, \mathcal{D}_t),$$

where Γ_0 is a design orbit, $\mathcal{L}_{\text{mag}}$ is the magnet lattice state, Φ_{RF} is the RF state, f_1, f_2 are beam phase-space distributions, $\mathcal{C}_t$ is a collision-event state when a crossing occurs, and $\mathcal{D}_t$ contains detector-response latent variables.

An observation protocol λ contains a frame, clock, calibration, spatial and temporal resolutions, trigger and selection rules, a deterministic response, a stochastic channel, and an estimator. Its typed chain is

$$X_t \xrightarrow{\mathcal{R}} \tilde{X}_t \xrightarrow{\mathcal{O}_\lambda} Z_t \xrightarrow{\mathcal{K}_\lambda} Y_t \xrightarrow{\hat{\theta}_\lambda} \hat{\theta}_t.$$

Here $\mathcal{K}_\lambda$ denotes a Markov kernel. It must not be merged with the deterministic map $\mathcal{O}_\lambda$, and $\hat{\theta}_\lambda$ must not be identified with the underlying measurand.



3 Reference machine passport

The numerical audit uses the CERN public Run 3 reference:

$$E_{\text{beam}} = 6.8 \text{ TeV}, \quad \sqrt{s} = 13.6 \text{ TeV}$$

for equal, head-on proton beams. The public circumference $C = 26659 \text{ m}$, roughly 2500 bunches per proton beam, and roughly 1.6×10^{11} protons per bunch are rounded or approximate operational descriptors, not defining constants.

Boundary 3.1 (Epoch). The LHC entered Long Shutdown 3 in June 2026. Therefore 6.8 TeV in this document is a Run 3 standard-operation reference, not a statement that a beam was circulating on 28 July 2026.

Every machine record must include

(facility, species, charge state, run, mode, epoch, status, value, unit, precision, source).

The design energy, an achieved beam energy, a special low-energy run, and an HL-LHC target are different records even when they share a unit.

4 SI-HEP representation boundary

Adopt $\eta = \text{diag}(1, -1, -1, -1)$. The SI momentum four-vector and energy four-vector are

$$p_{\text{SI}}^\mu = (E/c, \mathbf{p}_{\text{SI}}), \quad \Pi^\mu = c p_{\text{SI}}^\mu = (E, c\mathbf{p}_{\text{SI}}).$$

With energy anchor $\mathcal{E}_0 = 1 \text{ GeV}$, define the HEP coordinate

$$P_{\text{HEP}}^\mu = \frac{\Pi^\mu}{\mathcal{E}_0}.$$

The mass shell has three equivalent representations:

$$p_{\text{SI}}^2 = m^2 c^2, \quad \Pi^2 = (mc^2)^2, \quad P_{\text{HEP}}^2 = \left(\frac{mc^2}{\mathcal{E}_0} \right)^2.$$

The shorthand $E^2 = \mathbf{p}^2 + m^2$ is used only in the last chart. In the other charts, omitted factors of c change dimensions. The complete conversion contract is specified in the companion *SI-HEP Quantity Passport v0.5*.

5 Relativistic beam kinematics

Let

$$\mu_p := m_p c^2 = 0.93827208943(29) \text{ GeV}$$

from CODATA 2022, and let E denote proton total energy. Define the energy-valued momentum $\pi := pc$. Then

$$E^2 = \pi^2 + \mu_p^2, \quad \gamma = \frac{E}{\mu_p}, \quad \beta = \frac{\pi}{E}.$$

At high γ , direct subtraction of β from one loses floating-point significance. The stable identity is

$$1 - \beta = \frac{1 - \beta^2}{1 + \beta} = \frac{\gamma^{-2}}{1 + \beta}.$$

This quantity, rather than a rounded decimal for β , controls the speed deficit:

$$c - v = c(1 - \beta).$$

Proposition 5.1 (Equal head-on beams). *For two equal beams with four-momenta $\Pi_1 = (E, \pi)$ and $\Pi_2 = (E, -\pi)$,*

$$s = (\Pi_1 + \Pi_2)^2 = 4E^2, \quad \sqrt{s} = 2E.$$

Proof. The total spatial momentum vanishes, while the total energy is $2E$. $\square$

The result assumes equal, exactly head-on beams and neglects event-wise energy spread and crossing-angle corrections. Those effects belong to the event and beam-distribution passports.

6 Frame slices and light-cone observations

For a boost of rapidity Y along the beam axis,

$$\begin{aligned} ct' &= \gamma(ct - \beta z), & z' &= \gamma(z - \beta ct), \\ x' &= x, & y' &= y. \end{aligned}$$

The interval

$$\Delta s^2 = c^2 \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2$$

is Lorentz invariant. A rest-defined longitudinal proper length L_0 , evaluated between endpoints simultaneous in the laboratory, has laboratory equal-time length $L_{\parallel} = L_0/\gamma$. This is a relation between simultaneity slices, not a material compression law for a quantum proton.

In light-front coordinates

$$x^{\pm} = \frac{ct \pm z}{\sqrt{2}},$$

the same boost acts as

$$x^{+'} = e^{-Y} x^+, \quad x^{-'} = e^Y x^-.$$

A camera image is a different map. A source worldline $X^{\mu}(\xi, \tau)$ contributes to an image at observer event O^{μ} only if

$$(O - X(\xi, \tau_e))^2 = 0, \quad \tau_e < t_{\text{obs}}.$$

The incoming null direction is then projected to an image plane. Thus:

$$\text{equal-time boost} \neq \text{past-light-cone image}.$$

The Penrose–Terrell effect concerns the second map. It is not a detector-observable model for individual LHC protons.

7 Magnetic rigidity and path geometry

In SI,

$$\frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad \frac{dE}{dt} = q \mathbf{E} \cdot \mathbf{v}.$$

A transverse magnetic field bends the orbit but does no work. For local curvature radius ρ ,

$$|\mathbf{p}|_{\text{SI}} = |q| B \rho.$$

With $q = ze$, the exact declared-unit bridge is

$$\boxed{p[\text{GeV}/c] = 0.299792458 z B \rho[\text{T m}].}$$

For a Run 3 reference proton,

$$p = 6799.999935268 \text{ GeV}/c, \quad B \rho = 22682.358258 \text{ T m}.$$

Boundary 7.1 (Local versus global geometry). The circumference contains straight insertions and does not determine the dipole bending radius by $C/(2\pi)$. The local field-to-curvature map requires the lattice geometry. Conversely, the revolution frequency may use the total closed-orbit length.

8 RF phase and longitudinal state

For an idealized RF gap,

$$\Delta E = q V_{\text{eff}} \sin \phi_s.$$

For $q = ze$,

$$\Delta E[\text{eV}] = z V_{\text{eff}}[\text{V}] \sin \phi_s.$$

The unit conversion is exact; the effective-voltage and thin-gap model is not.

A common dimensionless longitudinal coordinate is

$$\delta = \frac{p - p_0}{p_0}.$$

The phase ϕ and δ are not direct positions and momenta until a canonical normalization is declared. Consequently, a plotted RF “bucket area” is not comparable across conventions without the Jacobian and reference energy. The observation protocol must record the RF harmonic, synchronous phase, voltage convention, longitudinal coordinate convention, and clock.

9 Transverse optics and emittance

In one transverse plane, linearized motion about the design orbit is

$$x''(s) + K(s)x(s) = 0, \quad [K] = L^{-2}.$$

Here x has dimension L , s has dimension L , and $x' = dx/ds$ is dimensionless. A first-order transport map $\mathbf{u}_2 = M\mathbf{u}_1$, $\mathbf{u} = (x, x')^\top$, is symplectic when

$$M^\top J M = J, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

For a centered beam with covariance matrix Σ , define the geometric RMS emittance

$$\varepsilon := \sqrt{\det \Sigma}, \quad \Sigma = \varepsilon \begin{pmatrix} \beta_T & -\alpha_T \\ -\alpha_T & \gamma_T \end{pmatrix}, \quad \beta_T \gamma_T - \alpha_T^2 = 1.$$

The dimensions are

$$[\beta_T] = L, \quad [\alpha_T] = 1, \quad [\gamma_T] = L^{-1}, \quad [\varepsilon] = L.$$

Proposition 9.1 (Qualified geometric-emittance invariance). *If $\Sigma_2 = M\Sigma_1 M^\top$ and M is a 2×2 symplectic map in the same canonical coordinate convention, then $\det \Sigma_2 = \det \Sigma_1$ and $\varepsilon_2 = \varepsilon_1$.*

Proof. A 2×2 symplectic matrix has $\det M = 1$. Hence $\det \Sigma_2 = (\det M)^2 \det \Sigma_1 = \det \Sigma_1$. $\square$

During ideal acceleration the geometric emittance in trace-space coordinates undergoes adiabatic damping. The corresponding normalized emittance is

$$\varepsilon_n = \beta_{\text{rel}} \gamma_{\text{rel}} \varepsilon.$$

Its invariance still requires ideal uncoupled transport and excludes scattering, radiation, noise, nonlinear filamentation, collective effects, and losses. The word “invariant” without this hypothesis list is rejected.

10 Bunches as hidden distributions

Let $u = (x, p_x, y, p_y, z, \delta)$ denote a declared six-dimensional coordinate chart and let

$$f_t(u) \geq 0, \quad \int f_t(u) d\mu_\Gamma(u) = 1$$

be a normalized hidden beam law with a typed phase-space measure. An intensity density uses a different normalization, $\int f_t d\mu_\Gamma = N_b$.

Profile monitors observe marginals or convolutions, for example

$$\rho_x^{\text{ideal}}(x) = \int f_t(u) d\mu_{\Gamma \setminus x}, \quad Y_x \sim \mathcal{K}_\lambda(\rho_x^{\text{ideal}}).$$

Equal one-dimensional marginals do not identify the joint law. Different transverse correlations, tails, halo populations, and loss risks can therefore agree at the profile level.

11 Collision and detector reconstruction

For reconstructed final-state energy four-vectors $\Pi_i^\mu = (E_i, \boldsymbol{\pi}_i)$,

$$\mu_{\text{inv}}^2 = \left(\sum_i E_i \right)^2 - \left\| \sum_i \boldsymbol{\pi}_i \right\|^2.$$

The same physical invariant in SI momentum coordinates is

$$m_{\text{inv}}^2 c^4 = \left(\sum_i E_i \right)^2 - c^2 \left\| \sum_i \mathbf{p}_i \right\|^2.$$

Detector data do not provide the exact Π_i^μ . Tracks, clusters, timing, calibration constants, object identification, and selections enter an estimator

$$\hat{\mu}_{\text{inv}} = \hat{\theta}_\lambda(Y), \quad \Sigma_{\hat{\mu}} \text{ or a likelihood interval.}$$

For polar angle θ ,

$$p_T = \|\mathbf{p}_\perp\|, \quad \eta = -\ln \tan \frac{\theta}{2}.$$

The logarithm is dimensionless. Under a longitudinal boost, p_T is unchanged. True rapidity

$$y = \frac{1}{2} \ln \frac{E + \pi_z}{E - \pi_z}$$

shifts additively, so rapidity differences are boost-invariant. Pseudorapidity equals rapidity only in the massless limit; it is not an unrestricted Lorentz scalar.

12 Transformation classes

Class	Object	Correct status
Lorentz	Π^2, μ_{inv}	invariant under proper Lorentz transformations
Longitudinal boost	$p_T, y, \Delta y$	p_T unchanged, y shifts, Δy invariant
Symplectic transport	$\varepsilon = \sqrt{\det \Sigma}$	invariant under the stated ideal map and coordinate convention
Adiabatic acceleration	ε_n	ideal normalized-emittance invariant under additional hypotheses
Unit chart	physical Q	same quantity; numerical coordinate transforms covariantly
Detector reconstruction	$\hat{\mu}_{\text{inv}}$	estimator with calibration and uncertainty, not an exact invariant
Magnet setting	$B\rho$	frame- and species-dependent rigidity relation, not Lorentz invariant

13 Non-identifiability

For a deterministic observation $y = \mathcal{O}_\lambda(x)$, the exact fiber is

$$\mathcal{F}_y = \{x : \mathcal{O}_\lambda(x) = y\}.$$

For a stochastic channel with likelihood $k_\lambda(y | x)$, the compatibility set becomes

$$\mathcal{F}_{y,\lambda} = \{x : k_\lambda(y | x) > 0\},$$

or, more usefully, a likelihood or posterior level set. A “size of the fiber” is undefined until a metric, measure, prior, threshold, and quotient by representation symmetries are specified.

This distinction blocks three invalid inferences:

- (i) a beam profile does not determine the full phase-space law;
- (ii) a reconstructed invariant mass does not determine a unique partonic history;
- (iii) an event display is not the collision event itself.

14 Reproducible Run 3 numerical audit

The following values use the exact SI definitions of c, h, e , the CODATA 2022 proton mass energy, the CERN public 6.8 TeV reference, and the rounded public circumference.

Quantity	Value	Status
Reference proton total energy	6800 GeV	operational reference
Proton mass energy	0.93827208943 GeV	CODATA 2022
Lorentz factor γ	$7.24736467876 \times 10^3$	derived
Speed ratio β	0.99999999048060	derived
Stable $1 - \beta$	$9.519404390069 \times 10^{-9}$	derived
Speed deficit $c - v$	$2.853845641 \text{ m s}^{-1}$	derived
Momentum	6799.999935268 GeV/ c	derived
Magnetic rigidity $B\rho$	22682.358258 T m	derived
Revolution frequency	11245.450135 Hz	rounded circumference
Symmetric head-on $\sqrt{s}$	13600 GeV	derived

The revolution-frequency value has more displayed digits only as a regression output; its physical precision is limited by the rounded circumference and the orbit definition. The approximate public bunch count and population would imply a beam-energy scale of about $4.36 \times 10^8 \text{ J}$, but this is not an event-by-event or fill-specific measurement.

15 Claim audit

Statement	Status	Required hypotheses or reason
Mass-shell equivalence across SI/HEP charts	theorem	explicit c -map and common signature
Run 3 numerical audit	derived check	declared source values and epoch
Geometric-emittance invariance	proposition	linear symplectic map, fixed canonical convention
Normalized-emittance invariance	ideal model	adiabatic acceleration and excluded nonideal effects
Equal-time longitudinal contraction	standard kinematics	inertial frames and declared simultaneity slices
Penrose–Terrell appearance	standard optical result	past-light-cone image of an extended source
Direct optical image of an LHC proton	rejected	no such detector channel is modelled
New LHC or particle-physics prediction	not claimed	reference/infrastructure module only

References

- [1] BIPM, *The International System of Units (SI)*, 9th ed., version 4.01. <https://www.bipm.org/documents/20126/41483022/SI-Brochure-9-EN.pdf>
- [2] P. J. Mohr et al., “CODATA recommended values of the fundamental physical constants: 2022,” 2025. <https://physics.nist.gov/cuu/pdf/JPCRD2022CODATA.pdf>
- [3] Particle Data Group, “Physical Constants,” in *Review of Particle Physics 2024*. <https://pdg.lbl.gov/2024/reviews/rpp2024-rev-phys-constants.pdf>
- [4] CERN, “Large Hadron Collider: facts and figures.” <https://home.cern/science/accelerators/large-hadron-collider/>
- [5] CERN, “CERN bids farewell to the LHC and enters Long Shutdown 3,” 29 June 2026. CERN news page, 29 June 2026.
- [6] O. Brüning et al. (eds.), *LHC Design Report*, Vol. I, CERN, 2004. <https://cds.cern.ch/record/782076/files/CERN-2004-003-V1-ft.pdf>
- [7] E. D. Courant and H. S. Snyder, “Theory of the alternating-gradient synchrotron,” *Annals of Physics* **3** (1958), 1–48. [https://doi.org/10.1016/0003-4916\(58\)90012-5](https://doi.org/10.1016/0003-4916(58)90012-5)
- [8] R. Penrose, “The apparent shape of a relativistically moving sphere,” *Math. Proc. Camb. Phil. Soc.* **55** (1959), 137–139. <https://doi.org/10.1017/S0305004100033776>
- [9] J. Terrell, “Invisibility of the Lorentz contraction,” *Physical Review* **116** (1959), 1041–1045. <https://doi.org/10.1103/PhysRev.116.1041>

Satellite Networks under Typed Frames and Temporal Observation Channels

Rigid-frame covariance, torus closures, line-of-sight graphs, sampling, and clock protocols

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Strict GO Core reference revision v1.2 — 28 July 2026

Status. This document supersedes the bilingual working note v1.1 and is an application-layer reference migrated to GO Core v0.2. It is not an orbit-determination method, an operational constellation model, a replacement for TLE/SGP4, or a new result in astrodynamics. Its contribution is a typed separation of hidden state, coordinate frame, sensor channel, temporal graph, sampling protocol, closure, and relativistic clock comparison.

Abstract

A satellite constellation is represented by one labeled dynamical state but by many coordinate traces and still more sensor records. The legacy note treated all of these as observer-frame drawings and then called pairwise distances, coordinate spectra, orbit closures, density statistics, and clock offsets “invariants”. This reference states the transformation group and protocol for every such claim. For a time-dependent rigid frame $\mathfrak{f} = (o, R) \in C^1(I, \text{SE}(3))$, the labeled Euclidean distance matrix is invariant at each common time, while component velocities, coordinate spectra, and single-object orbit closures generally are not. A distance matrix determines a labeled configuration only up to an element of $E(3)$, so chirality is lost. For a quasiperiodic phase flow, the exact phase closure is the coset subtorus determined by the integer resonance lattice; its image under a continuous observation map is channel-dependent and is only equivariant under a constant rigid isometry. Finite sampling introduces exact frequency aliases and cannot certify rational independence. Spherical-body line-of-sight and range gates define a temporal graph whose snapshot and time-respecting connectivity are separated. Proper time along a fixed worldline segment is a spacetime scalar, whereas comparing different clocks at equal coordinate time requires a declared synchronization and gravity model. These distinctions close the last frame-law defect in the Geometry of Observation corpus.

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1 Scope, hidden state, and the observation chain

Fix a coordinate-time interval $I \subset \mathbb{R}$, a finite label set $V = \{1, \dots, N\}$, and an Earth-centered inertial reference chart. The minimal spatial state is the labeled tuple

$$X(t) = (c_E(t), (r_i(t), v_i(t))_{i \in V}), \quad v_i = \dot{r}_i, \quad (1)$$

where c_E is the center of the occulting body. A set notation $\{r_i\}$ is insufficient: it discards labels and multiplicity, both of which are required for tracking and temporal edges.

Model 1.1 (Two-body benchmark). The reproducibility controls use

$$\dot{r}_i = v_i, \quad \dot{v}_i = -\mu_E \frac{r_i - c_E}{\|r_i - c_E\|^3}, \quad (2)$$

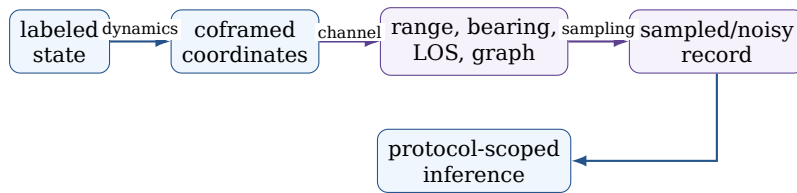
with a spherical exclusion radius R_E . This is an exact model definition, not an ephemeris accuracy claim.

Classical elements $(a, e, i, \Omega, \omega, M_0)$ are a local coordinate chart for model 1.1; they are singular at standard circular and equatorial limits and are not the hidden object. Likewise, a two-line element set is a model-specific mean-element record intended for a compatible SGP4-family propagator, not an arbitrary Kepler state. The former phrase “TLE-like benchmark” is therefore removed.

Definition 1.2 (Typed observation protocol). A protocol is the composite

$$\mathcal{X} \xrightarrow{\Phi_t} \mathcal{X}_t \xrightarrow{F_{\mathfrak{f}}} \mathcal{Y}_{\mathfrak{f}} \xrightarrow{\mathcal{Q}} \mathcal{D} \xrightarrow{\mathcal{S}_{\Delta t, T}} \mathcal{D}^M \xrightarrow{\mathcal{A}} \mathcal{I}, \quad (3)$$

where Φ_t is the declared dynamics, $F_{\mathfrak{f}}$ is an invertible coordinate change, $\mathcal{Q}$ is a possibly lossy sensor or graph channel, $\mathcal{S}_{\Delta t, T}$ samples over a finite horizon, and $\mathcal{A}$ is an estimator or diagnostic. Information loss is attributed to $\mathcal{Q}$ or $\mathcal{S}$, not to a frame change.



2 Time-dependent rigid frames and exact spatial invariants

Let $o : I \rightarrow \mathbb{R}^3$ be a frame origin and let $R : I \rightarrow \text{SO}(3)$ map frame components to inertial components. The coordinates of satellite i and of the Earth center are

$$y_i = R^T(r_i - o), \quad c_{\mathfrak{f}} = R^T(c_E - o). \quad (4)$$

Thus a frame is not merely an observer point. Position vectors may be composed across an Earth-Sun-barycentric tree only after all origins, orientations, epochs, and time scales are declared. In homogeneous coordinates the legal composition is matrix multiplication,

$$T_{A \leftarrow C}(t) = T_{A \leftarrow B}(t) T_{B \leftarrow C}(t), \quad (5)$$

not an untyped sum of vectors stored in different bases.

Let

$$\mathcal{G}_I = C^1(I, \text{SE}(3))$$

act diagonally on every node and on the central body through equation (4).

Theorem 2.1 (Distance-matrix invariance). *For every $\mathfrak{f} \in \mathcal{G}_I$, every common time t , and every $i, j \in V$,*

$$D_{ij}^{\mathfrak{f}}(t) = \|y_i(t) - y_j(t)\| = \|r_i(t) - r_j(t)\| = D_{ij}(t). \quad (6)$$

Consequently, every deterministic function of the labeled distance matrix is invariant under this declared action.

Proof. Translations cancel and $R^\top R = I$:

$$y_i - y_j = R^\top(r_i - r_j), \quad \left\| R^\top(r_i - r_j) \right\|^2 = (r_i - r_j)^\top R R^\top (r_i - r_j).$$

□

The corresponding velocity law is different. With $\Omega^\times = R^\top \dot{R} \in \mathfrak{so}(3)$,

$$\dot{y}_i = R^\top(v_i - \dot{o}) - \Omega^\times y_i. \quad (7)$$

Therefore coordinate velocities, accelerations, and component spectra are not invariants of $\mathcal{G}_I$. Even the norm of a relative velocity acquires the rotational term unless R is constant.

Theorem 2.2 (What the distance matrix identifies). *Let D be the labeled Euclidean distance matrix of N points, let $J = I_N - \frac{1}{N}\mathbf{1}\mathbf{1}^\top$, and define*

$$B = -\frac{1}{2}J(D \circ D)J. \quad (8)$$

Then B is the centered Gram matrix. If $B \succeq 0$ and $\text{rank } B \leq 3$, its factorization reconstructs the configuration up to $E(3)$. Pairwise distances do not distinguish translations, rotations, or reflections.

Proof. Centering the points gives $B = X_c X_c^\top$. Any two factorizations of the same positive semidefinite Gram matrix differ by an orthogonal map on the realized span. Restoring the centroid adds an arbitrary translation. □

Thus the distance channel is complete for a labeled snapshot modulo $E(3)$, but not modulo $\text{SE}(3)$: chirality requires an oriented volume or another parity-sensitive channel.

3 Phase closure, image closure, and frame covariance

For a quasiperiodic idealization, let

$$\theta(t) = \theta_0 + \omega t \pmod{2\pi}, \quad \theta \in \mathbb{T}^k, \quad (9)$$

and define the integer resonance lattice

$$L_\omega = \{m \in \mathbb{Z}^k : m \cdot \omega = 0\}. \quad (10)$$

Theorem 3.1 (Exact phase closure). *The closure of the phase orbit is the coset subtorus*

$$\Gamma_{\theta_0, \omega} = \{\theta \in \mathbb{T}^k : m \cdot (\theta - \theta_0) = 0 \pmod{2\pi} \text{ for every } m \in L_\omega\}, \quad (11)$$

with

$$\dim \Gamma_{\theta_0, \omega} = k - \text{rank}_{\mathbb{Z}} L_{\omega}.$$

For every continuous $F : \mathbb{T}^k \rightarrow \mathbb{R}^d$,

$$\text{cl}\{F(\theta(t)) : t \in \mathbb{R}\} = F(\Gamma_{\theta_0, \omega}). \quad (12)$$

The rationally commensurate and rationally independent cases are only the two endpoints of theorem 3.1; intermediate resonance rank is possible. Compactness and continuity justify equation (12). A noncompact secular drift is not silently treated as another torus angle.

Proposition 3.2 (Closure is equivariant, not invariant). *For a constant $g = (a, R) \in \text{SE}(3)$ and $F_g(\theta) = a + RF(\theta)$,*

$$K_{F_g} = gK_F. \quad (13)$$

Equality $K_{F_g} = K_F$ requires an additional symmetry of the set. For a time-dependent $g(t)$, even congruence may fail.

Proof. A homeomorphism commutes with closure, giving the first statement. For the second, take $F(\theta(t)) \equiv 0$ and $a(t) = (\cos \alpha t, \sin \alpha t, 0)$. The original closure is one point and the transformed closure is a circle. $\square$

The protocol-invariant object is therefore the phase-flow resonance lattice, when the dynamical model and time parameter are fixed. The single-object spatial closure K_F is a channel output.

4 Sampling, retarded observation, and non-identifiability

At times $t_m = t_0 + m\Delta t$, $m = 0, \dots, M-1$, the record is finite. Its support

$$\hat{K}_{M, \Delta t} = \{Q(X(t_m)) : 0 \leq m < M\} \quad (14)$$

is not the infinite-time closure in equation (12).

Proposition 4.1 (Exact sampling alias). *For any $q \in \mathbb{Z}$ and $t_m = m\Delta t$,*

$$e^{i(\omega + 2\pi q/\Delta t)t_m} = e^{i\omega t_m}. \quad (15)$$

Hence a sampled coordinate spectrum does not identify an unrestricted continuous frequency.

Finite noisy data also cannot decide whether a frequency ratio is exactly rational: arbitrarily close rational and irrational vectors produce arbitrarily close records on a fixed horizon. Statements about closure dimension therefore require either a declared exact model or uncertainty-bounded inference.

The distinction between state and sensor time is equally important. For a reception event $(t_r, o(t_r))$, a one-way ideal observation uses an emission time $t_{e,i}$ satisfying

$$t_r - t_{e,i} = \frac{\|r_i(t_{e,i}) - o(t_r)\|}{c}. \quad (16)$$

Different satellites are generally seen at different emission times. Pairwise distances between apparent retarded positions are not the simultaneous distance matrix in theorem 2.1.

Bearing-only data

$$b_i(t) = \frac{y_i(t)}{\|y_i(t)\|}$$

discard range. Even without noise, one viewpoint therefore does not recover $D(t)$ unless additional baselines, ranging, or dynamical hypotheses are supplied.

5 Line-of-sight and temporal network geometry

Let $p_i = r_i - c_E$, $d_{ij} = p_j - p_i$, and let the spherical occulting radius including a declared margin be $R_{\text{occ}} > 0$. For $i \neq j$, define

$$u_{ij}^* = \text{clip}\left(-\frac{p_i \cdot d_{ij}}{\|d_{ij}\|^2}, 0, 1\right), \quad (17)$$

$$h_{ij} = \|p_i + u_{ij}^* d_{ij}\|. \quad (18)$$

The degenerate case $p_i = p_j$ is handled separately. A strict spherical line of sight is declared when

$$\text{LOS}_{ij}(t) = \mathbf{1}\{h_{ij}(t) > R_{\text{occ}}\}. \quad (19)$$

Tangency belongs to the blocked class in this convention.

For link range $R_{\text{link}} > 0$, the symmetric snapshot graph has

$$A_{ij}(t) = \mathbf{1}\{i \neq j\} \mathbf{1}\{D_{ij}(t) \leq R_{\text{link}}\} \text{LOS}_{ij}(t). \quad (20)$$

Because the satellites, the Earth center, and the segment are all transformed together, equation (20) is invariant under $\mathcal{G}_I$. A fixed spatial bin or an untransformed occulting center would break that statement.

Under a vertex relabeling by a permutation matrix P ,

$$A \mapsto PAP^T, \quad L_G = \text{diag}(A\mathbf{1}) - A \mapsto PL_G P^T.$$

Connected-component counts and $\text{spec}(L_G)$ are label-invariant, while node-specific routing outputs are equivariant under the same relabeling.

Definition 5.1 (Time-respecting journey). Let an active edge $e_k = (v_{k-1}, v_k)$ be entered at t_k and have a declared nonnegative transmission delay $\ell_{e_k}(t_k)$. A journey satisfies

$$t_{k+1} \geq t_k + \ell_{e_k}(t_k) \quad (21)$$

for every consecutive edge. Temporal reachability and earliest arrival are defined by such journeys, not by connectivity of the time-aggregated graph.

The range-only graph $\mathbf{1}\{D_{ij} \leq \rho\}$ is called a *proximity graph*. It is not a conjunction-risk calculation. Operational collision assessment additionally requires predicted relative states, covariances, object sizes, maneuver hypotheses, and a time-window protocol.

6 Density and entropy require transported partitions

Let $K_\ell : \mathbb{R}^3 \rightarrow [0, \infty)$ satisfy $\int_{\mathbb{R}^3} K_\ell(x) dx = 1$, and let $w_i \geq 0$, $\sum_i w_i = 1$. The normalized occupancy density is

$$\rho_f(x, t) = \sum_{i=1}^N w_i K_\ell(x - y_i(t)). \quad (22)$$

For a radial kernel and a rigid transform $g = (a, R)$, $\rho_g(x, t) = \rho(R^T(x - a), t)$. This is covariance of a field, not pointwise invariance at one fixed coordinate x .

For a finite measurable partition $\Pi = \{B_1, \dots, B_M\}$, define

$$p_m = \int_{B_m} \rho(x, t) dx, \quad H_{\Pi}^{(2)} = - \sum_{p_m > 0} p_m \log_2 p_m.$$

The entropy is preserved when Π is transported with the rigid frame. With fixed angular or Cartesian bins it is a protocol-dependent diagnostic. Kernel scale, partition, logarithm base, horizon, and empty-total policy are mandatory fields.

7 Proper time and the synchronization boundary

Use metric signature $(-+++)$. Along a fixed timelike worldline segment γ_i with fixed endpoint events,

$$\tau_i[\gamma_i] = \frac{1}{c} \int_{\gamma_i} \sqrt{-g_{\mu\nu} dx^\mu dx^\nu}. \quad (23)$$

This scalar is invariant under coordinate changes. In flat spacetime, the corresponding event interval is invariant under the Poincaré group. Equal-time Euclidean distances are not Lorentz invariants because a boost changes simultaneity.

In a declared stationary weak-field Earth chart with $\rho_i = \|r_i - c_E\|$, $\Phi_i = -\mu_E/\rho_i$, and $v \ll c$,

$$\frac{d\tau_i}{dt} = 1 + \frac{\Phi_i}{c^2} - \frac{\|v_i\|^2}{2c^2} + O(c^{-4}). \quad (24)$$

Relative to an ideal coordinate-stationary reference clock at $R_\star$,

$$\delta_i^{(t, \Phi)} = \frac{d\tau_i}{dt} - \frac{d\tau_\star}{dt} \simeq \frac{\mu_E}{c^2} \left(\frac{1}{R_\star} - \frac{1}{\rho_i} \right) - \frac{\|v_i\|^2}{2c^2}. \quad (25)$$

For a circular two-body orbit,

$$\delta_{\text{circ}}(r) = \frac{\mu_E}{c^2} \left(\frac{1}{R_\star} - \frac{3}{2r} \right). \quad (26)$$

Proposition 7.1 (A spatial locus does not determine proper time). *An unparameterized spatial image $K = \{r(t)\}$ does not determine τ .*

Proof. The same circle can be traversed with two different speed histories. The spatial image is unchanged, while the kinetic term in equation (24) changes. A time-stamped trajectory in a known chart and a declared metric contains strictly more information than its image. $\square$

The comparison $\tau_i(T) - \tau_j(T)$ for endpoints selected by “the same coordinate time T ” is protocol-dependent: the time coordinate, slicing, reference potential, and synchronization convention select the endpoint events. This does not contradict scalar invariance of each fixed worldline integral.

8 Benchmark controls and claim firewall

The reproducibility bundle propagates a synthetic labeled network under model 1.1. It checks time-dependent rigid-frame distance invariance, Euclidean-distance-matrix reconstruction, spherical segment clearance, graph invariance under co-transformed geometry, Laplacian spectral invariance under relabeling, exact sampling aliases, temporal earliest arrival, and weak-field clock rates. The numerical values are regression controls, not calibrated constellation predictions.

Table 1: Typed status of the principal legacy quantities.

Object	Transformation or protocol	Correct status
Distance matrix $D(t)$	$C^1(I, \text{SE}(3))$, common time	Invariant when every node is transformed diagonally.
Coordinate trace $y_i(t)$	chosen frame $\mathfrak{f}$	Covariant representation, not invariant.
Spatial closure K_F	constant rigid isometry	Equivariant up to congruence; generally changes for $g(t)$.
Phase closure $\Gamma_{\theta_0, \omega}$	fixed dynamics and time parameter	Model-defined coset subtorus.
Coordinate spectrum	frame plus sampling schedule	Protocol-dependent and aliased.
LOS/link graph	co-transformed body, fixed thresholds	Snapshot invariant; temporal reachability needs ordered contacts.
Proper time $\tau[\gamma]$	fixed worldline endpoints and metric	Spacetime scalar.
Clock difference at $t = T$	declared synchronization and potential	Protocol-dependent comparison.

Boundary 8.1 (Claim boundary). This reference proves elementary transformation, closure, sampling, and graph statements inside declared models. It does not provide operational orbit determination, high-fidelity force models, covariance propagation, collision probability, communication-link budgets, clock steering, or a new theory of satellite networks.

The minimum reproducible protocol records:

- (i) labeled state variables, dynamics, constants, force model, epoch, reference frame, orientation convention, and time scale;
- (ii) observer origin and attitude, sensor type, light-time policy, occulting-body model, range/elevation masks, and noise model;
- (iii) sampling cadence, horizon, missing-data rule, quantizer or partition, estimator, tolerances, and uncertainty statement;
- (iv) for clocks, metric signature, potential convention, coordinate time, synchronization, endpoint events, and retained order in c^{-1} .

References

- [1] G. Petit and B. Luzum (eds.), *IERS Conventions (2010)*, IERS Technical Note 36, especially Chapters 5 and 10.
- [2] CCSDS, *Orbit Data Messages*, CCSDS 502.0-B-3, Issue 3, April 2023.
- [3] D. A. Vallado, P. Crawford, R. Hujsak, and T. S. Kelso, “Revisiting Spacetrack Report #3,” AIAA 2006-6753, <https://doi.org/10.2514/6.2006-6753>.
- [4] GPS Directorate, *Navstar GPS Space Segment/Navigation User Segment Interfaces*, IS-GPS-200N, 1 August 2022.

- [5] N. Ashby, "Relativity in the Global Positioning System," *Living Reviews in Relativity* 6, 1 (2003), <https://doi.org/10.12942/lrr-2003-1>.
- [6] D. Kempe, J. Kleinberg, and A. Kumar, "Connectivity and Inference Problems for Temporal Networks," STOC 2000, 504–513, <https://doi.org/10.1145/335305.335364>.
- [7] I. J. Schoenberg, "Remarks to Maurice Frechet's Article," *Annals of Mathematics* 36 (1935), 724–732, <https://www.jstor.org/stable/1968654>.
- [8] V. I. Arnold, *Mathematical Methods of Classical Mechanics*, 2nd ed., Springer, 1989.